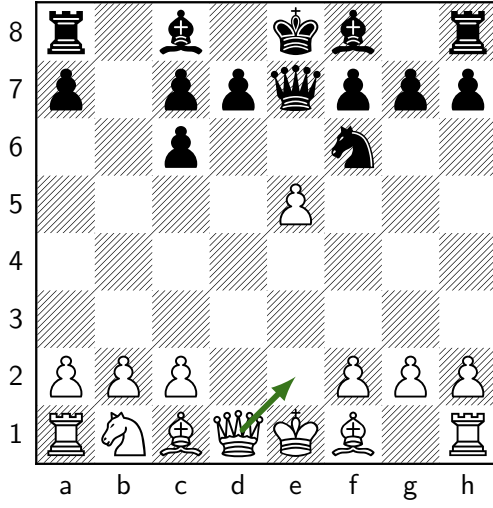


Melanie-ANALOGY — Scotch Game

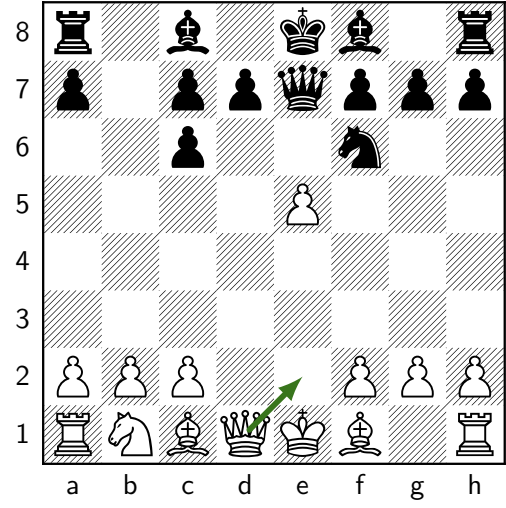
Melanie (Deep_Vision)

Abstract

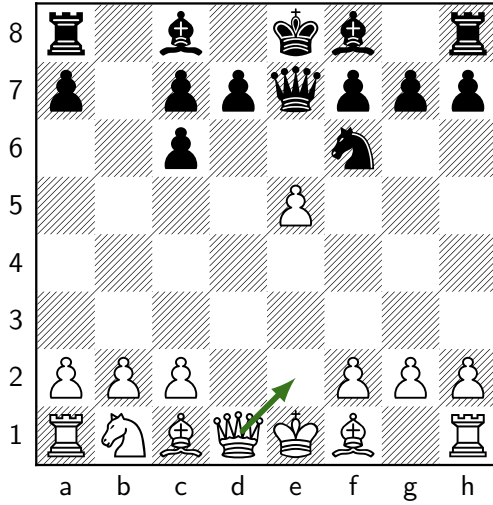
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



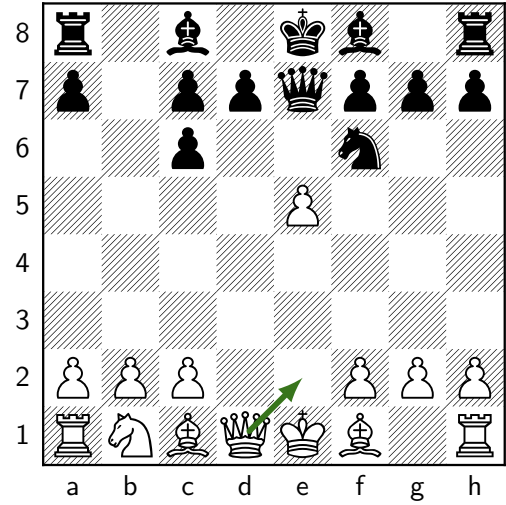
(a)



(b)

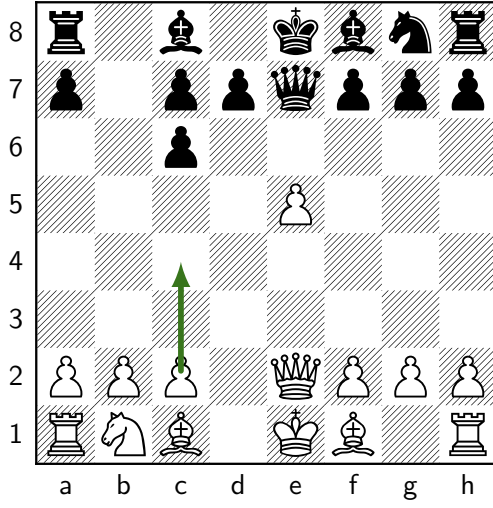


(c)

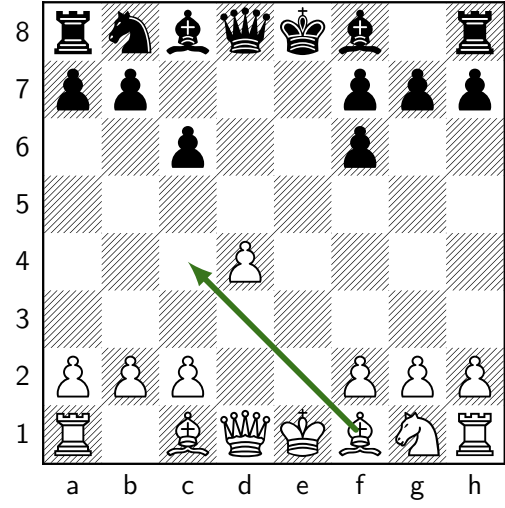


(d)

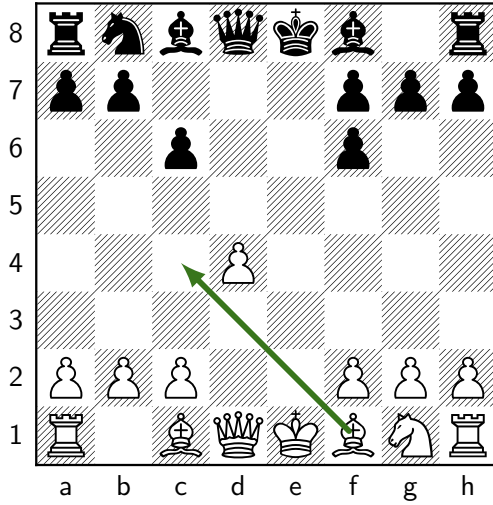
Figure 1: **Move 7, White: Qe2**. (a) Our position; Melanie plays Qe2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3924` (ply 12) [structural distance $d=0$, QAP obj=104]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3925` (ply 12) [structural distance $d=0$, QAP obj=104]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#733` (ply 12) [structural distance $d=0$, QAP obj=104]: the master played Qe2; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pf2 as their Nb1, our Pc2 as their Bc1; transferred move d1e2 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Qe2 (summed QAP objective 2077, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



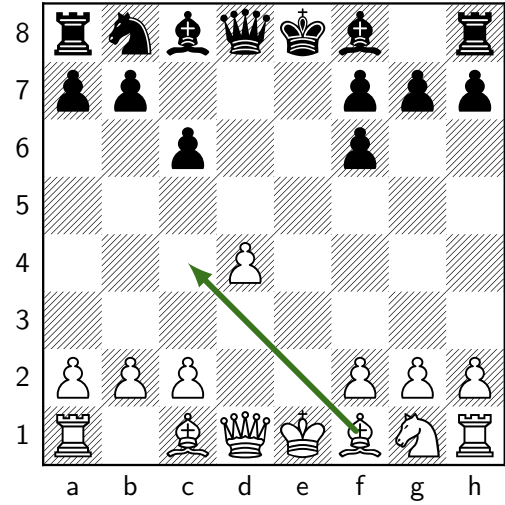
(a)



(b)

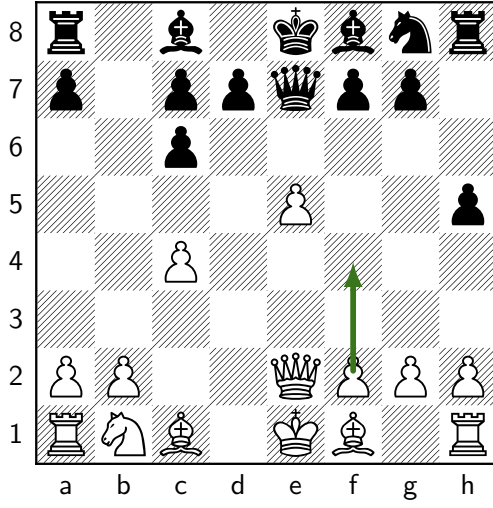


(c)

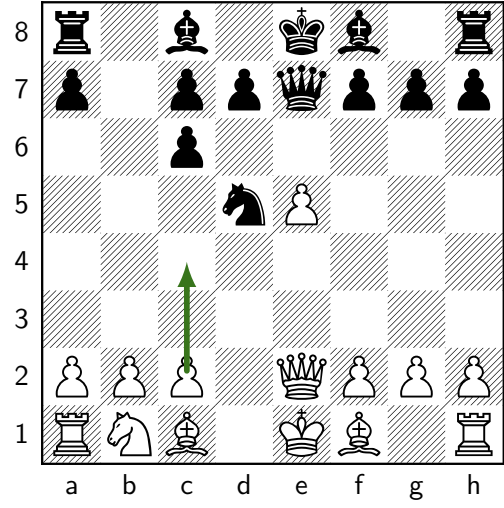


(d)

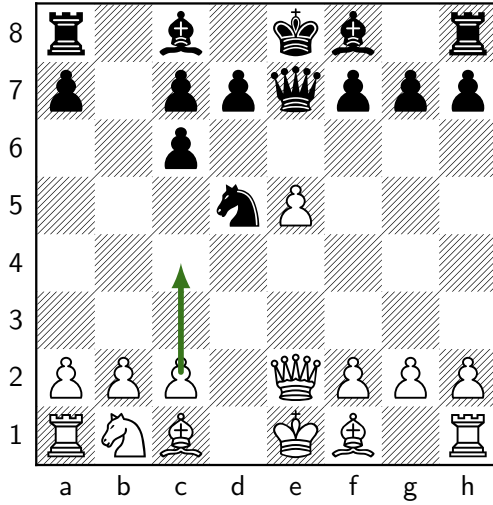
Figure 2: **Move 8, White: c4.** (a) Our position; Melanie plays c4 (c2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3988 (ply 10) [structural distance $d=44$, QAP obj=35]: the master played Bc4; under μ our Pc2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Ra1, our Pf2 as their Bc1; transferred move c2c4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3989 (ply 10) [structural distance $d=44$, QAP obj=35]: the master played Bc4; under μ our Pc2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Ra1, our Pf2 as their Bc1; transferred move c2c4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#284 (ply 10) [structural distance $d=44$, QAP obj=35]: the master played Bc4; under μ our Pc2 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Bf1 as their Ra1, our Pf2 as their Bc1; transferred move c2c4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to c4 (summed QAP objective 278, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



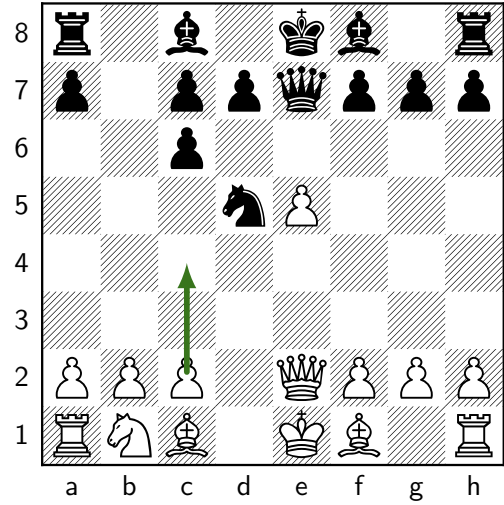
(a)



(b)

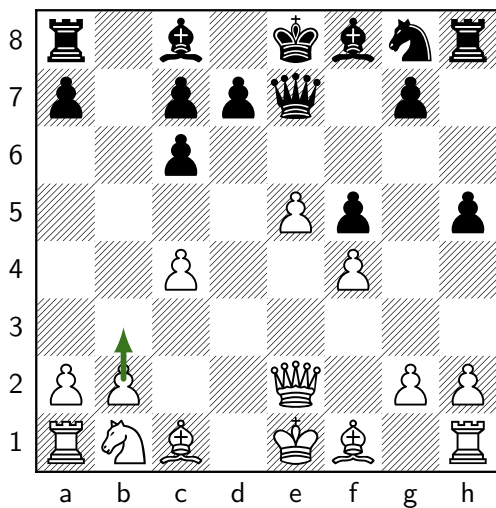


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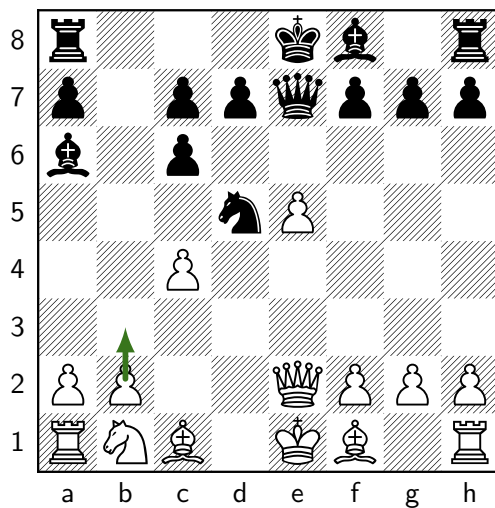


(d)

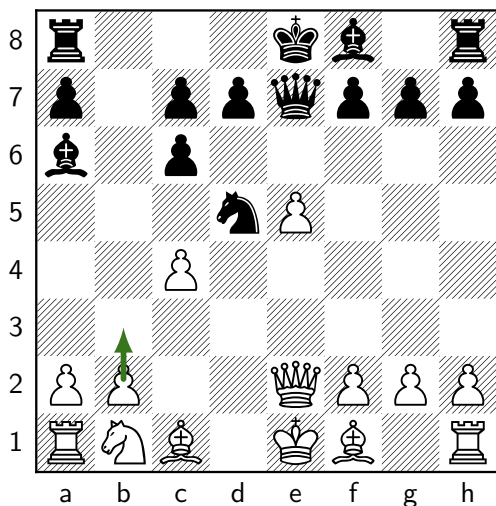
Figure 3: **Move 9, White: f4 (f2f4).** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3924` (ply 14) [structural distance $d=20$, QAP obj=60]: the master played c4; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Ra1*, our *Pc4* as their *Nb1*; transferred move f2f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3925` (ply 14) [structural distance $d=20$, QAP obj=60]: the master played c4; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Ra1*, our *Pc4* as their *Nb1*; transferred move f2f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#733` (ply 14) [structural distance $d=20$, QAP obj=60]: the master played c4; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Bc1* as their *Ra1*, our *Pc4* as their *Nb1*; transferred move f2f4 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to f4 (summed QAP objective 1084, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



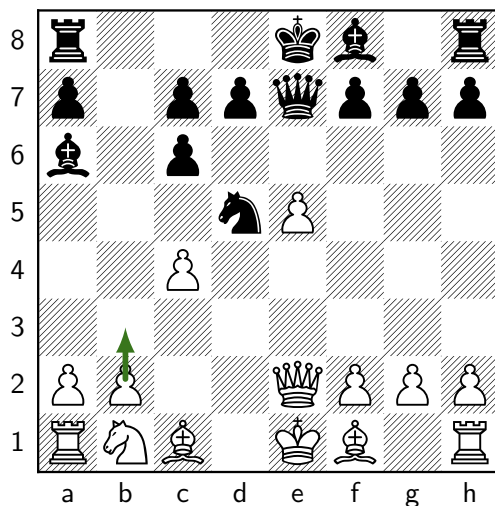
(a)



(b)

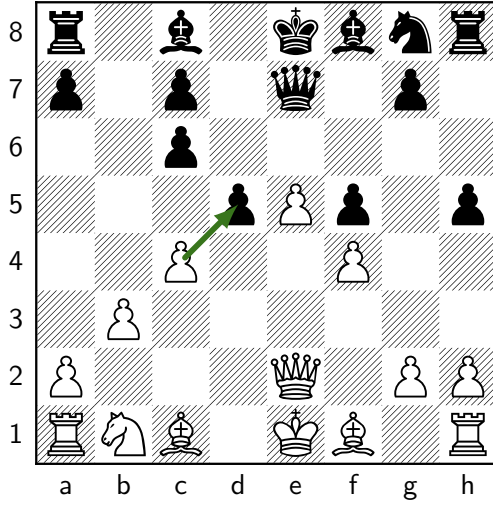


(c)

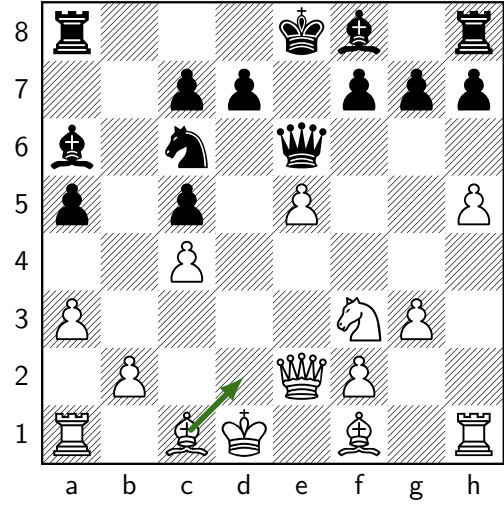


(d)

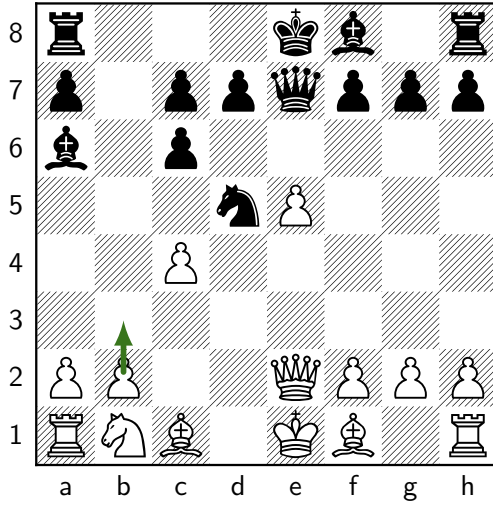
Figure 4: **Move 10, White: b3.** (a) Our position; Melanie plays **b3** (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1432` (ply 16) [structural distance $d=40$, QAP obj=53]: the master played **b3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph2* as their *Ra1*, our *Bf1* as their *Nb1*; transferred move **b2b3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1433` (ply 16) [structural distance $d=40$, QAP obj=53]: the master played **b3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph2* as their *Ra1*, our *Bf1* as their *Nb1*; transferred move **b2b3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2657` (ply 16) [structural distance $d=40$, QAP obj=53]: the master played **b3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Ph2* as their *Ra1*, our *Bf1* as their *Nb1*; transferred move **b2b3** (**endorses this move**). Synthesis: 14 retrieved analogs transferred to **b3** (summed QAP objective 669, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



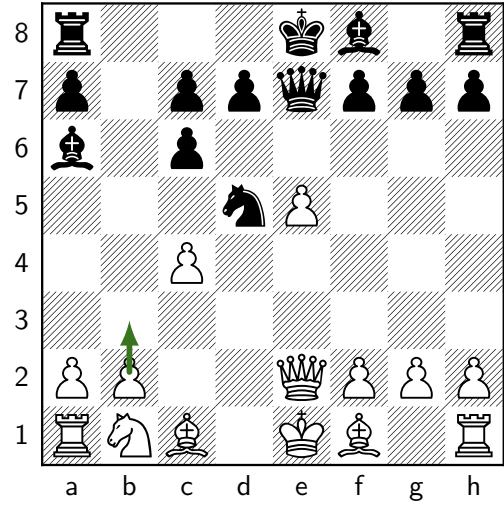
(a)



(b)

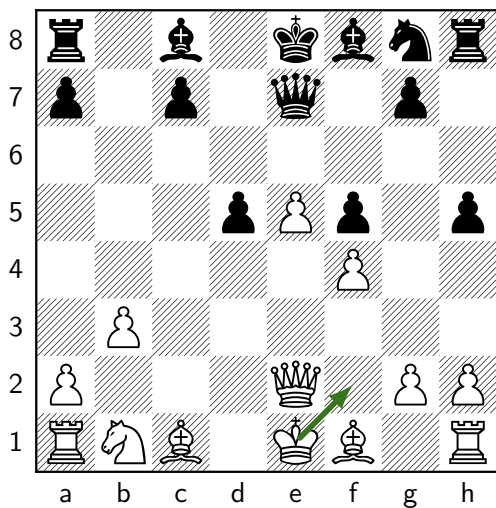


(c)

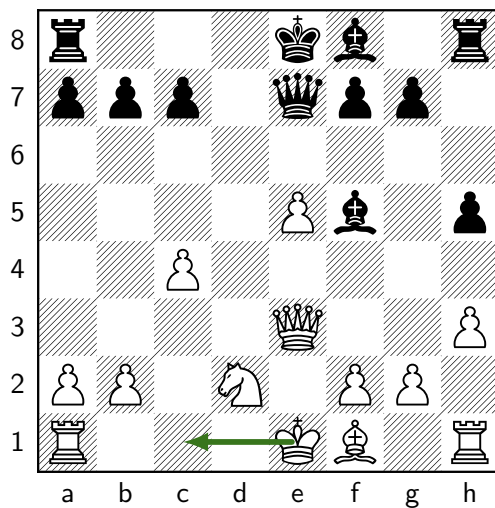


(d)

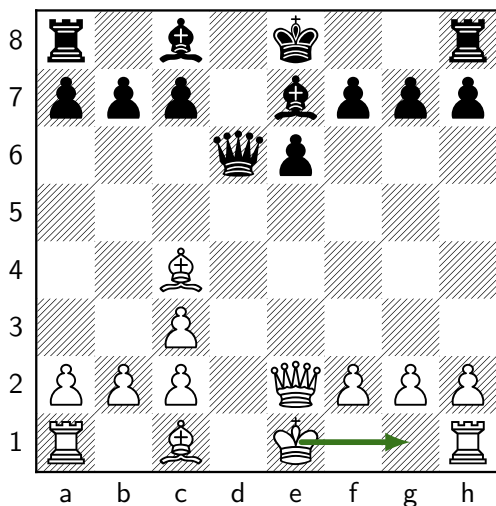
Figure 5: **Move 11, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1131` (ply 30) [structural distance $d=68$, QAP obj=66]: the master played Bd2; under μ our $Pc4$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Bf1$ as their $Ra1$, our $Pa2$ as their $Bf1$; transferred move c4d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1432` (ply 16) [structural distance $d=68$, QAP obj=52]: the master played b3; under μ our $Pc4$ plays the role of that game’s $Pb2$; cross-type roles: our $Ph2$ as their $Ra1$, our $rh8$ as their $Bc1$; transferred move c4d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1433` (ply 16) [structural distance $d=68$, QAP obj=52]: the master played b3; under μ our $Pc4$ plays the role of that game’s $Pb2$; cross-type roles: our $Ph2$ as their $Ra1$, our $rh8$ as their $Bc1$; transferred move c4d5 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to cxd5 (summed QAP objective 821, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



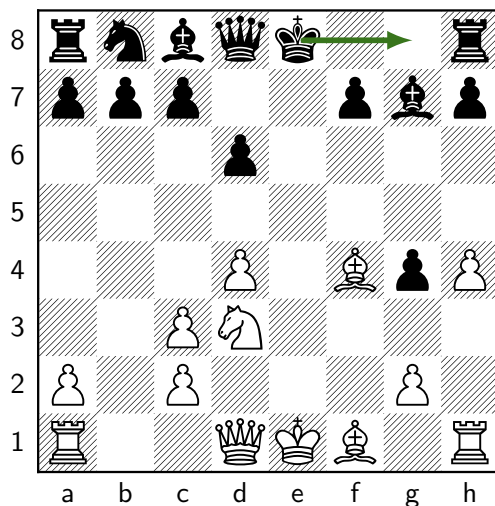
(a)



(b)

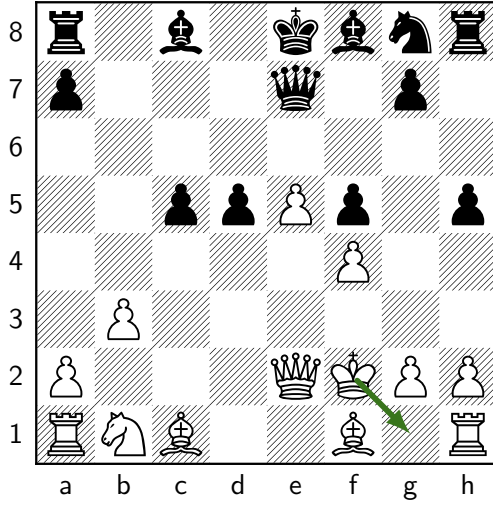


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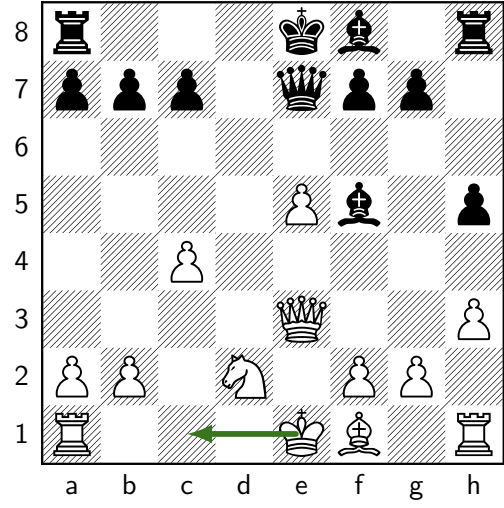


(d)

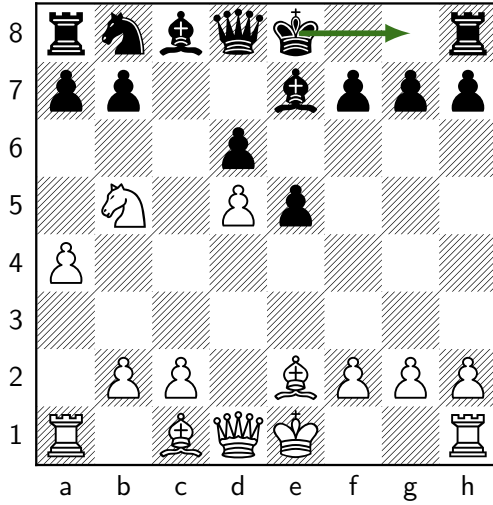
Figure 6: **Move 12, White: Kf2.** (a) Our position; Melanie plays Kf2 (e1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2408` (ply 28) [structural distance $d=52$, QAP obj=86]: the master played 0-0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *Nb1* as their *Bf1*, our *Pb3* as their *Rh1*; transferred move e1f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#87` (ply 20) [structural distance $d=65$, QAP obj=50]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *Bf1* as their *Ra1*, our *Ph2* as their *Bc1*; transferred move e1f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2287` (ply 19) [structural distance $d=70$, QAP obj=44]: the master played 0-0; under μ our *Ke1* plays the role of that game's *Ke1*; cross-type roles: our *Bc1* as their *Qd8*, our *Pg2* as their *Rh8*; transferred move e1f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 258, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



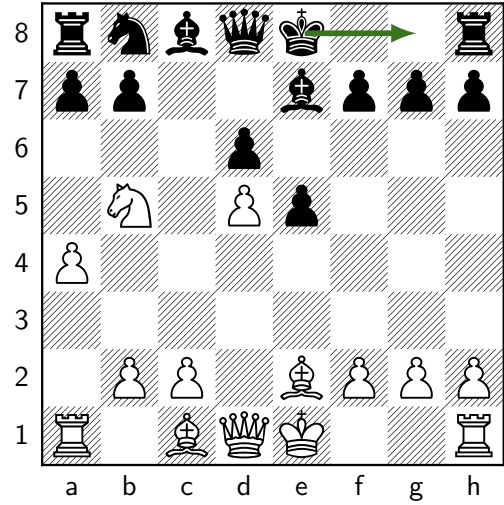
(a)



(b)

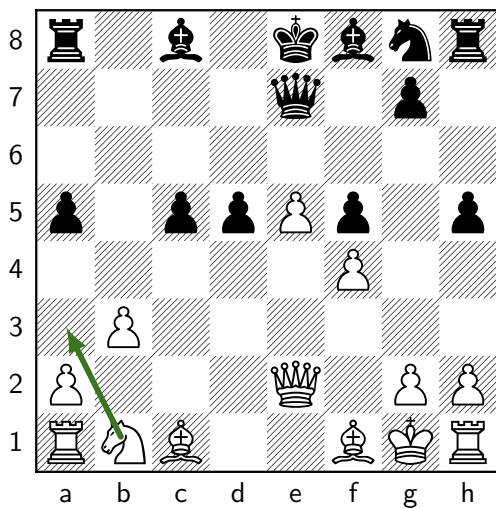


(c)

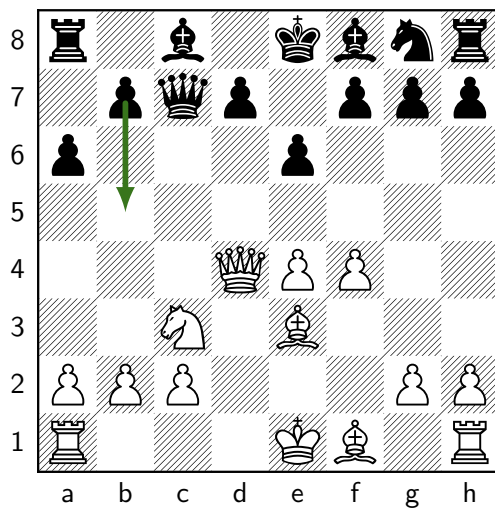


(d)

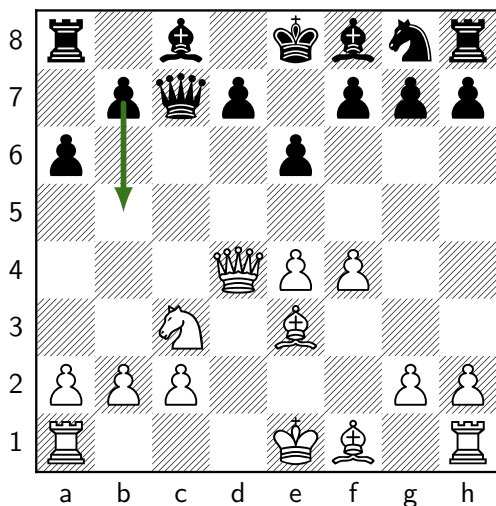
Figure 7: **Move 13, White: Kg1 (f2g1).** (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2408` (ply 28) [structural distance $d=58$, QAP obj=83]: the master played 0-0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Bf1*, our *pf5* as their *Rh1*; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1784` (ply 19) [structural distance $d=50$, QAP obj=37]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Pf4* as their *Ra8*, our *pc5* as their *Nb8*; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#985` (ply 19) [structural distance $d=50$, QAP obj=37]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *Pf4* as their *Ra8*, our *pc5* as their *Nb8*; transferred move f2g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



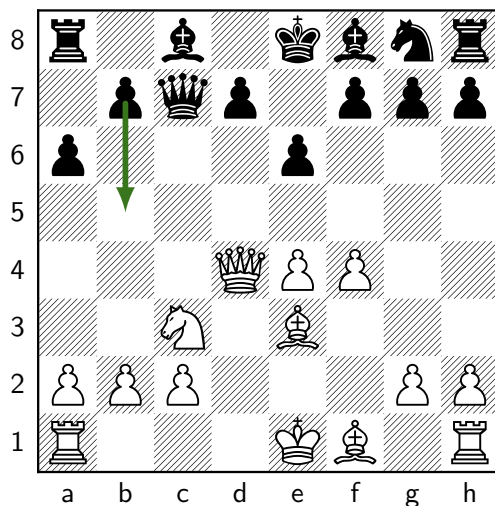
(a)



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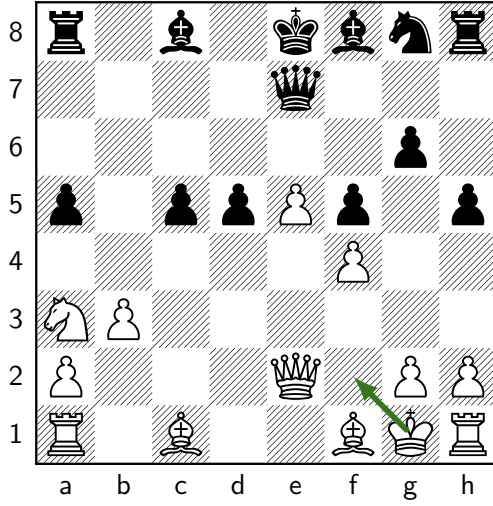


(c)

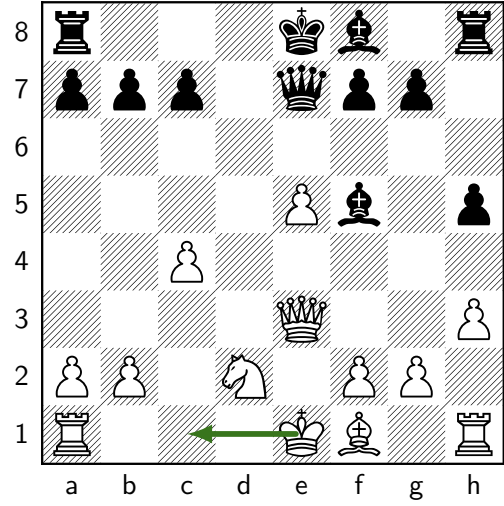


(d)

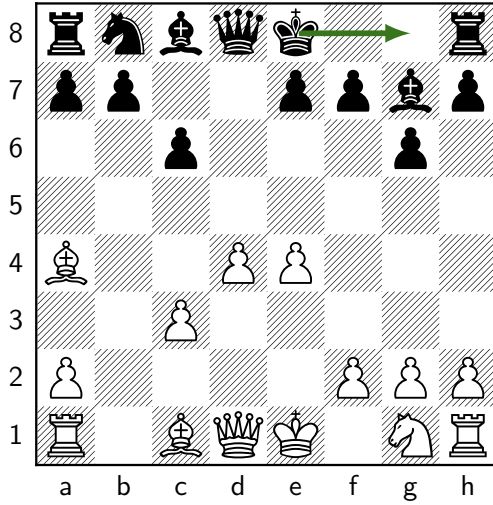
Figure 8: **Move 14, White: Na3**. (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2071` (ply 15) [structural distance $d=66$, QAP obj=47]: the master played b5; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bc8, our Pb3 as their Bf8; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2072` (ply 15) [structural distance $d=66$, QAP obj=47]: the master played b5; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bc8, our Pb3 as their Bf8; transferred move b1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2807` (ply 15) [structural distance $d=66$, QAP obj=47]: the master played b5; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Bc8, our Pb3 as their Bf8; transferred move b1a3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Na3 (summed QAP objective 374, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



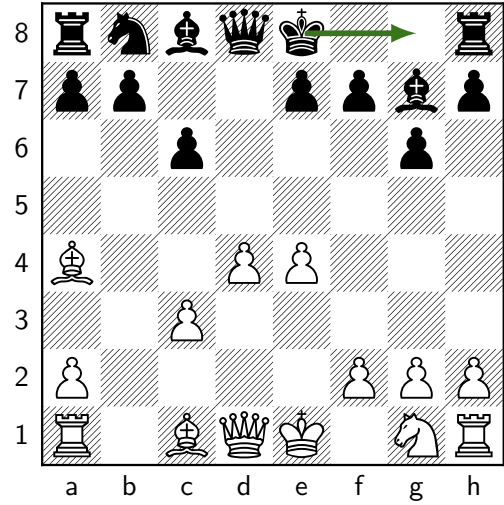
(a)



(b)

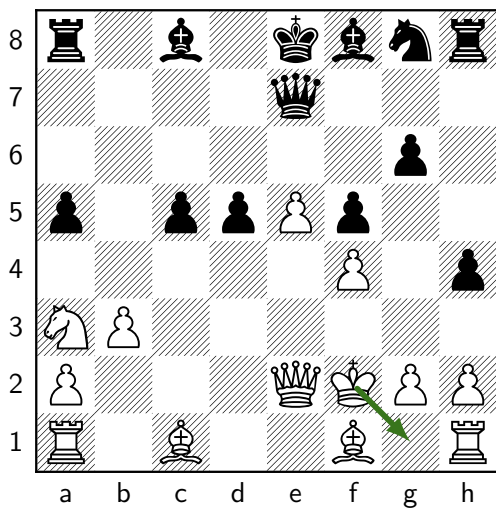


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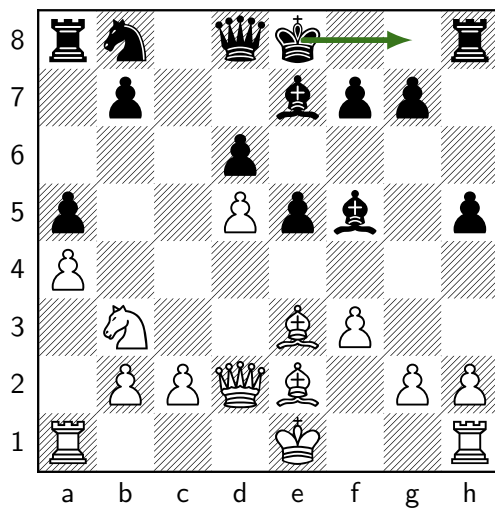


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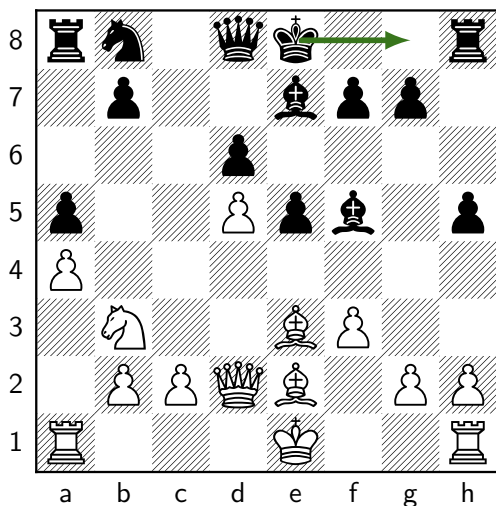
Figure 9: **Move 15, White: Kf2.** (a) Our position; Melanie plays Kf2 (g1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2408` (ply 28) [structural distance $d=73$, QAP obj=81]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Ph2* as their *Bf1*, our *Pf4* as their *Rh1*; transferred move g1f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3093` (ply 15) [structural distance $d=77$, QAP obj=60]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *ra8* as their *Bc8*, our *Bf1* as their *Rh8*; transferred move g1f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3094` (ply 15) [structural distance $d=77$, QAP obj=60]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *ra8* as their *Bc8*, our *Bf1* as their *Rh8*; transferred move g1f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 320, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



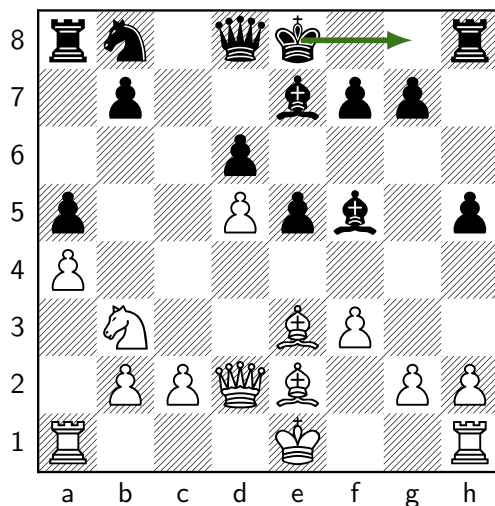
(a)



(b)

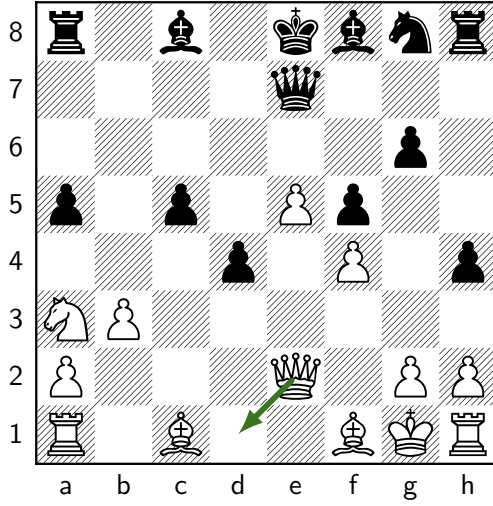


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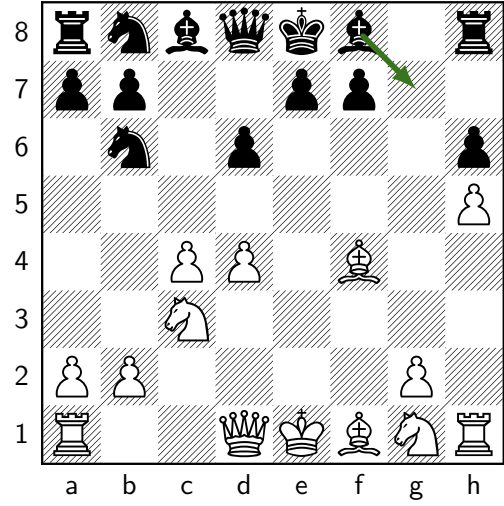


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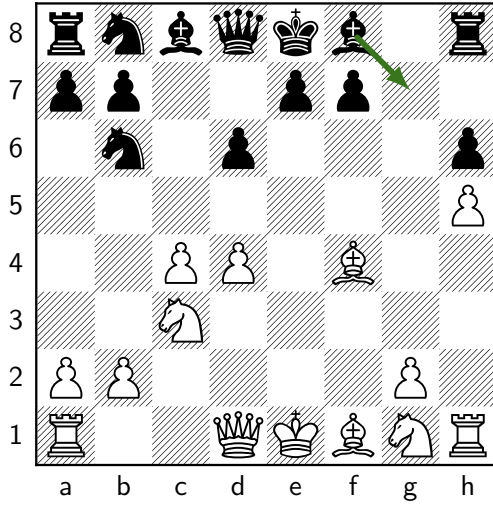
Figure 10: **Move 16, White: Kg1 (f2g1).** (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1392` (ply 25) [structural distance $d=73$, QAP obj=36]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pd5* as their *Nb8*, our *Bc1* as their *Pb7*; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1798` (ply 25) [structural distance $d=73$, QAP obj=36]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pd5* as their *Nb8*, our *Bc1* as their *Pb7*; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1799` (ply 25) [structural distance $d=73$, QAP obj=36]: the master played 0-0; under μ our *Kf2* plays the role of that game’s *Ke1*; cross-type roles: our *pd5* as their *Nb8*, our *Bc1* as their *Pb7*; transferred move f2g1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg1 (summed QAP objective 252, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



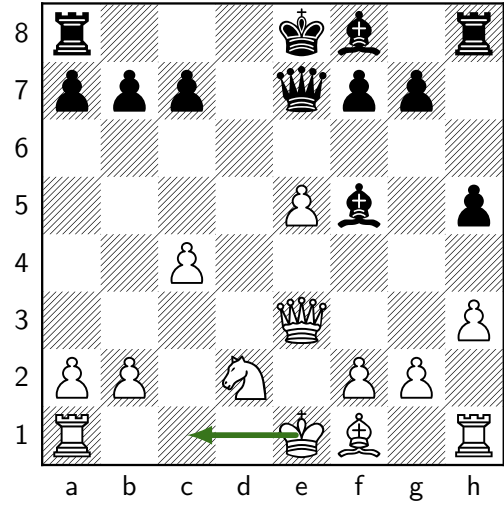
(a)



(b)

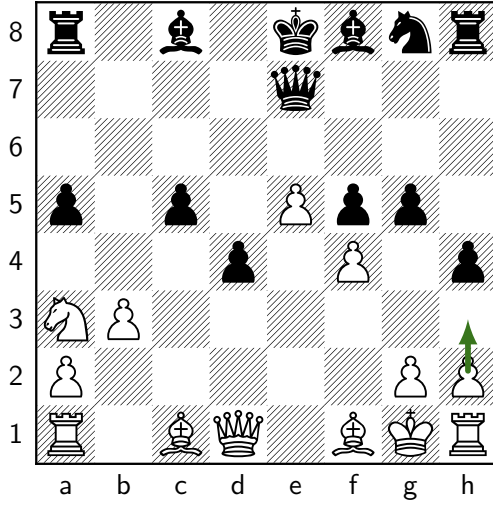


(c)

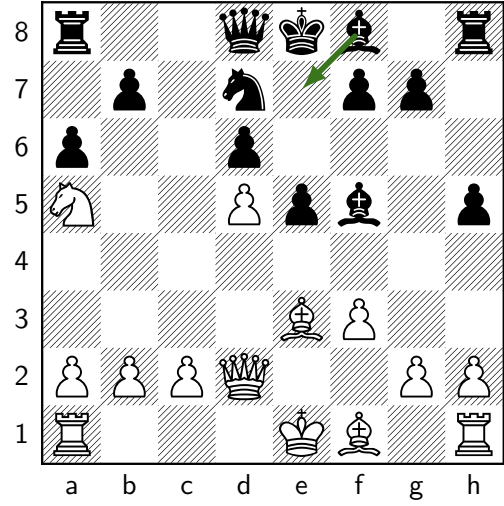


(d)

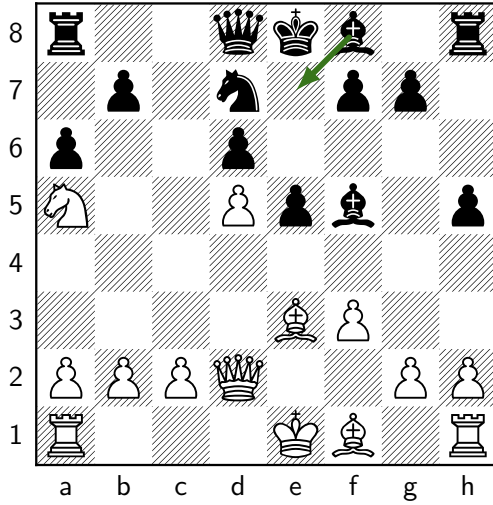
Figure 11: **Move 17, White: Qd1**. (a) Our position; Melanie plays Qd1 (e2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#168` (ply 19) [structural distance $d=76$, QAP obj=64]: the master played Bg7; under μ our Qe2 plays the role of that game's Bf1 (a **cross-type** correspondence); cross-type roles: our Na3 as their Bc8, our Rh1 as their Qd8; transferred move e2d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#169` (ply 19) [structural distance $d=76$, QAP obj=64]: the master played Bg7; under μ our Qe2 plays the role of that game's Bf1 (a **cross-type** correspondence); cross-type roles: our Na3 as their Bc8, our Rh1 as their Qd8; transferred move e2d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2408` (ply 28) [structural distance $d=77$, QAP obj=68]: the master played O-O-O; under μ our Kg1 plays the role of that game's Ke1; cross-type roles: our Ph2 as their Bf1, our Na3 as their Rh1; transferred move g1f2. Synthesis: 2 retrieved analogs transferred to Qd1 (summed QAP objective 128, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



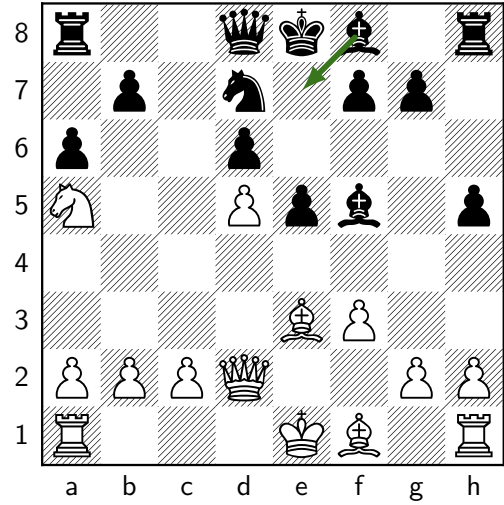
(a)



(b)

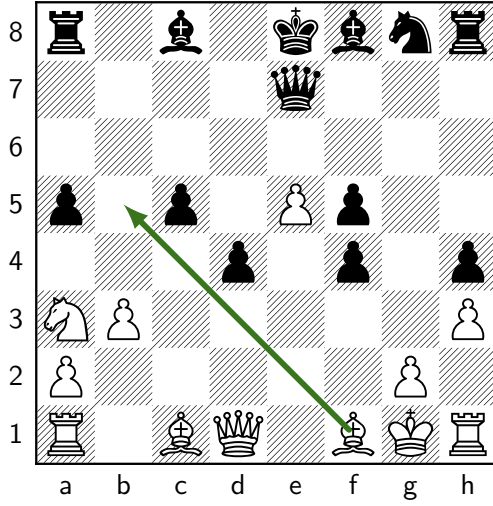


(c)

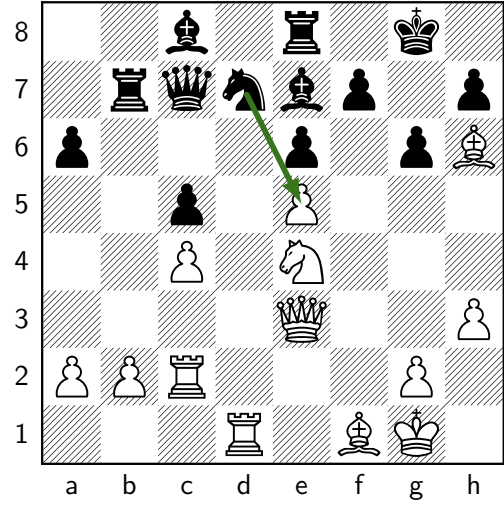


(d)

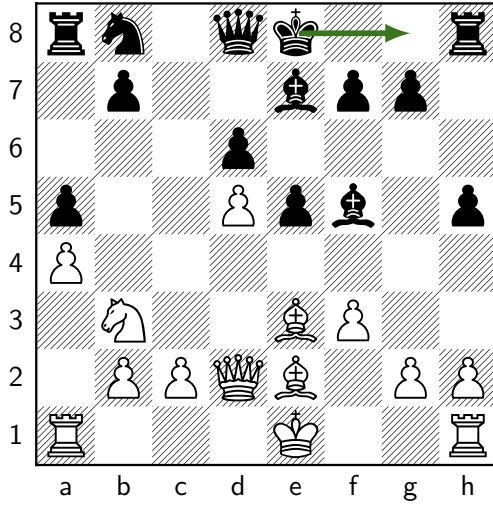
Figure 12: **Move 18, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#406` (ply 23) [structural distance $d=69$, QAP obj=33]: the master played **Be7**; under μ our *Ph2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Na3* as their *Ra8*, our *Pf4* as their *Qd8*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2582` (ply 23) [structural distance $d=69$, QAP obj=33]: the master played **Be7**; under μ our *Ph2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Na3* as their *Ra8*, our *Pf4* as their *Qd8*; transferred move **h2h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2586` (ply 23) [structural distance $d=69$, QAP obj=33]: the master played **Be7**; under μ our *Ph2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Na3* as their *Ra8*, our *Pf4* as their *Qd8*; transferred move **h2h3** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **h3** (summed QAP objective 295, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



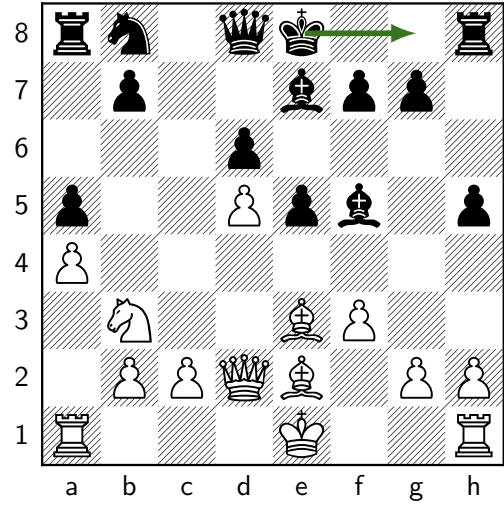
(a)



(b)

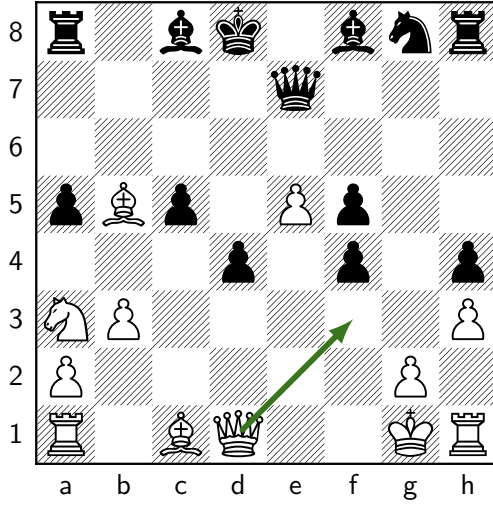


(c)

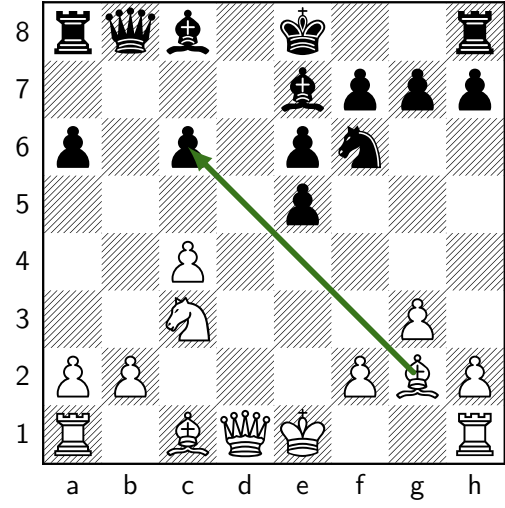


(d)

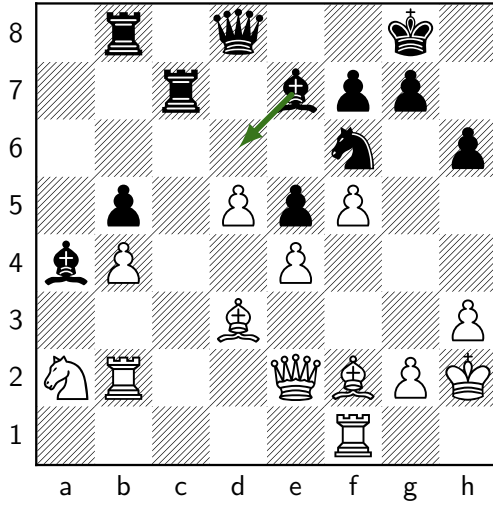
Figure 13: **Move 19, White: Bb5+.** (a) Our position; Melanie plays Bb5+ (f1b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#169` (ply 45) [structural distance $d=64$, QAP obj=69]: the master played Nxe5; under μ our *Pe5* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Bc1* as their *Rb7*, our *Ph3* as their *Be7*; transferred move e5e6. (c) Retrieved from `ccr1_top_p0000#1392` (ply 25) [structural distance $d=71$, QAP obj=45]: the master played O-O; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pa5* as their *Ra8*, our *Ph3* as their *Nb8*; transferred move g1h2. (d) Retrieved from `ccr1_top_p0002#1798` (ply 25) [structural distance $d=71$, QAP obj=45]: the master played O-O; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pa5* as their *Ra8*, our *Ph3* as their *Nb8*; transferred move g1h2. Synthesis: no retrieved analog transferred to Bb5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



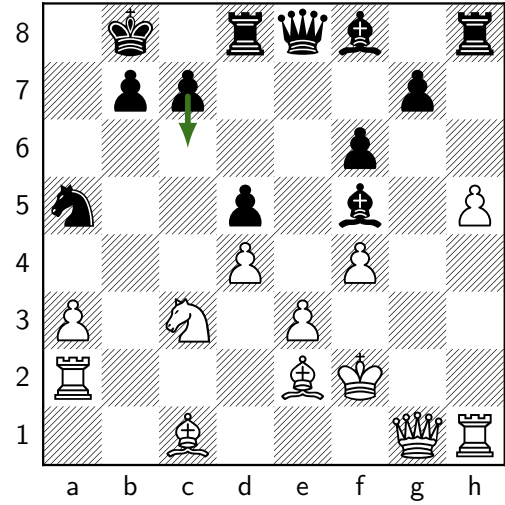
(a)



(b)

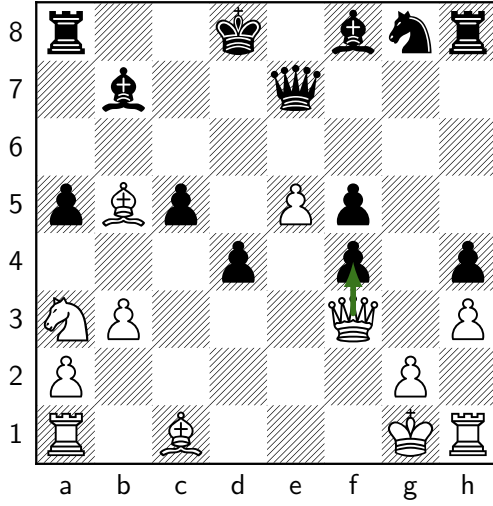


(c)

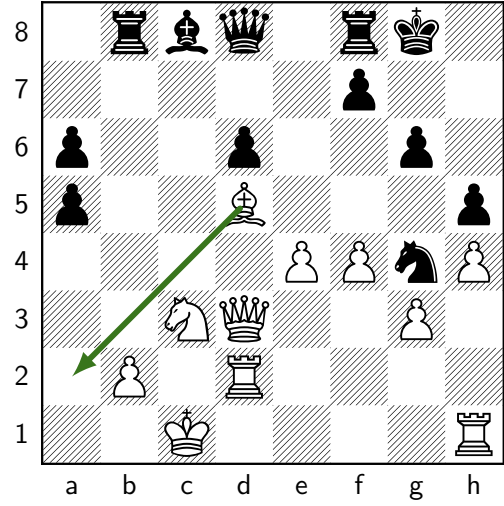


(d)

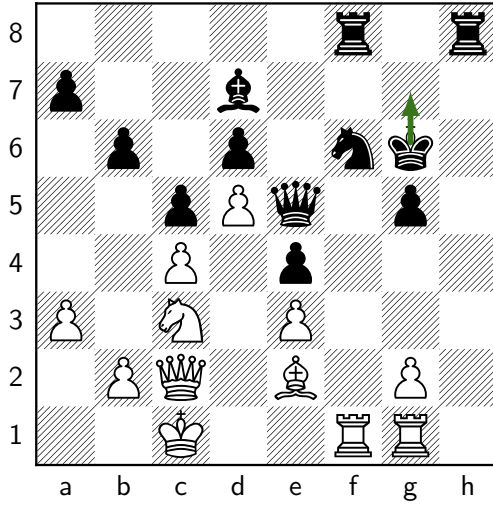
Figure 14: **Move 20, White: Qf3.** (a) Our position; Melanie plays Qf3 (d1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1916` (ply 24) [structural distance $d=80$, QAP obj=34]: the master played Bxc6+; under μ our Qd1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Ph3 as their Bc1, our Bc1 as their Qd1; transferred move d1f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2420` (ply 49) [structural distance $d=80$, QAP obj=90]: the master played Bd6; under μ our Bc1 plays the role of that game’s Be2; cross-type roles: our pa5 as their Rb8, our Qd1 as their Nf6; transferred move c1b2. (d) Retrieved from `ccr1_top_p0000#1260` (ply 45) [structural distance $d=75$, QAP obj=86]: the master played c6; under μ our Pg2 plays the role of that game’s Pc2; cross-type roles: our Qd1 as their Rh8, our Pa2 as their Na5; transferred move g2g4. Synthesis: 1 retrieved analog transferred to Qf3 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



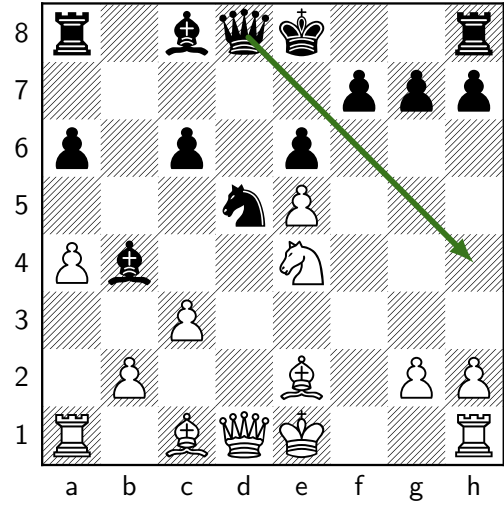
(a)



(b)

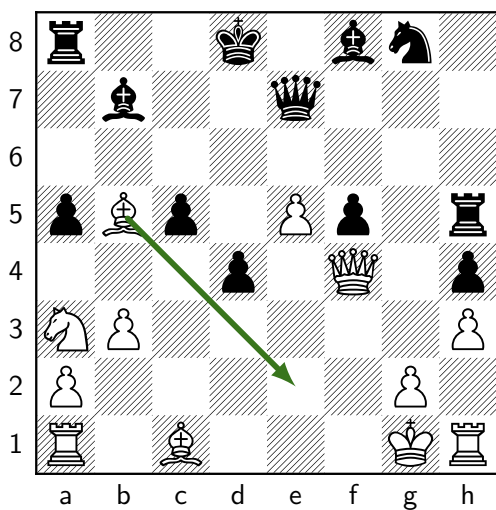


(c)

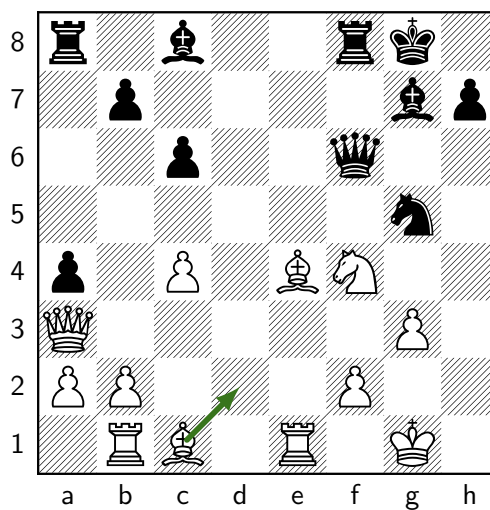


(d)

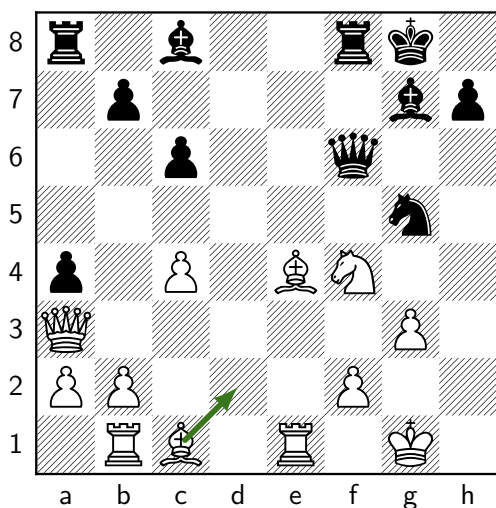
Figure 15: **Move 21, White: Qxf4.** (a) Our position; Melanie plays Qxf4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0002#563` (ply 50) [structural distance $d=92$, QAP obj=25]: the master played Ba2; under μ our Qf3 plays the role of that game’s Bd5 (a **cross-type** correspondence); cross-type roles: our pa5 as their Rh1, our rh8 as their Pb2; transferred move f3f4 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#1842` (ply 57) [structural distance $d=91$, QAP obj=49]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kg3; cross-type roles: our ph4 as their Rf8, our ng8 as their Rh8; transferred move g1h2. (d) Retrieved from `ccr1_top.p0002#2203` (ply 25) [structural distance $d=90$, QAP obj=49]: the master played Qh4+; under μ our Rh1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra8, our rh8 as their Bc8; transferred move h1h2. Synthesis: 1 retrieved analog transferred to Qxf4 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



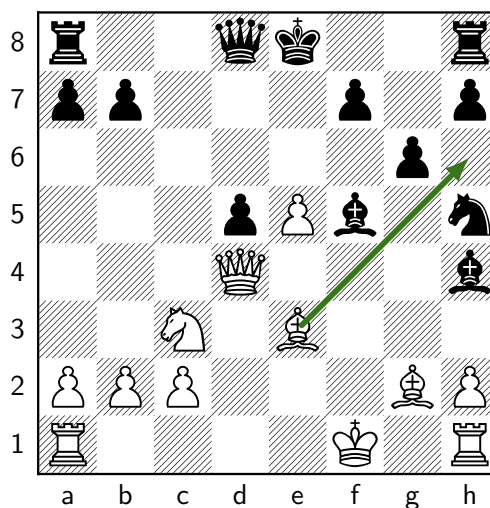
(a)



(b)

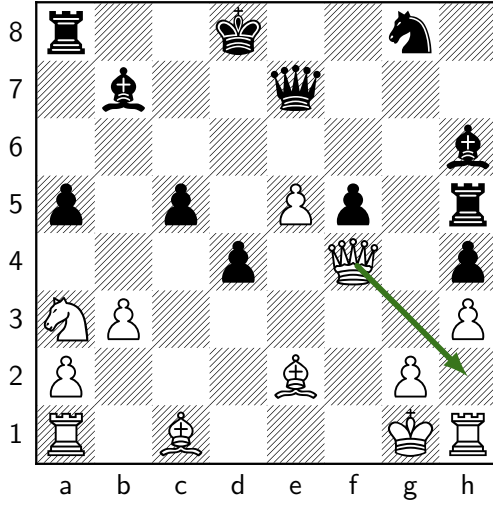


(c)

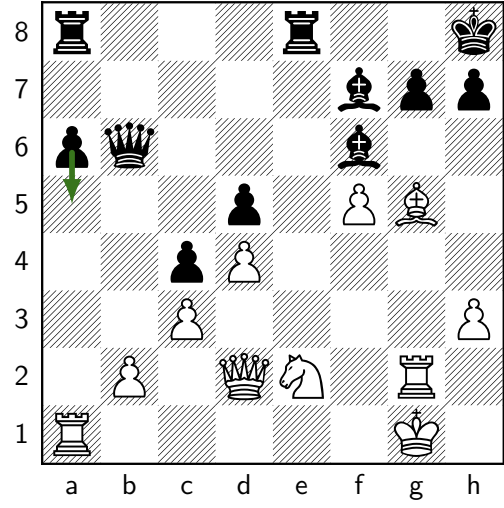


(d)

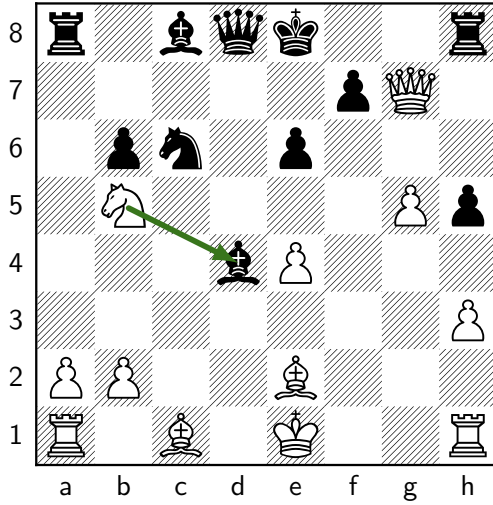
Figure 16: **Move 22, White: Be2.** (a) Our position; Melanie plays Be2 (b5e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3281` (ply 38) [structural distance $d=85$, QAP obj=58]: the master played Bd2; under μ our Bb5 plays the role of that game’s Bc1; cross-type roles: our Bc1 as their Pa2, our Qf4 as their Be4; transferred move b5e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3282` (ply 38) [structural distance $d=85$, QAP obj=58]: the master played Bd2; under μ our Bb5 plays the role of that game’s Bc1; cross-type roles: our Bc1 as their Pa2, our Qf4 as their Be4; transferred move b5e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3580` (ply 28) [structural distance $d=78$, QAP obj=85]: the master played Bh6; under μ our Pg2 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Pa2, our ng8 as their Pb2; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to Be2 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



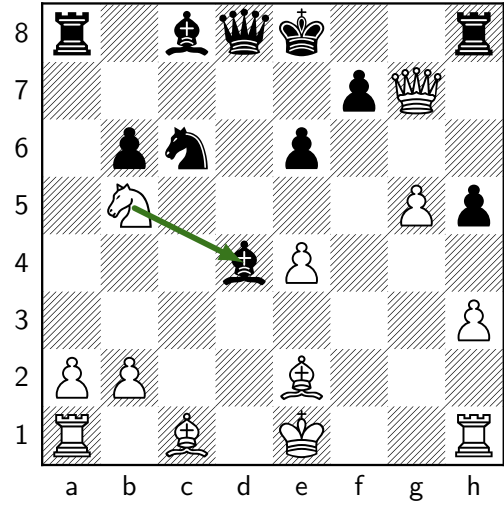
(a)



(b)

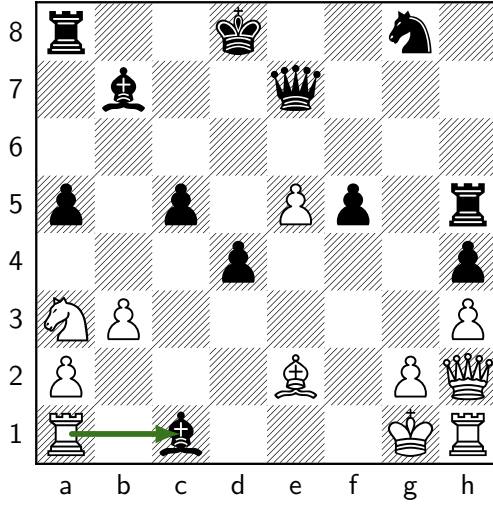


(c)

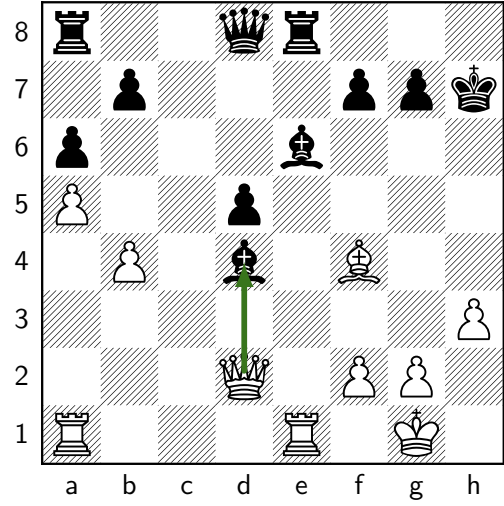


(d)

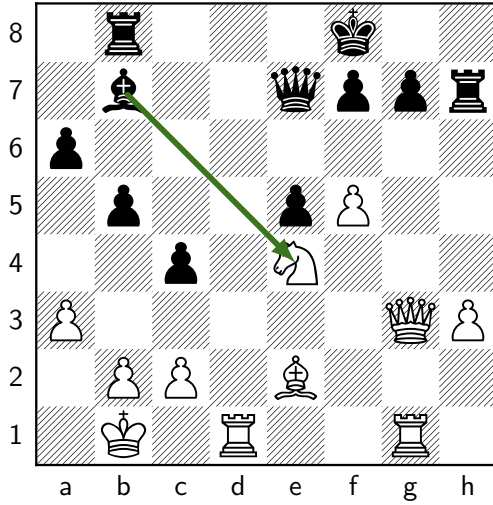
Figure 17: **Move 23, White: Qh2.** (a) Our position; Melanie plays Qh2 (f4h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3174` (ply 51) [structural distance $d=87$, QAP obj=51]: the master played a5; under μ our *Pg2* plays the role of that game’s *Pa3*; cross-type roles: our *Be2* as their *Ra8*, our *pd4* as their *Bf7*; transferred move g2g3. (c) Retrieved from `ccr1_top_p0000#500` (ply 38) [structural distance $d=72$, QAP obj=47]: the master played Nxd4; under μ our *Be2* plays the role of that game’s *Nb5* (a **cross-type** correspondence); cross-type roles: our *rh5* as their *Bc1*, our *pd4* as their *Be2*; transferred move e2g4. (d) Retrieved from `ccr1_top_p0000#501` (ply 38) [structural distance $d=72$, QAP obj=47]: the master played Nxd4; under μ our *Be2* plays the role of that game’s *Nb5* (a **cross-type** correspondence); cross-type roles: our *rh5* as their *Bc1*, our *pd4* as their *Be2*; transferred move e2g4. Synthesis: no retrieved analog transferred to Qh2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



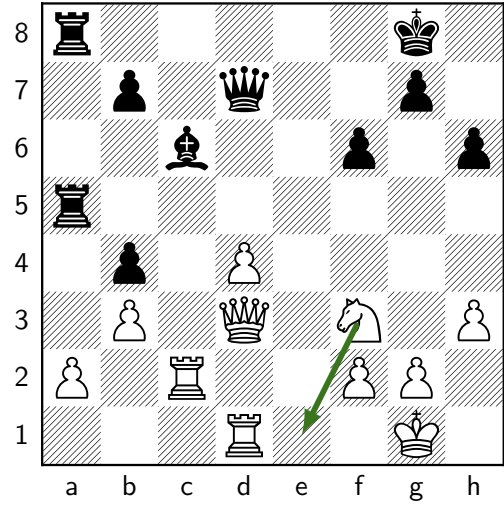
(a)



(b)

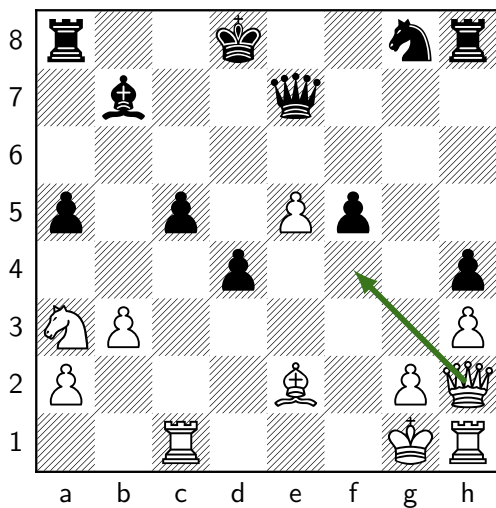


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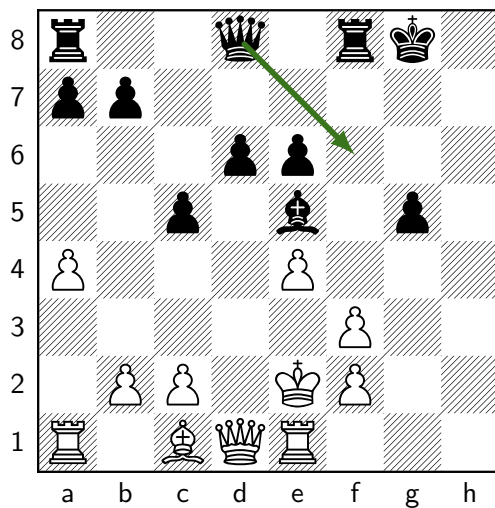


(d)

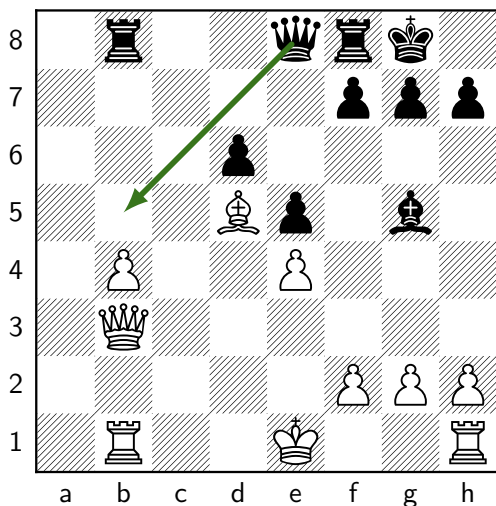
Figure 18: **Move 24, White: Rxc1.** (a) Our position; Melanie plays Rxc1 (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#755` (ply 38) [structural distance $d=82$, QAP obj=77]: the master played Qxd4; under μ our *Ra1* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *Na3* as their *Ra1*, our *qe7* as their *Re1*; transferred move a1c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2855` (ply 47) [structural distance $d=84$, QAP obj=49]: the master played Bxe4; under μ our *Be2* plays the role of that game’s *Bb2*; cross-type roles: our *ng8* as their *Rh7*, our *Na3* as their *Pc4*; transferred move e2d3. (d) Retrieved from `ccr1_top_p0003#2302` (ply 60) [structural distance $d=77$, QAP obj=48]: the master played Ne1; under μ our *Na3* plays the role of that game’s *Nf3*; cross-type roles: our *ng8* as their *Rd1*, our *Be2* as their *Ph3*; transferred move a3b1. Synthesis: 1 retrieved analog transferred to Rxc1 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



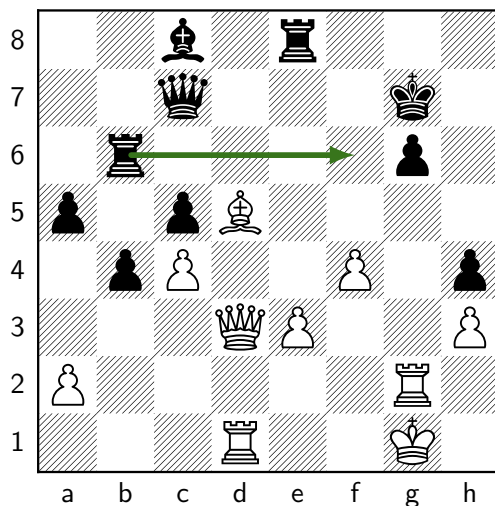
(a)



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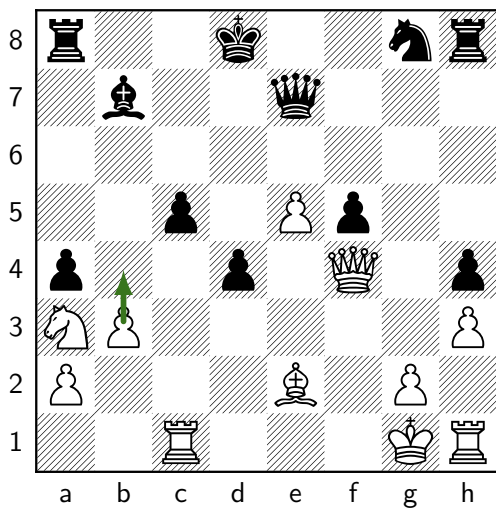


(c)

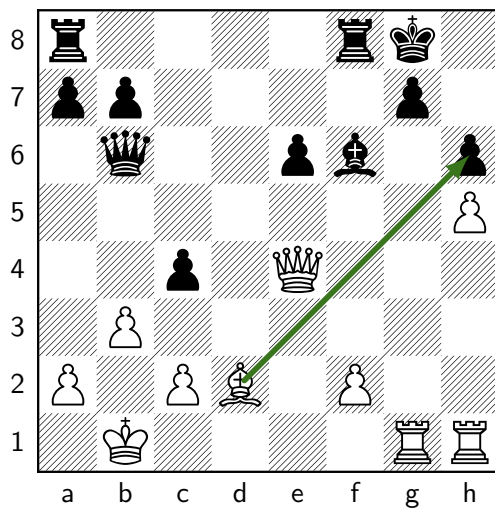


(d)

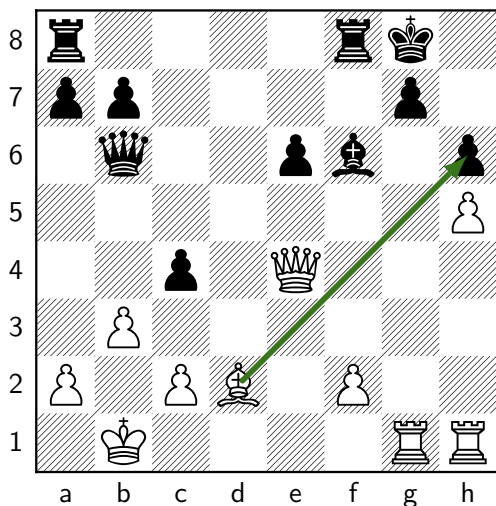
Figure 19: **Move 25, White: Qf4**. (a) Our position; Melanie plays Qf4 (h2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#452 (ply 39) [structural distance $d=77$, QAP obj=61]: the master played Qf6; under μ our Qh2 plays the role of that game’s Qd1; cross-type roles: our Pa2 as their Ra8, our Rh1 as their Pa7; transferred move h2f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2778 (ply 41) [structural distance $d=80$, QAP obj=61]: the master played Qb5; under μ our Qh2 plays the role of that game’s Qe1; cross-type roles: our bb7 as their Rb8, our Be2 as their Pg7; transferred move h2f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3979 (ply 65) [structural distance $d=83$, QAP obj=60]: the master played Rf6; under μ our Qh2 plays the role of that game’s Rb3 (a **cross-type** correspondence); cross-type roles: our ng8 as their bd5, our Na3 as their Pb4; transferred move h2f4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qf4 (summed QAP objective 228, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



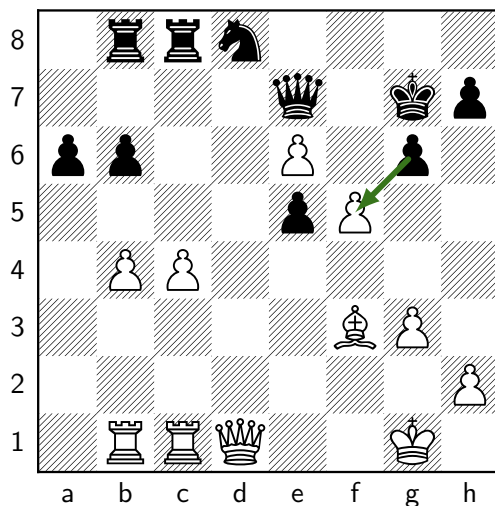
(a)



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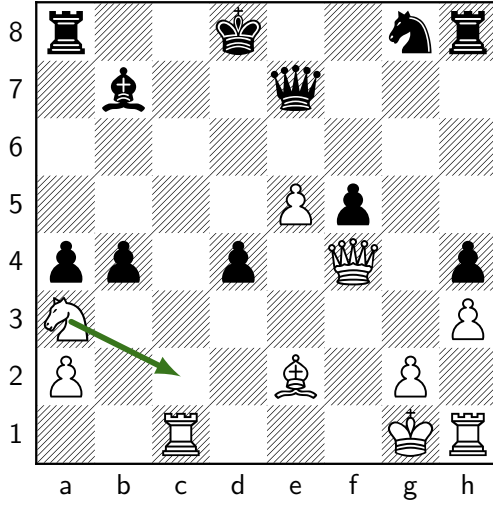


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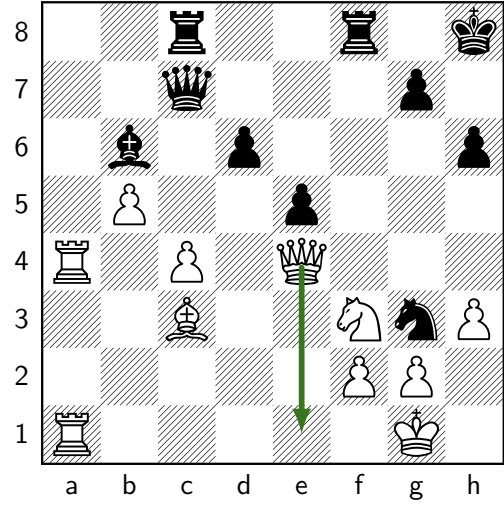


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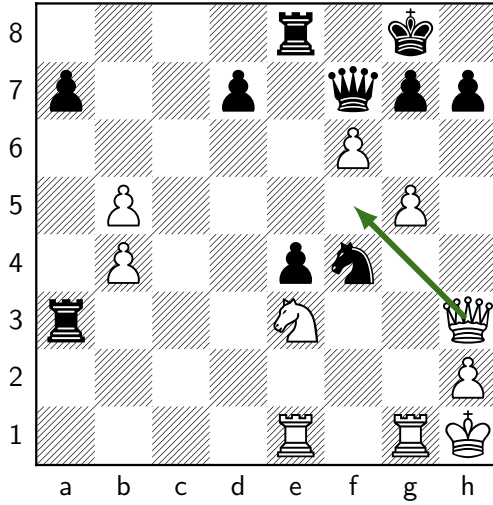
Figure 20: **Move 26, White: b4.** (a) Our position; Melanie plays b4 (b3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1164` (ply 44) [structural distance $d=74$, QAP obj=90]: the master played $Bxh6$; under μ our $Pb3$ plays the role of that game’s $Bd2$ (a **cross-type** correspondence); cross-type roles: our $bb7$ as their $Rg1$, our $Na3$ as their $Rh1$; transferred move $b3b4$ (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2668` (ply 44) [structural distance $d=74$, QAP obj=90]: the master played $Bxh6$; under μ our $Pb3$ plays the role of that game’s $Bd2$ (a **cross-type** correspondence); cross-type roles: our $bb7$ as their $Rg1$, our $Na3$ as their $Rh1$; transferred move $b3b4$ (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1845` (ply 57) [structural distance $d=73$, QAP obj=86]: the master played $gxf5$; under μ our $Pb3$ plays the role of that game’s $Pg3$; cross-type roles: our $ng8$ as their $Rb8$, our $qe7$ as their $Rc8$; transferred move $b3b4$ (**endorses this move**). Synthesis: 7 retrieved analogs transferred to $b4$ (summed QAP objective 496, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



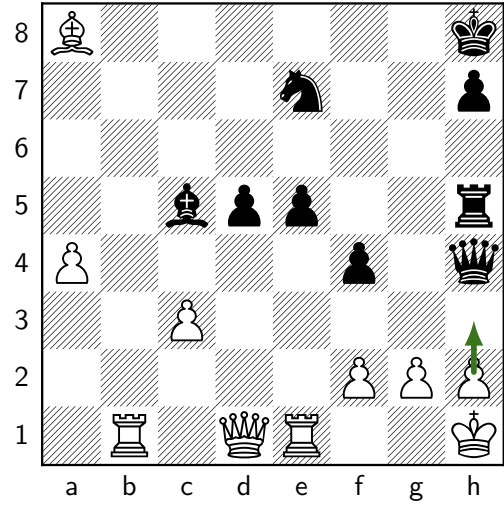
(a)



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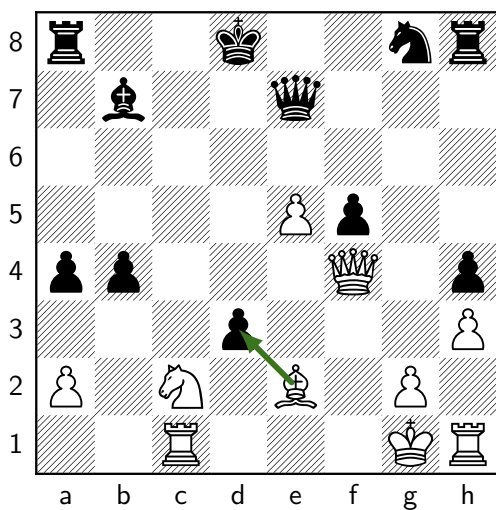


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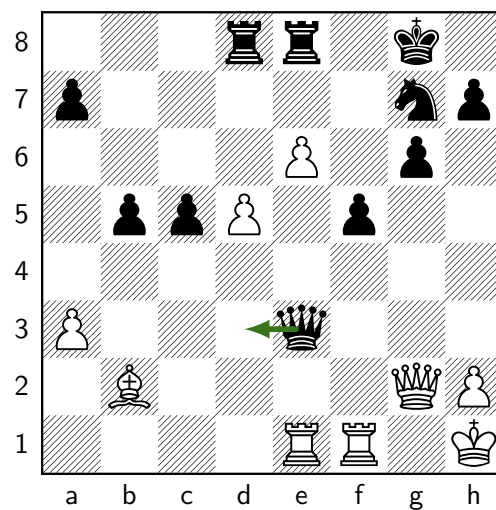


(d)

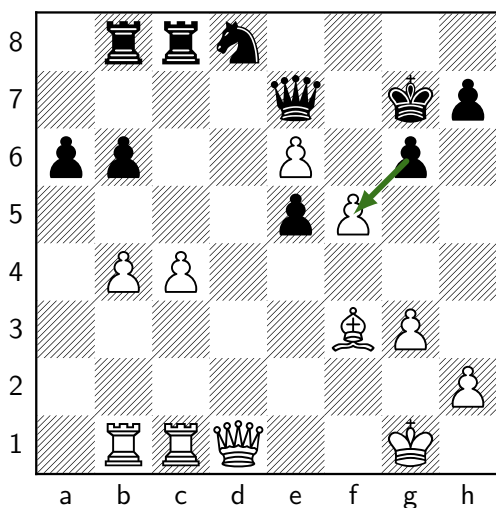
Figure 21: **Move 27, White: Nc2.** (a) Our position; Melanie plays Nc2 (a3c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3683` (ply 64) [structural distance $d=75$, QAP obj=40]: the master played Qe1; under μ our Na3 plays the role of that game’s Qe4 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Ra1, our Qf4 as their Bc3; transferred move a3c2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2636` (ply 58) [structural distance $d=79$, QAP obj=34]: the master played Qf5; under μ our Na3 plays the role of that game’s Qh3 (a **cross-type** correspondence); cross-type roles: our pa4 as their Re1, our qe7 as their ra3; transferred move a3c2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1507` (ply 60) [structural distance $d=78$, QAP obj=55]: the master played h3; under μ our Pe5 plays the role of that game’s Ph2; cross-type roles: our bb7 as their Re1, our Be2 as their Pa4; transferred move e5e6. Synthesis: 2 retrieved analogs transferred to Nc2 (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



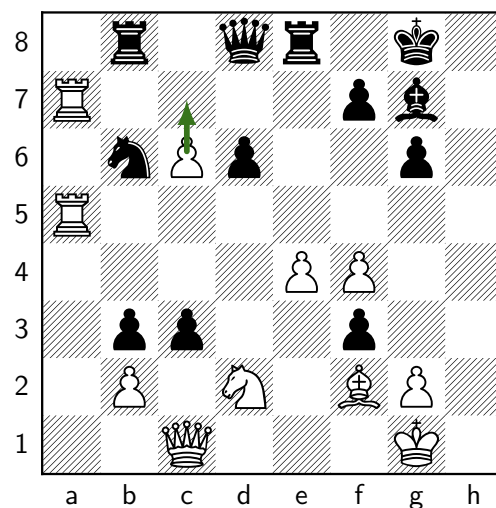
(a)



(b)

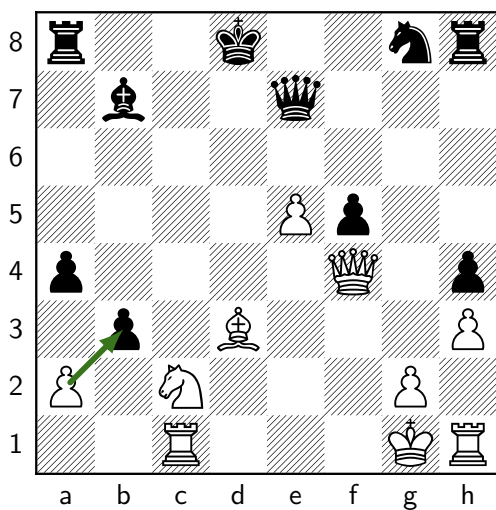


(c)

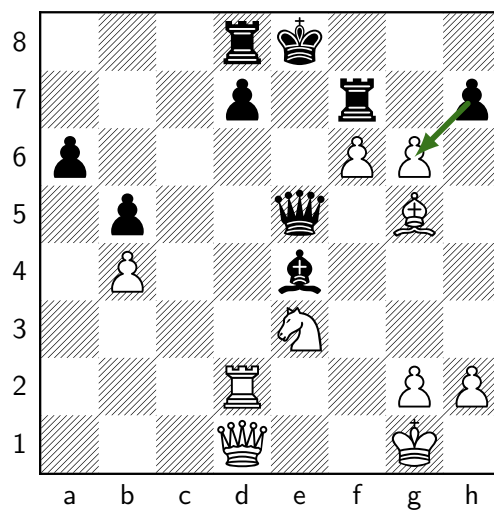


(d)

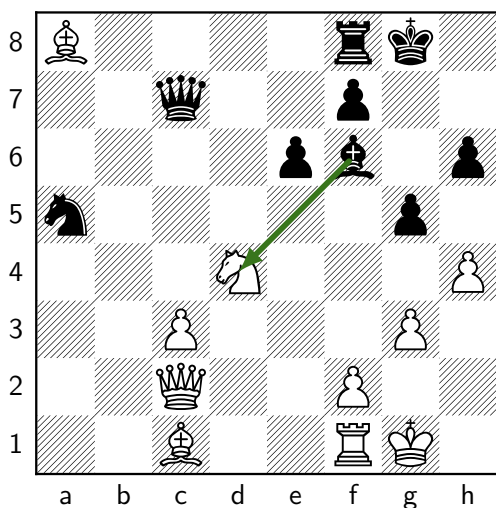
Figure 22: **Move 28, White: Bxd3**. (a) Our position; Melanie plays Bxd3 (e2d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2117` (ply 71) [structural distance $d=82$, QAP obj=37]: the master played Qd3; under μ our *Be2* plays the role of that game’s *Qe6* (a **cross-type** correspondence); cross-type roles: our *Nc2* as their *Rd8*, our *Rc1* as their *Pa7*; transferred move e2d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1845` (ply 57) [structural distance $d=69$, QAP obj=34]: the master played gxf5; under μ our *Be2* plays the role of that game’s *Pg3* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *Rb8*, our *qe7* as their *Rc8*; transferred move e2d3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1451` (ply 70) [structural distance $d=84$, QAP obj=65]: the master played c7; under μ our *Rc1* plays the role of that game’s *Pc6* (a **cross-type** correspondence); cross-type roles: our *Be2* as their *Qc1*, our *Be2* as their *Pb2*; transferred move c1d1. Synthesis: 2 retrieved analogs transferred to Bxd3 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



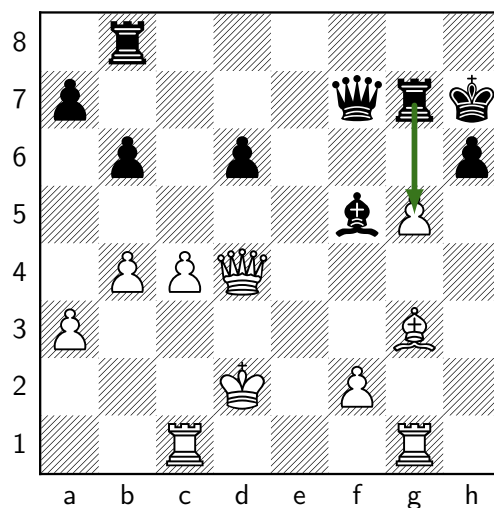
(a)



(b)

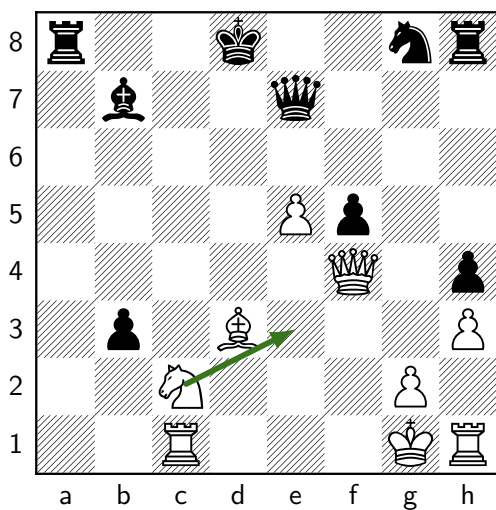


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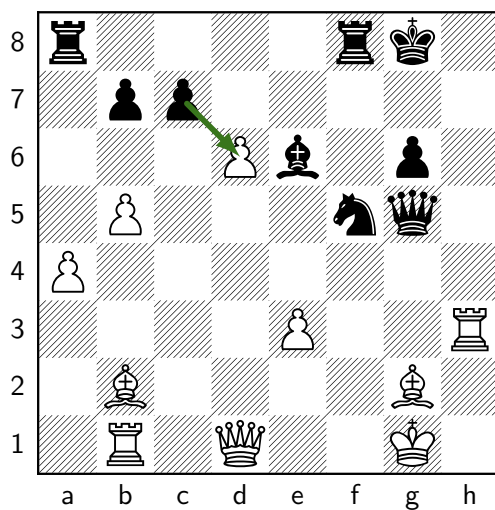


(d)

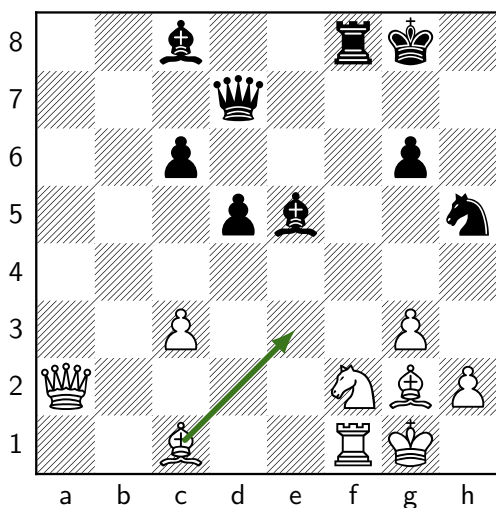
Figure 23: **Move 29, White: axb3.** (a) Our position; Melanie plays axb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#3657 (ply 61) [structural distance $d=85$, QAP obj=58]: the master played hxg6; under μ our *Pa2* plays the role of that game’s *Ph2*; cross-type roles: our *Nc2* as their *Rf7*, our *Bd3* as their *Pa6*; transferred move a2b3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1677 (ply 49) [structural distance $d=85$, QAP obj=43]: the master played Bxd4; under μ our *Pa2* plays the role of that game’s *Bf3* (a **cross-type** correspondence); cross-type roles: our *qe7* as their *ba8*, our *bb7* as their *Qc7*; transferred move a2b3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3377 (ply 57) [structural distance $d=78$, QAP obj=33]: the master played Rxf6; under μ our *Pa2* plays the role of that game’s *Rg2* (a **cross-type** correspondence); cross-type roles: our *Bd3* as their *Rb8*, our *Nc2* as their *Ph6*; transferred move a2b3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to axb3 (summed QAP objective 194, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



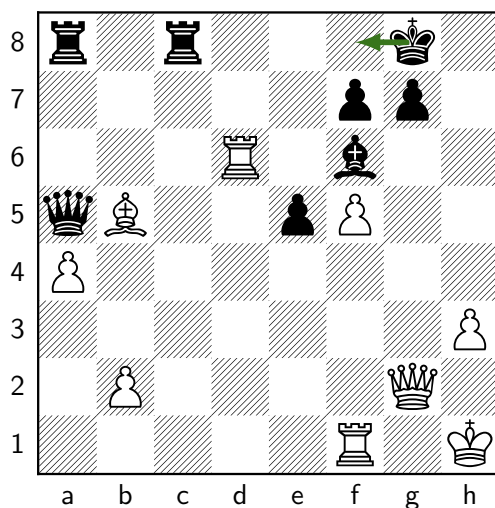
(a)



(b)

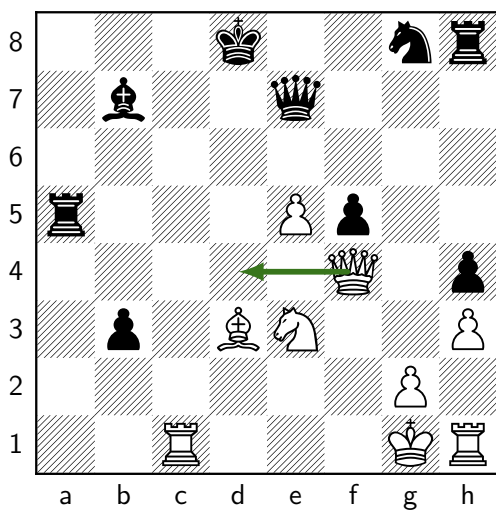


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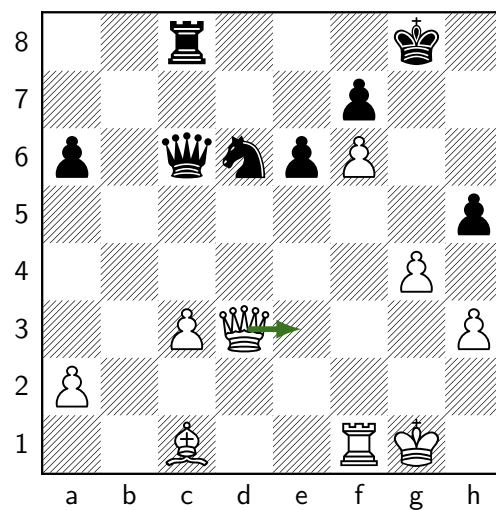


(d)

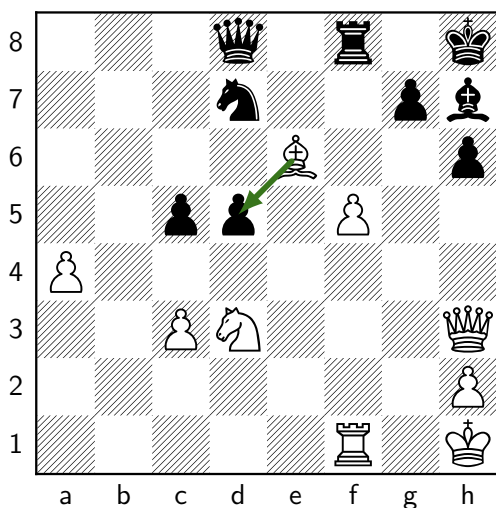
Figure 24: **Move 30, White: Ne3.** (a) Our position; Melanie plays Ne3 (c2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2165` (ply 55) [structural distance $d=77$, QAP obj=22]: the master played `cx d6`; under μ our `Nc2` plays the role of that game’s `Pc2` (a **cross-type** correspondence); cross-type roles: our `bb7` as their `Ra8`, our `Rc1` as their `Be6`; transferred move `c2e3` (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3122` (ply 52) [structural distance $d=74$, QAP obj=14]: the master played `Be3`; under μ our `Nc2` plays the role of that game’s `Bc1` (a **cross-type** correspondence); cross-type roles: our `Pg2` as their `Nf2`, our `qe7` as their `be5`; transferred move `c2e3` (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2405` (ply 53) [structural distance $d=73$, QAP obj=52]: the master played `Kf8`; under μ our `Kg1` plays the role of that game’s `Kg1`; cross-type roles: our `Pe5` as their `Ra8`, our `rh8` as their `Pf7`; transferred move `g1f2`. Synthesis: 2 retrieved analogs transferred to Ne3 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



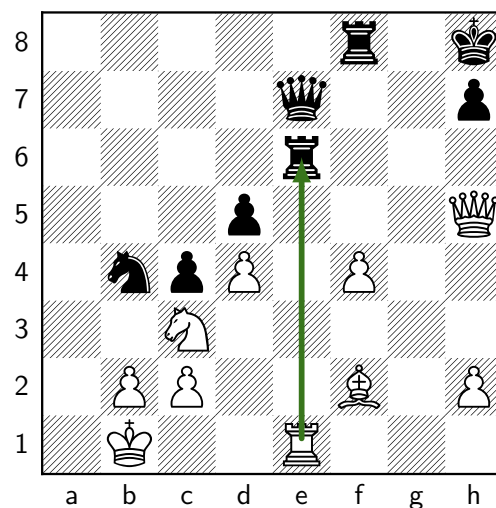
(a)



(b)

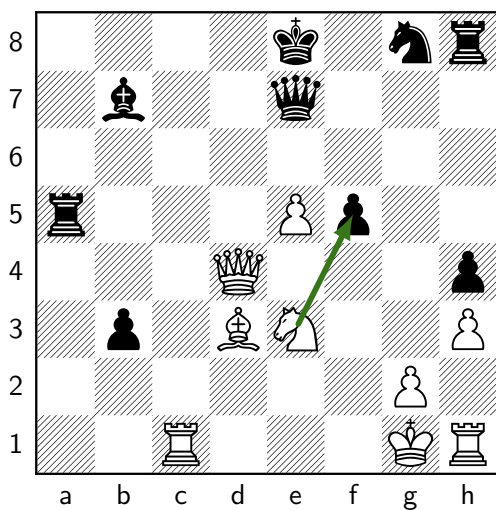


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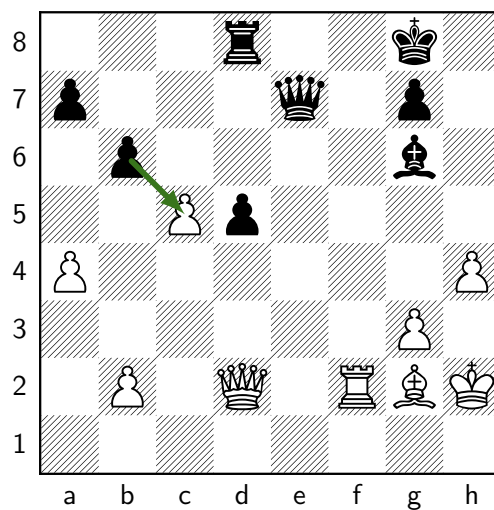


(d)

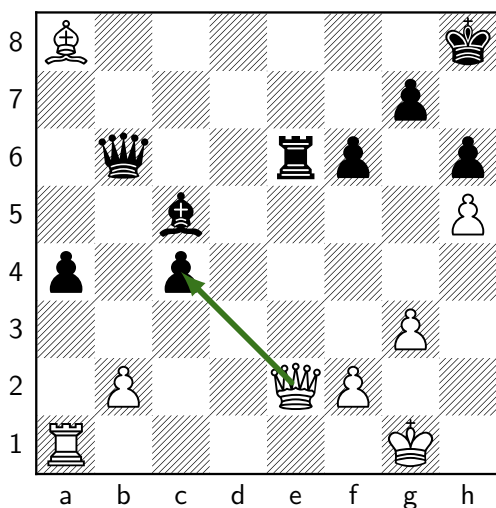
Figure 25: **Move 31, White: Qd4+.** (a) Our position; Melanie plays Qd4+ (f4d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#731` (ply 60) [structural distance $d=79$, QAP obj=64]: the master played Qe3; under μ our Qf4 plays the role of that game’s Qd3; cross-type roles: our Rc1 as their Pa2, our Ne3 as their Ph3; transferred move f4d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3116` (ply 70) [structural distance $d=71$, QAP obj=57]: the master played Bxd5; under μ our Qf4 plays the role of that game’s Be6 (a **cross-type** correspondence); cross-type roles: our Ne3 as their Pc3, our Bd3 as their Nd3; transferred move f4d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2768` (ply 72) [structural distance $d=76$, QAP obj=54]: the master played Rxe6; under μ our Qf4 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Ne3 as their Bf2, our bb7 as their Ph2; transferred move f4d4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Qd4+ (summed QAP objective 366, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



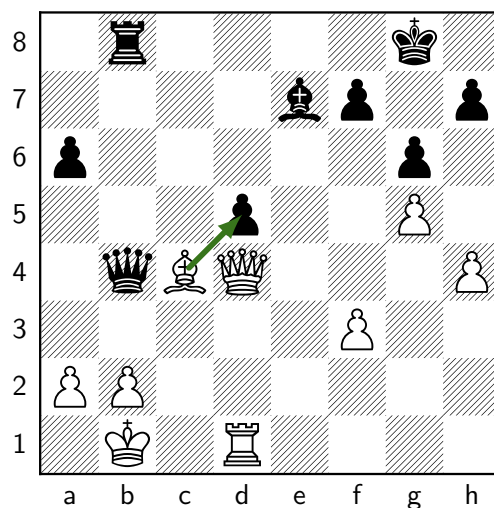
(a)



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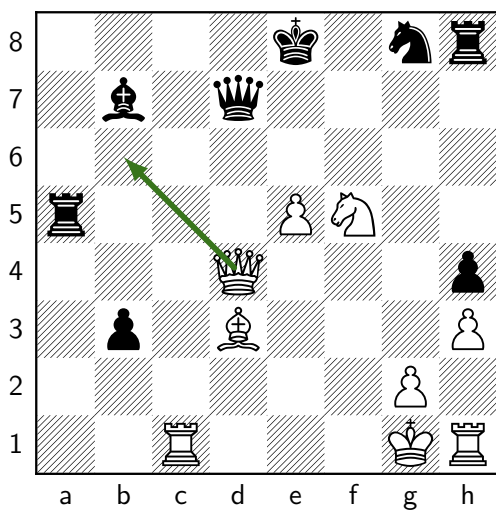


(c)

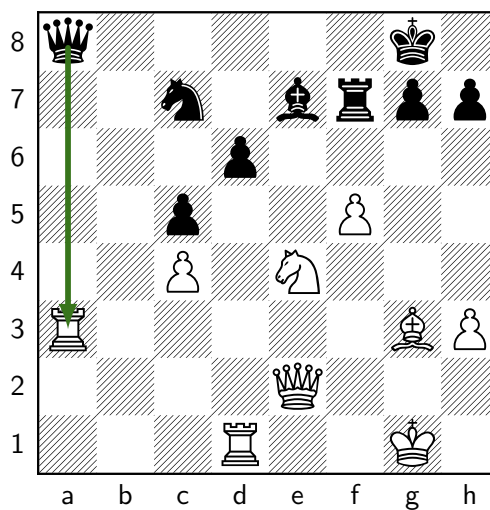


(d)

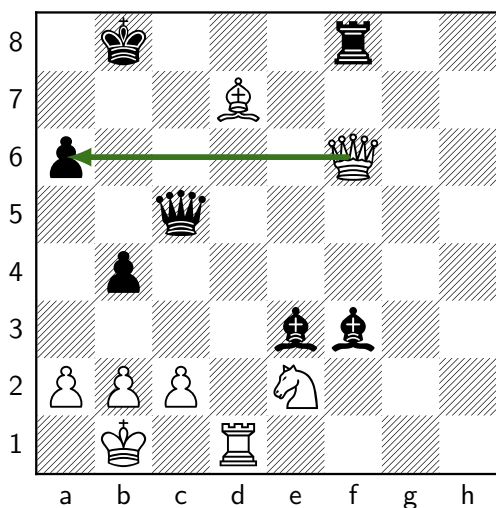
Figure 26: **Move 32, White: Nxf5.** (a) Our position; Melanie plays Nxf5 (e3f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1442` (ply 69) [structural distance $d=59$, QAP obj=59]: the master played bxc5; under μ our *Ne3* plays the role of that game’s *Pb3* (a **cross-type** correspondence); cross-type roles: our *Qd4* as their *Rd8*, our *Rc1* as their *Pa7*; transferred move e3f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3379` (ply 54) [structural distance $d=75$, QAP obj=21]: the master played Qxc4; under μ our *Ne3* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *ng8* as their *Pg3*; transferred move e3f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#519` (ply 56) [structural distance $d=76$, QAP obj=12]: the master played Bxd5; under μ our *Ne3* plays the role of that game’s *Bc4* (a **cross-type** correspondence); cross-type roles: our *ng8* as their *Pf3*, our *Bd3* as their *Qd4*; transferred move e3f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxf5 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



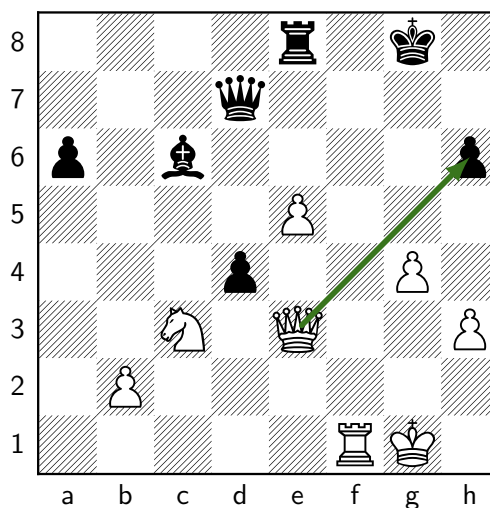
(a)



(b)

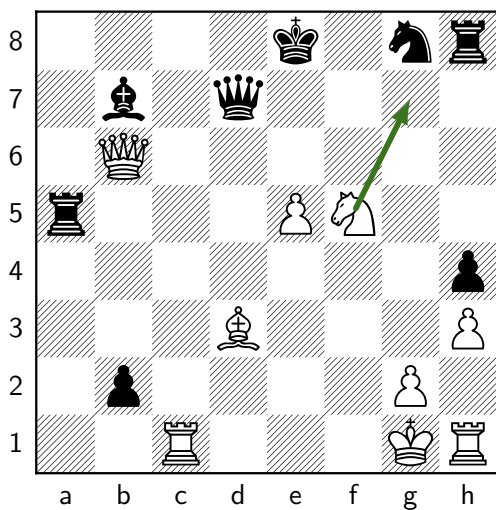


(c)

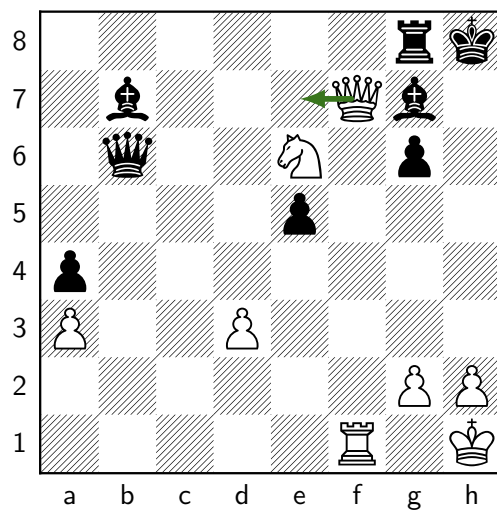


(d)

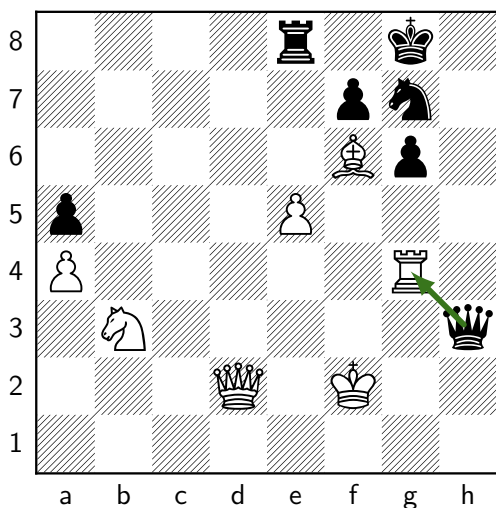
Figure 27: **Move 33, White: Qb6.** (a) Our position; Melanie plays Qb6 (d4b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#147` (ply 63) [structural distance $d=77$, QAP obj=41]: the master played Qxa3; under μ our Qd4 plays the role of that game’s Qa1; cross-type roles: our Pg2 as their Rf7, our Rh1 as their Ph7; transferred move d4b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#504` (ply 56) [structural distance $d=78$, QAP obj=39]: the master played Qxa6; under μ our Qd4 plays the role of that game’s Qf6; cross-type roles: our Rh1 as their Pc2, our Qd4 as their Ne2; transferred move d4b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3740` (ply 56) [structural distance $d=68$, QAP obj=32]: the master played Qxh6; under μ our Qd4 plays the role of that game’s Qe3; cross-type roles: our Bd3 as their Pb2, our qd7 as their pd4; transferred move d4b6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qb6 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



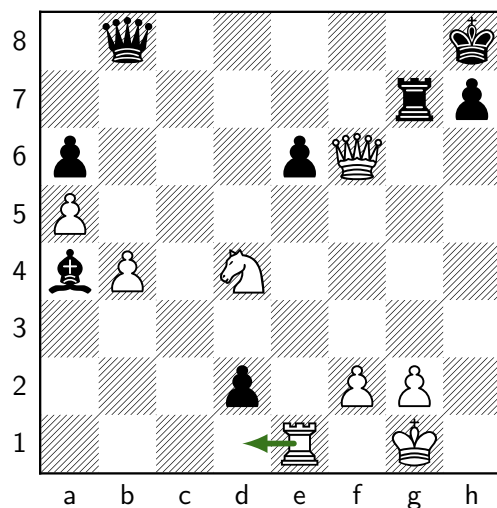
(a)



(b)

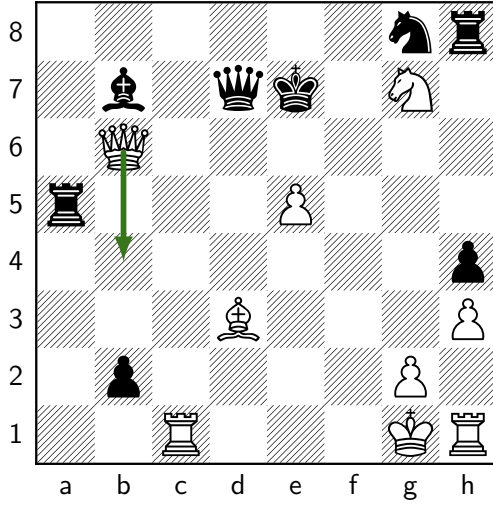


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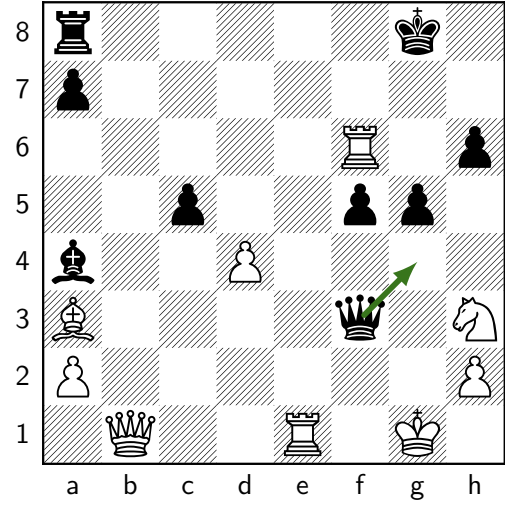


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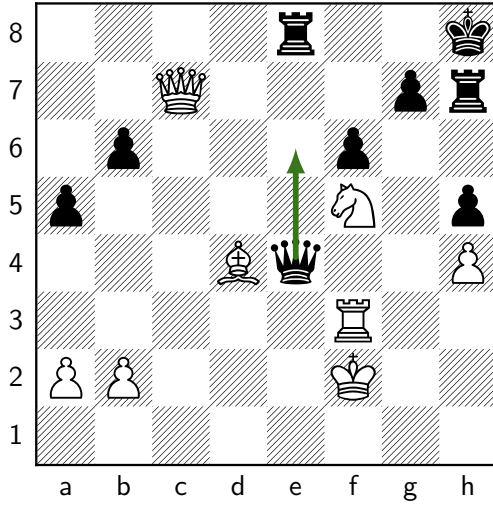
Figure 28: **Move 34, White: Ng7+.** (a) Our position; Melanie plays Ng7+ (f5g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0000#2198` (ply 64) [structural distance $d=82$, QAP obj=69]: the master played Qe7; under μ our Qb6 plays the role of that game’s Qf7; cross-type roles: our Bd3 as their Ph2, our Nf5 as their Pa3; transferred move b6a5. (c) Retrieved from `ccr1_top-p0003#1936` (ply 81) [structural distance $d=88$, QAP obj=67]: the master played Qxg4; under μ our Qb6 plays the role of that game’s Qh6; cross-type roles: our Bd3 as their Pf7, our Rc1 as their Ng7; transferred move b6a5. (d) Retrieved from `ccr1_top-p0002#850` (ply 74) [structural distance $d=78$, QAP obj=63]: the master played Rd1; under μ our Rc1 plays the role of that game’s Re1; cross-type roles: our Bd3 as their Pf2, our rh8 as their ba4; transferred move c1a1. Synthesis: no retrieved analog transferred to Ng7+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



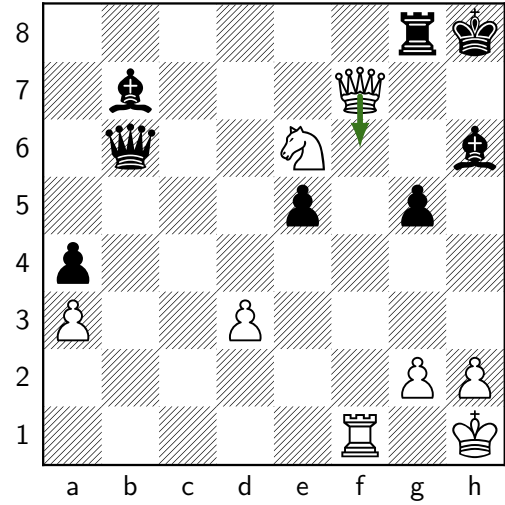
(a)



(b)

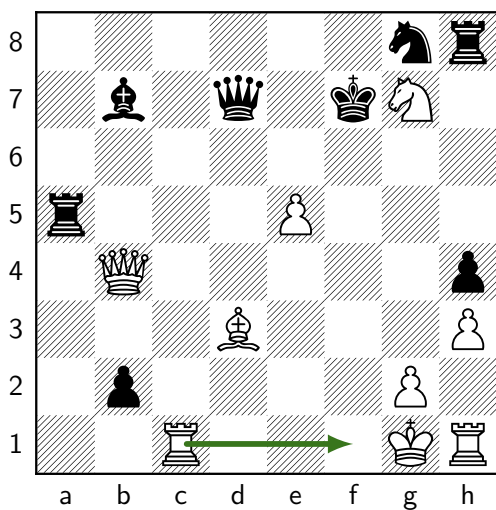


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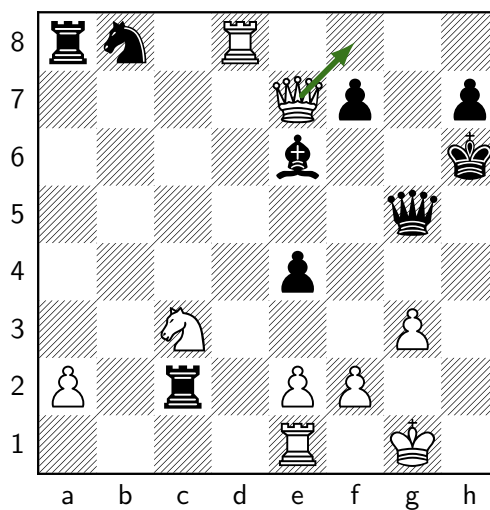


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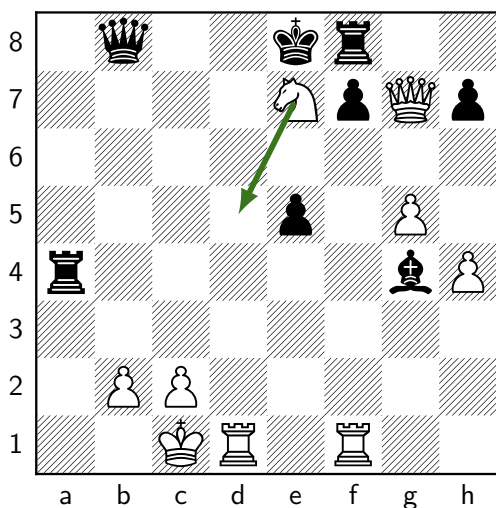
Figure 29: **Move 35, White: Qb4+.** (a) Our position; Melanie plays Qb4+ (b6b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#189` (ply 55) [structural distance $d=82$, QAP obj=22]: the master played Qg4+; under μ our Qb6 plays the role of that game’s Qf6; cross-type roles: our Pg2 as their Ra8, our Ng7 as their Pa7; transferred move b6b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3194` (ply 85) [structural distance $d=84$, QAP obj=81]: the master played Qe6; under μ our Qb6 plays the role of that game’s Qe5; cross-type roles: our Ng7 as their Re8, our bb7 as their qc7; transferred move b6c5. (d) Retrieved from `ccr1_top_p0000#2198` (ply 68) [structural distance $d=89$, QAP obj=64]: the master played Qf6+; under μ our Qb6 plays the role of that game’s Qf7; cross-type roles: our Rc1 as their Ph2, our Ng7 as their Pd3; transferred move b6c5. Synthesis: 1 retrieved analog transferred to Qb4+ (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



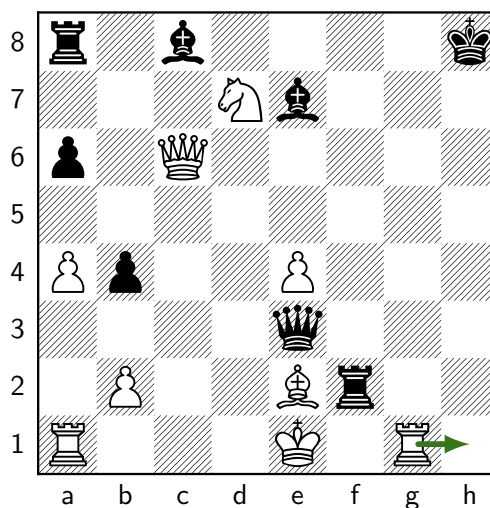
(a)



(b)

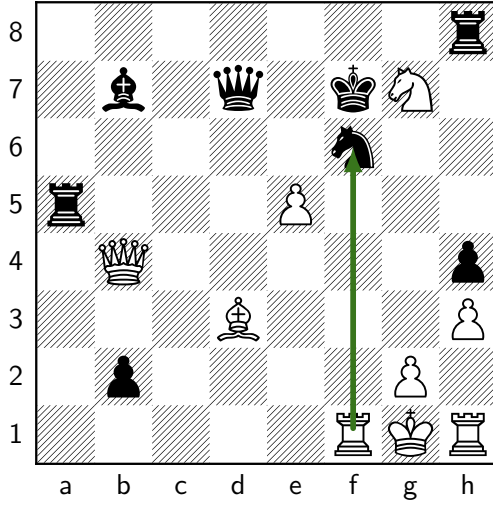


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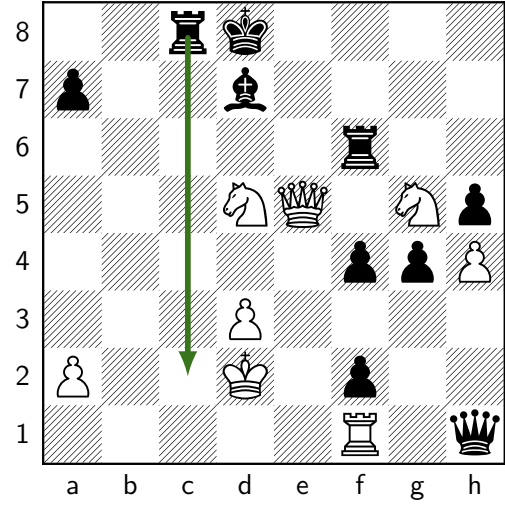


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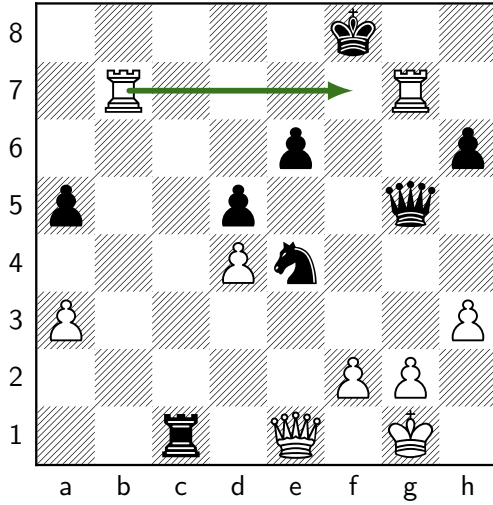
Figure 30: **Move 36, White: Rf1+.** (a) Our position; Melanie plays Rf1+ (c1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3278` (ply 62) [structural distance $d=85$, QAP obj=55]: the master played Qf8+; under μ our Qb4 plays the role of that game's Qe7; cross-type roles: our Bd3 as their Pa2, our kf7 as their rc2; transferred move b4b6. (c) Retrieved from `ccrl_top_p0002#1770` (ply 54) [structural distance $d=74$, QAP obj=54]: the master played Nd5; under μ our Ng7 plays the role of that game's Ne7; cross-type roles: our bb7 as their Rf1, our Bd3 as their Pb2; transferred move g7f5. (d) Retrieved from `ccrl_top_p0002#1683` (ply 70) [structural distance $d=93$, QAP obj=51]: the master played Rh1+; under μ our Rh1 plays the role of that game's Rg1; cross-type roles: our Bd3 as their Ra1, our Ng7 as their Be2; transferred move h1h2. Synthesis: no retrieved analog transferred to Rf1+ – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



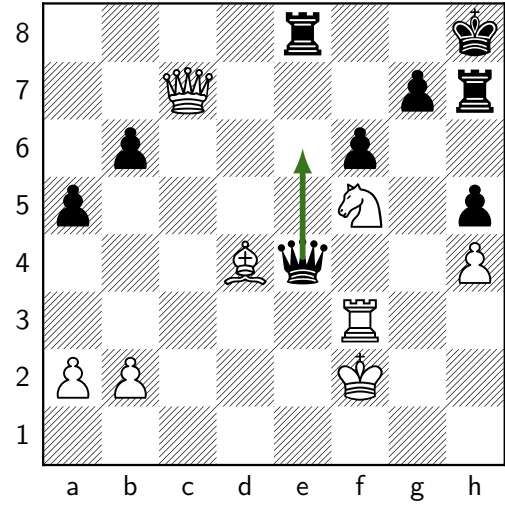
(a)



(b)



(c)



(d)

Figure 31: **Move 37, White: Rxf6+.** (a) Our position; Melanie plays Rxf6+ (f1f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1026` (ply 61) [structural distance $d=89$, QAP obj=39]: the master played Rc2+; under μ our Rf1 plays the role of that game's Rc1; cross-type roles: our Qb4 as their nd5, our Rh1 as their ng5; transferred move f1f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#749` (ply 70) [structural distance $d=105$, QAP obj=35]: the master played Rb7+; under μ our Rf1 plays the role of that game's Rb7; cross-type roles: our Qb4 as their rc1, our ra5 as their Qe1; transferred move f1f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3194` (ply 85) [structural distance $d=101$, QAP obj=65]: the master played Qe6; under μ our Qb4 plays the role of that game's Qe5; cross-type roles: our Ng7 as their Re8, our Rf1 as their qc7; transferred move b4f4. Synthesis: 2 retrieved analogs transferred to Rxf6+ (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

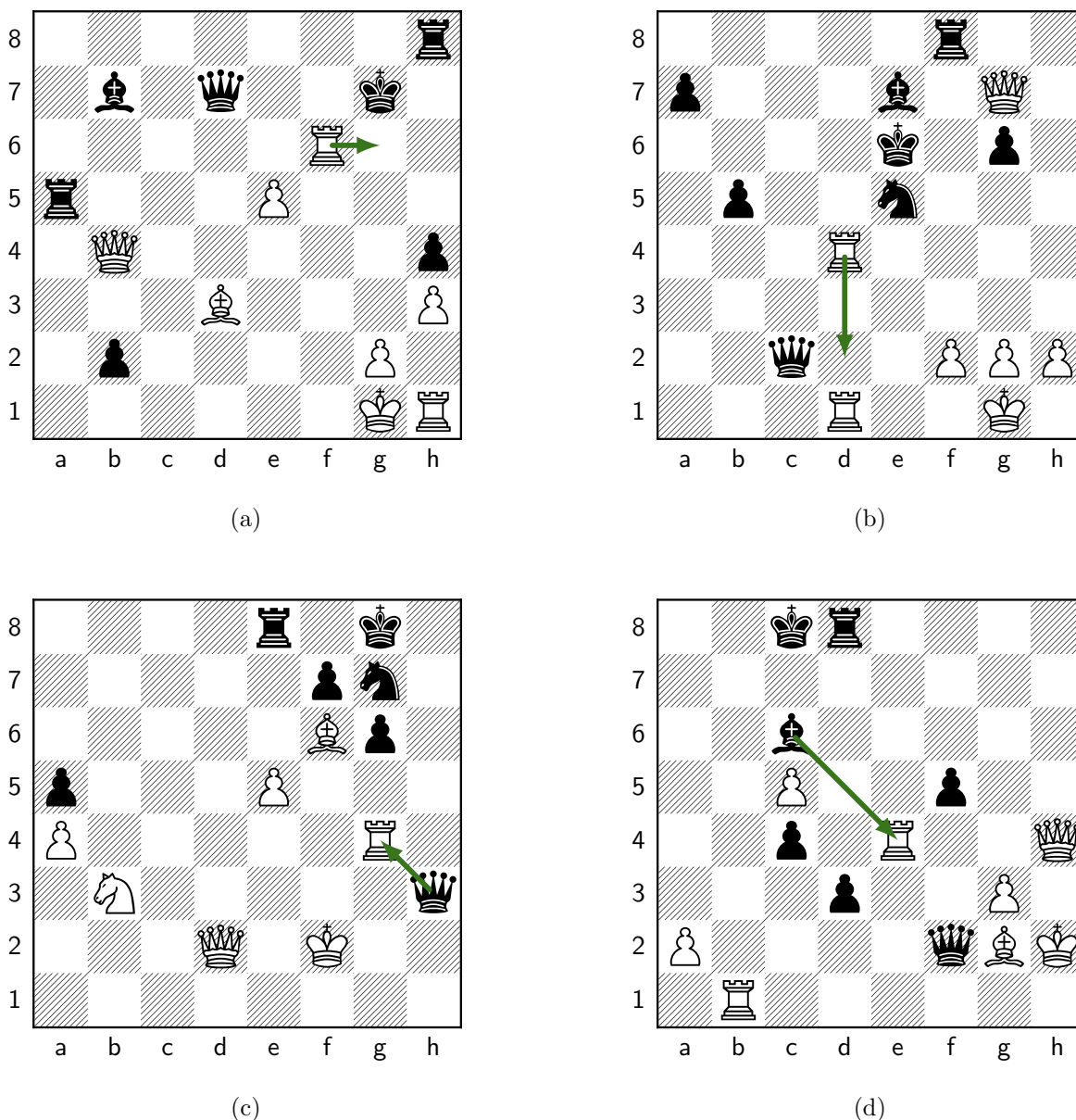
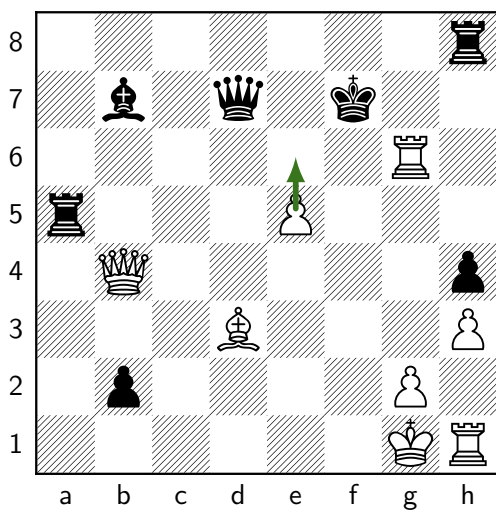
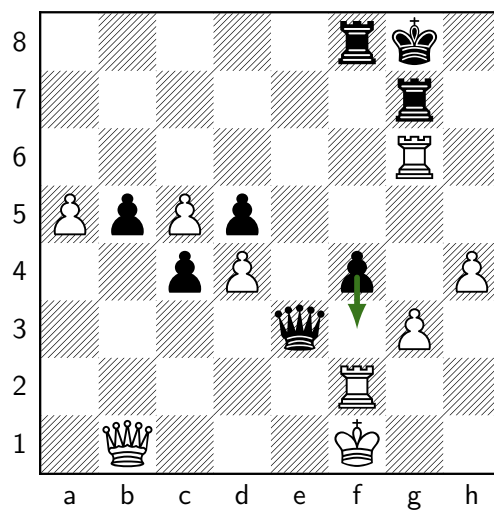


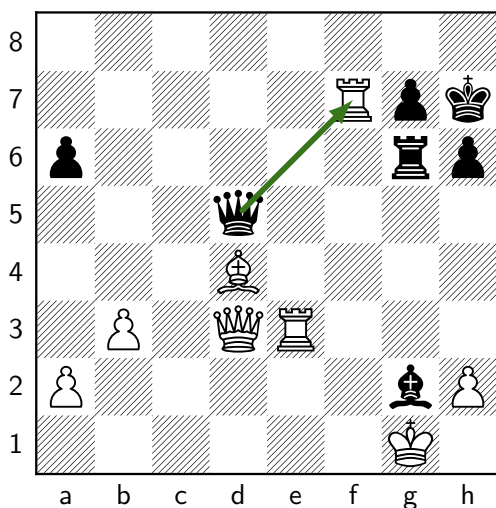
Figure 32: **Move 38, White: Rg6+.** (a) Our position; Melanie plays Rg6+ (f6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1262` (ply 72) [structural distance $d=63$, QAP obj=60]: the master played R4d2; under μ our *Pe5* plays the role of that game’s *Rd4* (a **cross-type** correspondence); cross-type roles: our *kg7* as their *qc2*, our *Bd3* as their *Ph2*; transferred move *e5e6*. (c) Retrieved from `ccr1_top_p0003#1936` (ply 81) [structural distance $d=61$, QAP obj=55]: the master played Qxg4; under μ our *Qb4* plays the role of that game’s *Qh6*; cross-type roles: our *Rf6* as their *Ng7*, our *kg7* as their *bf6*; transferred move *b4a5*. (d) Retrieved from `ccr1_top_p0001#1526` (ply 59) [structural distance $d=58$, QAP obj=54]: the master played Bxe4; under μ our *Qb4* plays the role of that game’s *Bc3* (a **cross-type** correspondence); cross-type roles: our *Qb4* as their *Pf5*, our *bb7* as their *pg3*; transferred move *b4e4*. Synthesis: no retrieved analog transferred to Rg6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



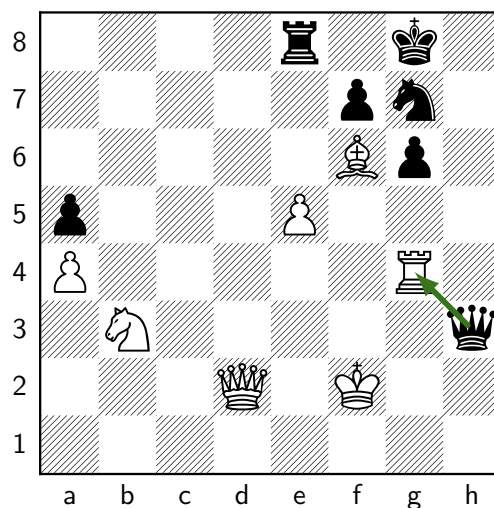
(a)



(b)

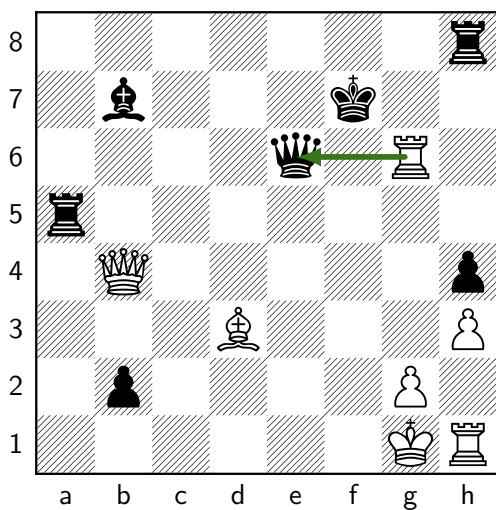


(c)

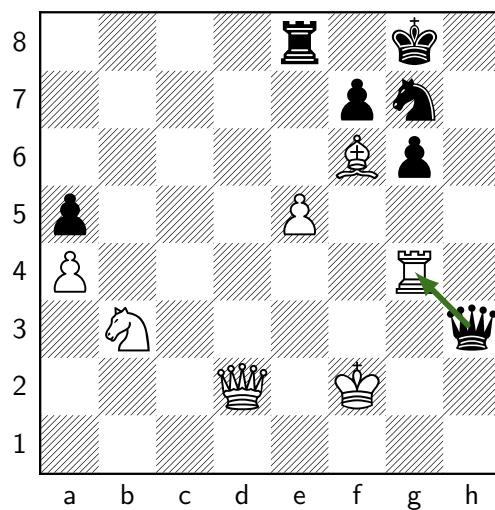


(d)

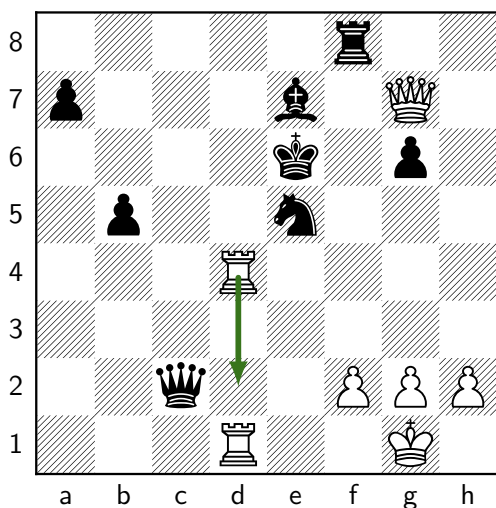
Figure 33: **Move 39, White: e6+.** (a) Our position; Melanie plays e6+ (e5e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#163` (ply 87) [structural distance $d=76$, QAP obj=40]: the master played f3; under μ our *Pe5* plays the role of that game’s *Pf5*; cross-type roles: our *Pg2* as their *Rf8*, our *kf7* as their *rg6*; transferred move e5e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#836` (ply 79) [structural distance $d=74$, QAP obj=57]: the master played Qxf7; under μ our *Qb4* plays the role of that game’s *Qd4*; cross-type roles: our *Bd3* as their *Rg6*, our *ra5* as their *bd4*; transferred move b4b2. (d) Retrieved from `ccr1_top_p0003#1936` (ply 81) [structural distance $d=70$, QAP obj=55]: the master played Qxg4; under μ our *Qb4* plays the role of that game’s *Qh6*; cross-type roles: our *Rg6* as their *Ng7*, our *kf7* as their *bf6*; transferred move b4a5. Synthesis: 1 retrieved analog transferred to e6+ (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



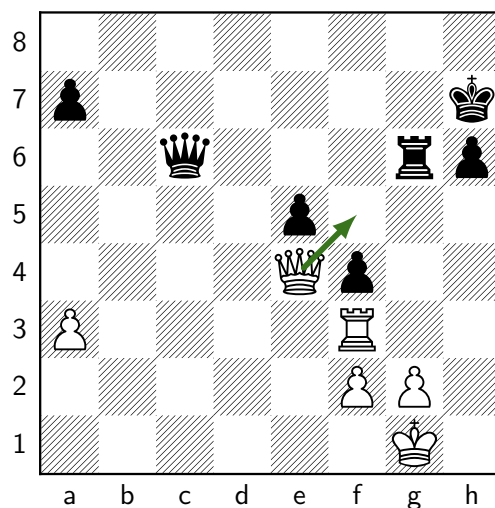
(a)



(b)

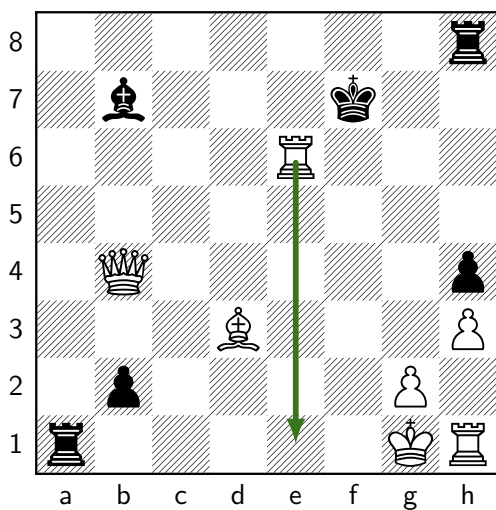


(c)

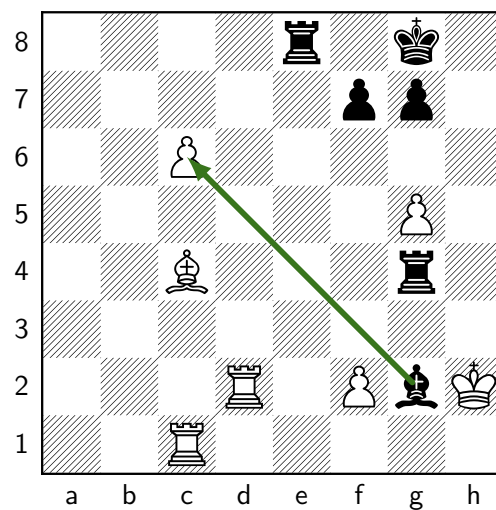


(d)

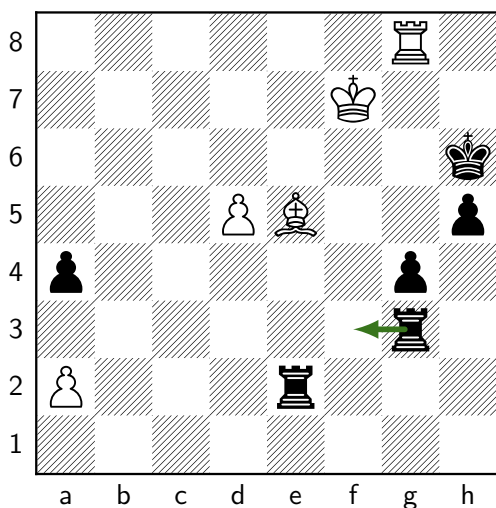
Figure 34: **Move 40, White: Rxe6.** (a) Our position; Melanie plays Rxe6 (g6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1936` (ply 81) [structural distance $d=74$, QAP obj=42]: the master played Qxg4; under μ our Qb4 plays the role of that game’s Qh6; cross-type roles: our qe6 as their Ng7, our Rg6 as their bf6; transferred move b4a5. (c) Retrieved from `ccrl_top_p0002#1262` (ply 72) [structural distance $d=58$, QAP obj=41]: the master played R4d2; under μ our Bd3 plays the role of that game’s Rd4 (a **cross-type** correspondence); cross-type roles: our bb7 as their Pf2, our Rh1 as their Ph2; transferred move d3e2. (d) Retrieved from `ccrl_top_p0001#2284` (ply 82) [structural distance $d=76$, QAP obj=34]: the master played Qf5; under μ our Rg6 plays the role of that game’s Qe4 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pg2, our Bd3 as their Pa3; transferred move g6g5. Synthesis: no retrieved analog transferred to Rxe6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



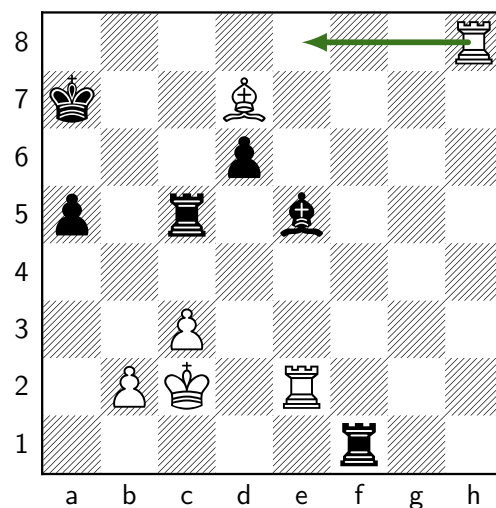
(a)



(b)

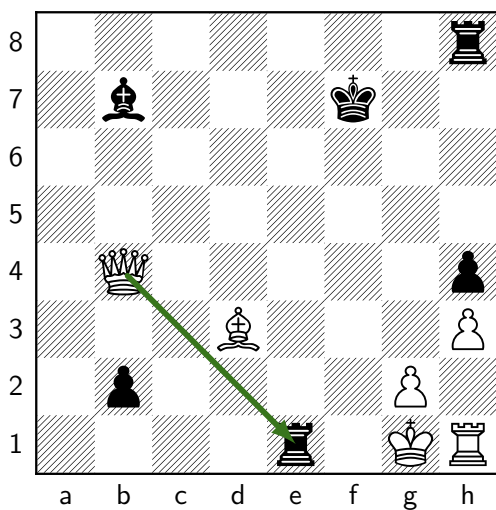


(c)

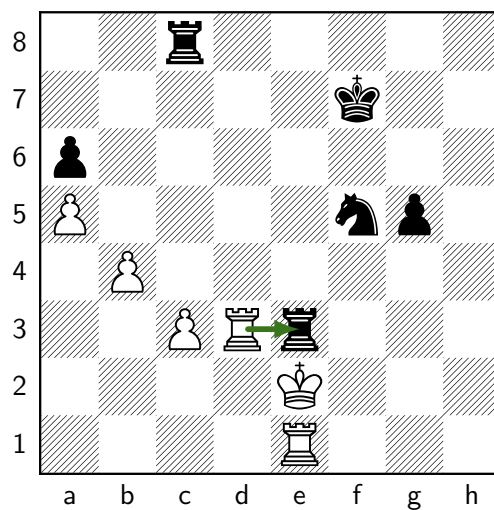


(d)

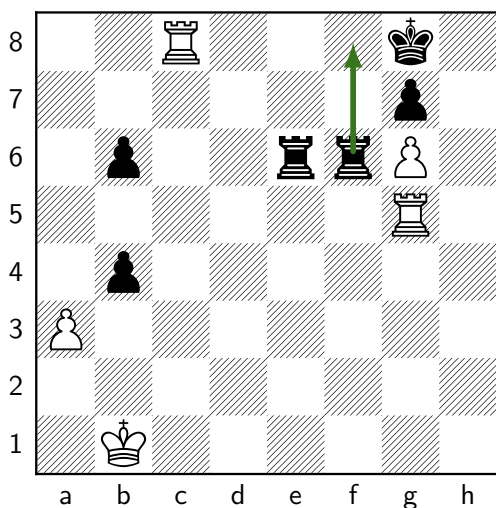
Figure 35: **Move 41, White: Re1**. (a) Our position; Melanie plays Re1 (e6e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2677` (ply 85) [structural distance $d=41$, QAP obj=37]: the master played Bxc6; under μ our Re6 plays the role of that game’s Bg7 (a **cross-type** correspondence); cross-type roles: our Bd3 as their pc6, our Qb4 as their Rg4; transferred move e6e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3358` (ply 97) [structural distance $d=54$, QAP obj=17]: the master played Rf3+; under μ our Re6 plays the role of that game’s Rg6; cross-type roles: our bb7 as their rg8, our ra1 as their pd5; transferred move e6e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#136` (ply 94) [structural distance $d=46$, QAP obj=16]: the master played Re8; under μ our Re6 plays the role of that game’s Rh8; cross-type roles: our Rh1 as their Pb2, our Qb4 as their Re2; transferred move e6e1 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Re1 (summed QAP objective 152, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



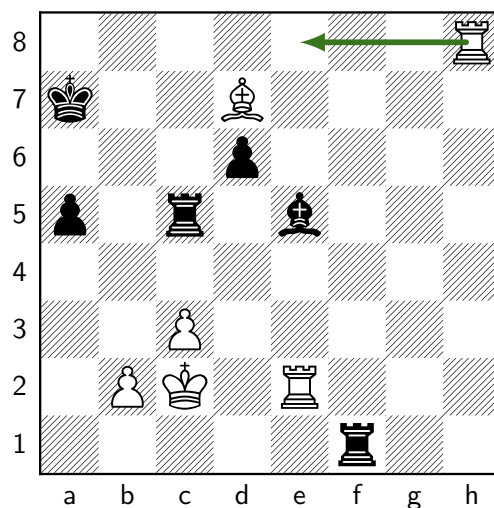
(a)



(b)



(c)



(d)

Figure 36: **Move 42, White: Qxe1**. (a) Our position; Melanie plays **Qxe1** (**b4e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3819` (ply 96) [structural distance $d=56$, QAP obj=44]: the master played **Rxe3**; under μ our Qb_4 plays the role of that game's Rd_3 (a **cross-type** correspondence); cross-type roles: our Bd_3 as their Pc_3 , our bb_7 as their nf_5 ; transferred move **b4e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1982` (ply 77) [structural distance $d=52$, QAP obj=36]: the master played **Rf8**; under μ our Qb_4 plays the role of that game's Rf_3 (a **cross-type** correspondence); cross-type roles: our Rh_1 as their Pb_6 , our Bd_3 as their Re_6 ; transferred move **b4e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#136` (ply 94) [structural distance $d=56$, QAP obj=13]: the master played **Re8**; under μ our Qb_4 plays the role of that game's Rh_8 (a **cross-type** correspondence); cross-type roles: our rh_8 as their Pb_2 , our Bd_3 as their Re_2 ; transferred move **b4e1** (**endorses this move**). Synthesis: 13 retrieved analogs transferred to **Qxe1** (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

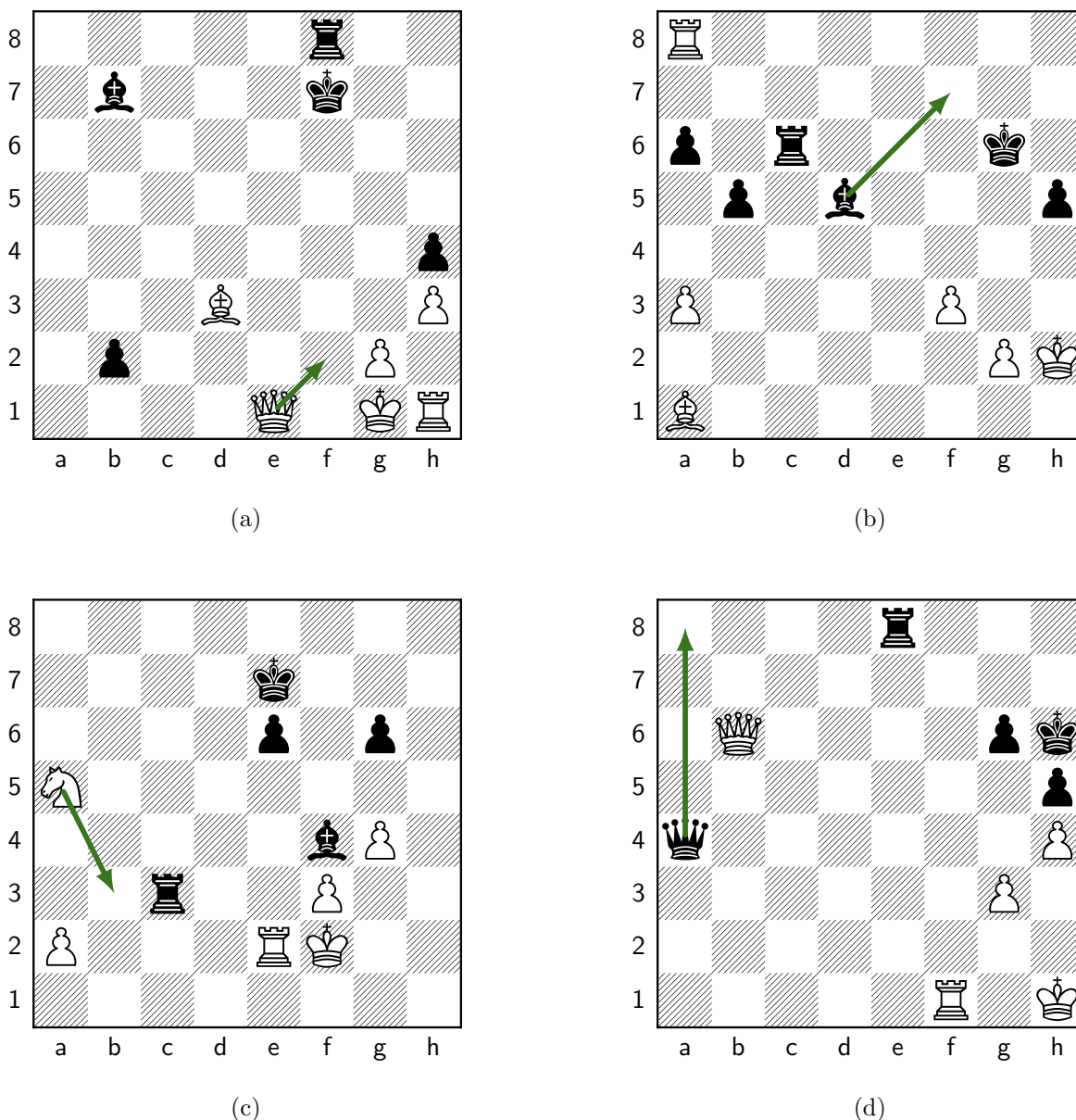
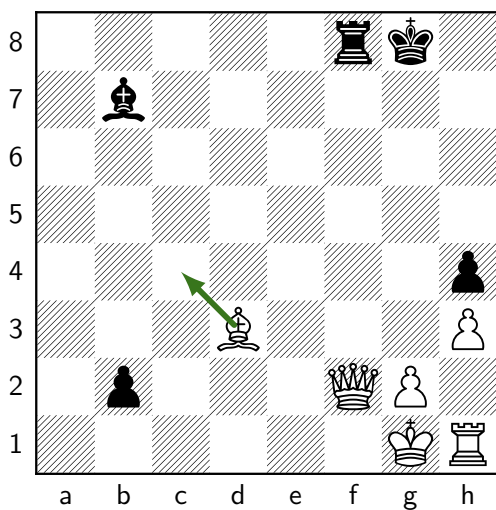
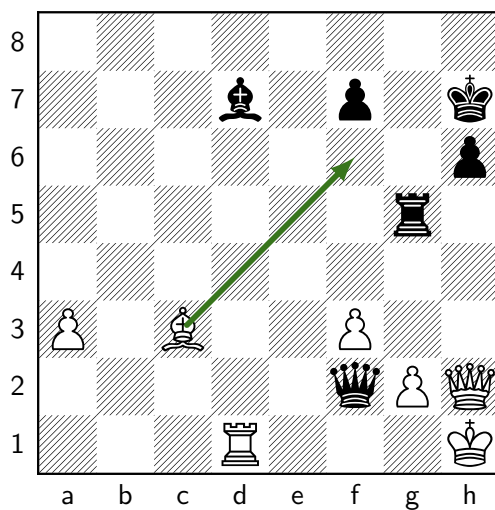


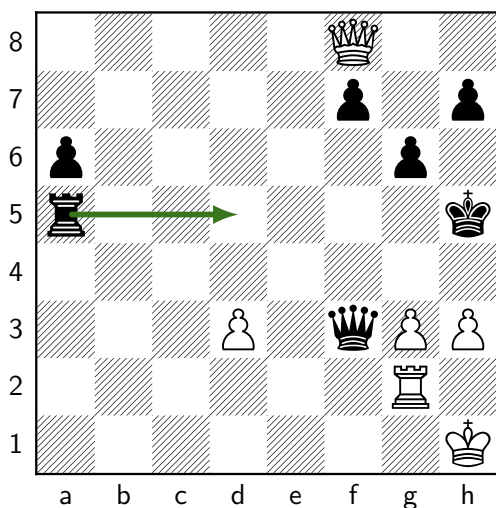
Figure 37: **Move 43, White: Qf2+.** (a) Our position; Melanie plays Qf2+ (e1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3873 (ply 95) [structural distance $d=39$, QAP obj=21]: the master played Bf7; under μ our Qe1 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our bb7 as their ra8, our Bd3 as their Ph5; transferred move e1f2 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2954 (ply 90) [structural distance $d=41$, QAP obj=46]: the master played Nb3; under μ our Bd3 plays the role of that game’s Na5 (a **cross-type** correspondence); cross-type roles: our bb7 as their Re2, our Qe1 as their rc3; transferred move d3e2. (d) Retrieved from ccr1_top_p0002#1557 (ply 87) [structural distance $d=36$, QAP obj=45]: the master played Qa8+; under μ our Qe1 plays the role of that game’s Qa5; cross-type roles: our bb7 as their qb6; transferred move e1b1. Synthesis: 2 retrieved analogs transferred to Qf2+ (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



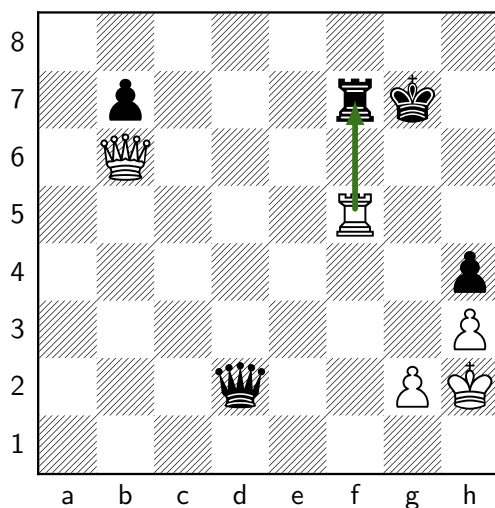
(a)



(b)

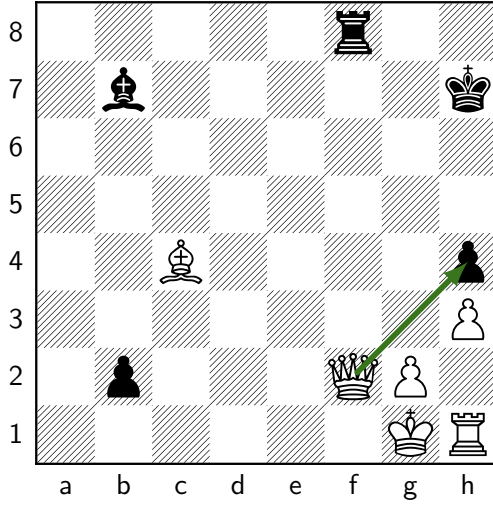


(c)

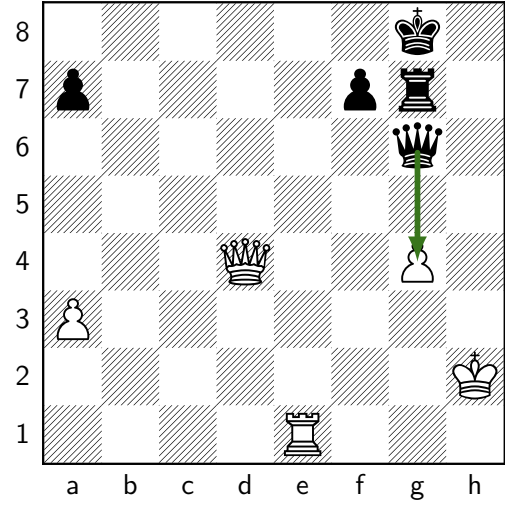


(d)

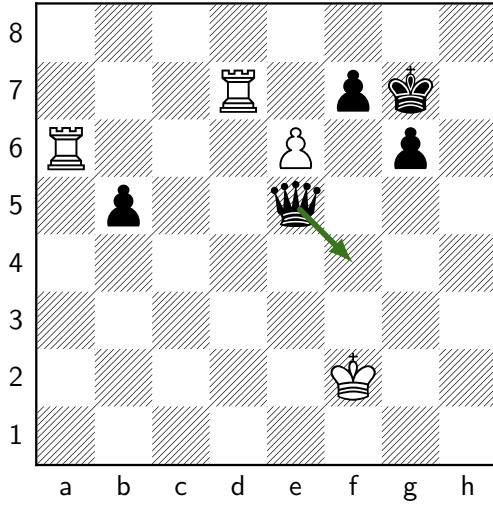
Figure 38: **Move 44, White: Bc4+.** (a) Our position; Melanie plays Bc4+ (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2482` (ply 70) [structural distance $d=55$, QAP obj=27]: the master played Bf6; under μ our Bd3 plays the role of that game’s Bc3; cross-type roles: our Qf2 as their Rd1, our rf8 as their qf2; transferred move d3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2903` (ply 75) [structural distance $d=53$, QAP obj=8]: the master played Rd5; under μ our Bd3 plays the role of that game’s Ra4 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Ph7, our bb7 as their Pa6; transferred move d3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#51` (ply 102) [structural distance $d=57$, QAP obj=37]: the master played Rxf7+; under μ our Qf2 plays the role of that game’s Rf5 (a **cross-type** correspondence); cross-type roles: our bb7 as their qd2, our Bd3 as their Qb6; transferred move f2f8. Synthesis: 2 retrieved analogs transferred to Bc4+ (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



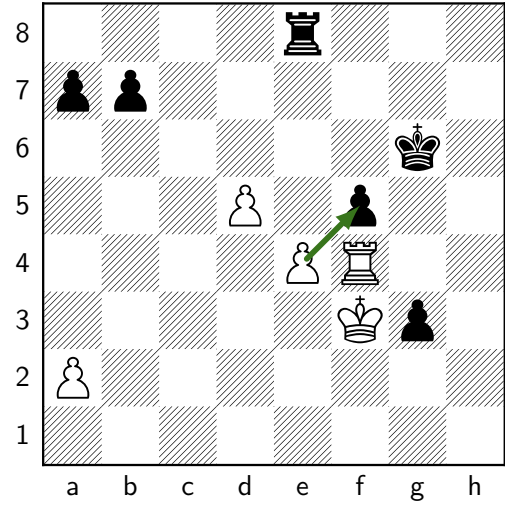
(a)



(b)

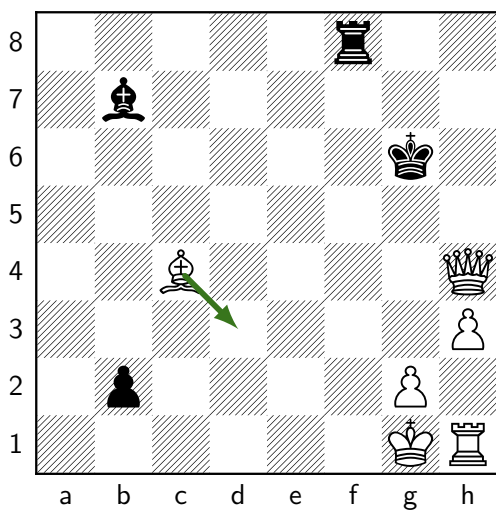


(c)

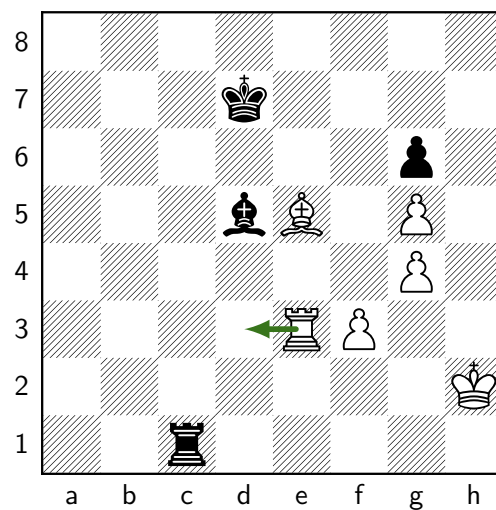


(d)

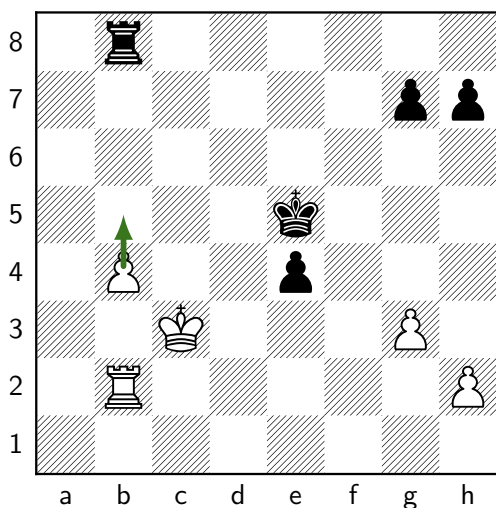
Figure 39: **Move 45, White: Qxh4+.** (a) Our position; Melanie plays Qxh4+ (f2h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1268` (ply 87) [structural distance $d=49$, QAP obj=22]: the master played Qxg4; under μ our Qf2 plays the role of that game’s Qg3; cross-type roles: our Rh1 as their Pa7, our Qf2 as their Rg7; transferred move f2h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3094` (ply 99) [structural distance $d=49$, QAP obj=35]: the master played Qf4+; under μ our Qf2 plays the role of that game’s Qe4; cross-type roles: our Qf2 as their Pf7, our bb7 as their ra6; transferred move f2c5. (d) Retrieved from `ccr1_top_p0003#683` (ply 84) [structural distance $d=54$, QAP obj=33]: the master played exf5+; under μ our Qf2 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa2, our Qf2 as their Rf4; transferred move f2f4. Synthesis: 1 retrieved analog transferred to Qxh4+ (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



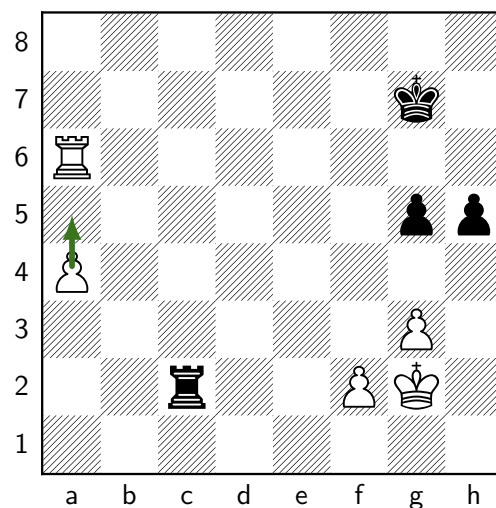
(a)



(b)

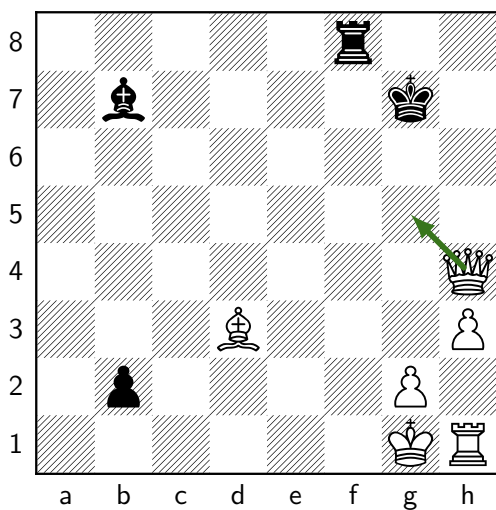


(c)

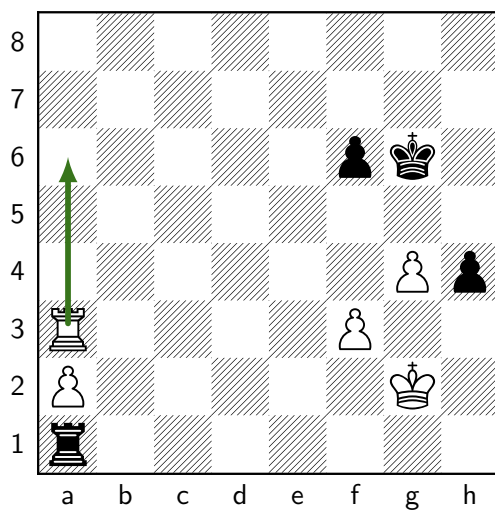


(d)

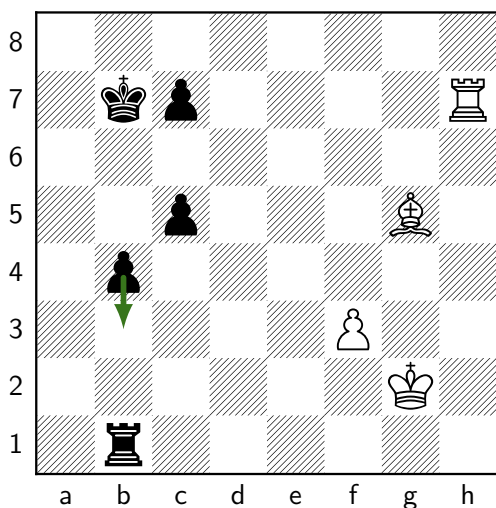
Figure 40: **Move 46, White: Bd3+.** (a) Our position; Melanie plays Bd3+ (c4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2254` (ply 92) [structural distance $d=33$, QAP obj=11]: the master played Rd3; under μ our *Bc4* plays the role of that game’s *Re3* (a **cross-type** correspondence); cross-type roles: our *Rh1* as their *Be5*, our *Qh4* as their *pg6*; transferred move c4d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1483` (ply 80) [structural distance $d=36$, QAP obj=34]: the master played b5; under μ our *Pg2* plays the role of that game’s *Pb4*; cross-type roles: our *Bc4* as their *Rb2*, our *Qh4* as their *Pg3*; transferred move g2g3. (d) Retrieved from `ccr1_top_p0000#139` (ply 102) [structural distance $d=38$, QAP obj=27]: the master played a5; under μ our *Bc4* plays the role of that game’s *Pa4* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *rc2*, our *Qh4* as their *pg5*; transferred move c4b5. Synthesis: 1 retrieved analog transferred to Bd3+ (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



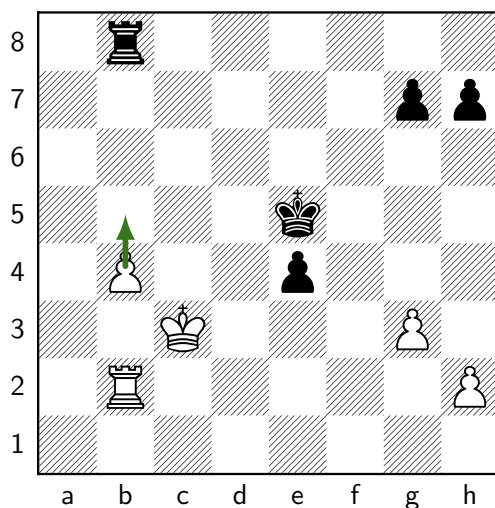
(a)



(b)



(c)



(d)

Figure 41: **Move 47, White: Qg5+.** (a) Our position; Melanie plays Qg5+ (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#2539 (ply 88) [structural distance $d=32$, QAP obj=1]: the master played Ra6; under μ our Qh4 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pf3, our Bd3 as their Pg4; transferred move h4g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#1031 (ply 89) [structural distance $d=33$, QAP obj=42]: the master played b3; under μ our Bd3 plays the role of that game’s Pb5 (a **cross-type** correspondence); cross-type roles: our bb7 as their rh7, our Ph3 as their bg5; transferred move d3b5. (d) Retrieved from ccr1_top_p0002#1483 (ply 80) [structural distance $d=28$, QAP obj=35]: the master played b5; under μ our Pg2 plays the role of that game’s Pb4; cross-type roles: our Bd3 as their Rb2, our Qh4 as their Pg3; transferred move g2g3. Synthesis: 1 retrieved analog transferred to Qg5+ (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

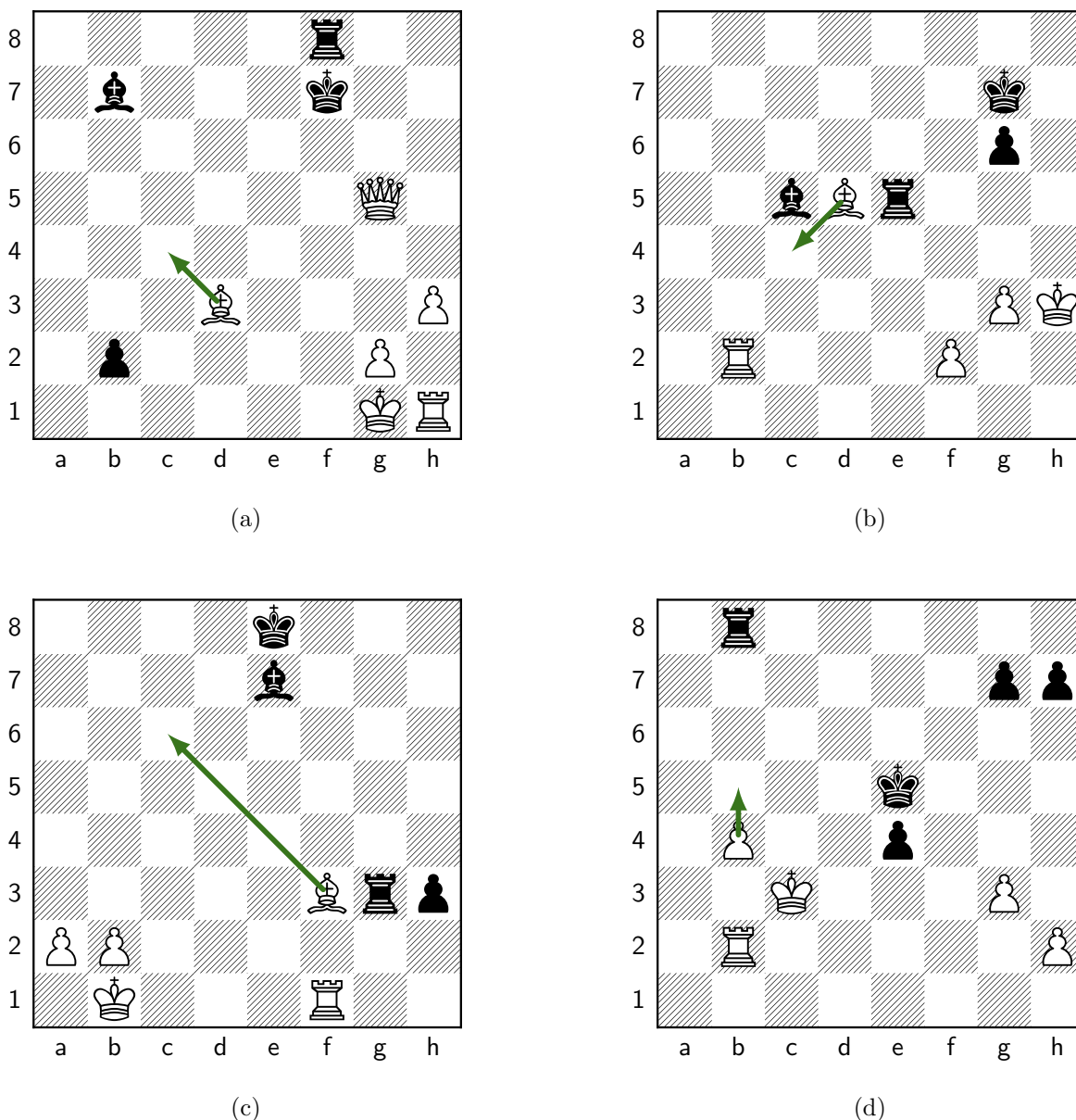
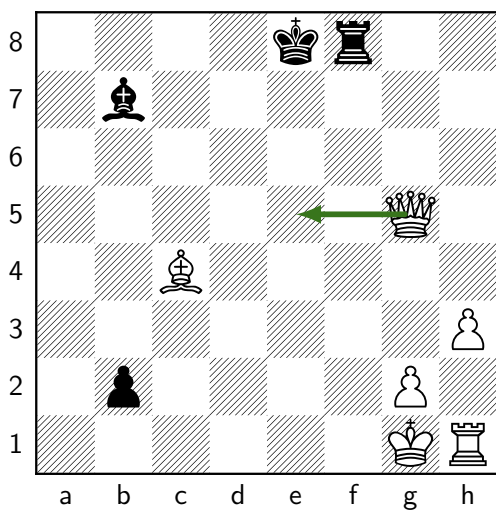
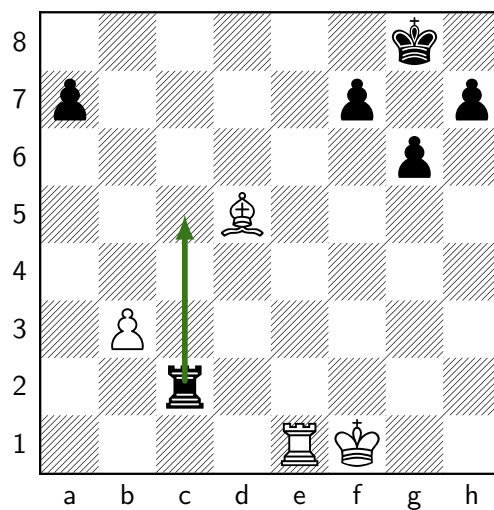


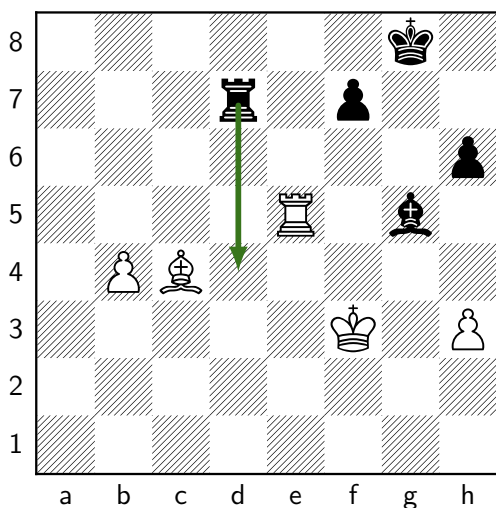
Figure 42: **Move 48, White: Bc4+.** (a) Our position; Melanie plays Bc4+ (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#100` (ply 92) [structural distance $d=30$, QAP obj=2]: the master played Bc4; under μ our Bd3 plays the role of that game’s Bd5; cross-type roles: our Qg5 as their Rb2, our pb2 as their re5; transferred move d3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2106` (ply 90) [structural distance $d=32$, QAP obj=2]: the master played Bc6+; under μ our Bd3 plays the role of that game’s Bf3; cross-type roles: our Qg5 as their Rf1, our Rh1 as their Pa2; transferred move d3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1483` (ply 80) [structural distance $d=30$, QAP obj=35]: the master played b5; under μ our Pg2 plays the role of that game’s Pb4; cross-type roles: our Qg5 as their Rb2, our Bd3 as their Pg3; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to Bc4+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



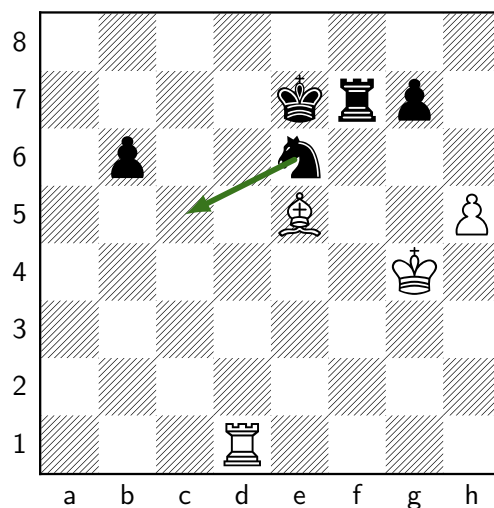
(a)



(b)

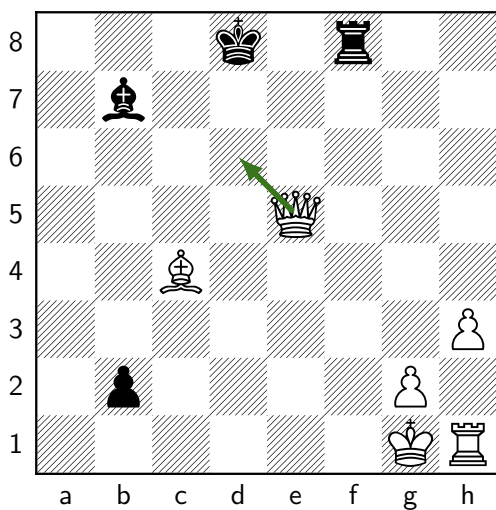


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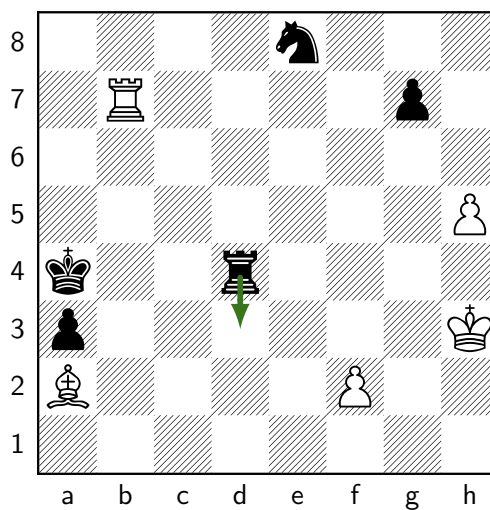


(d)

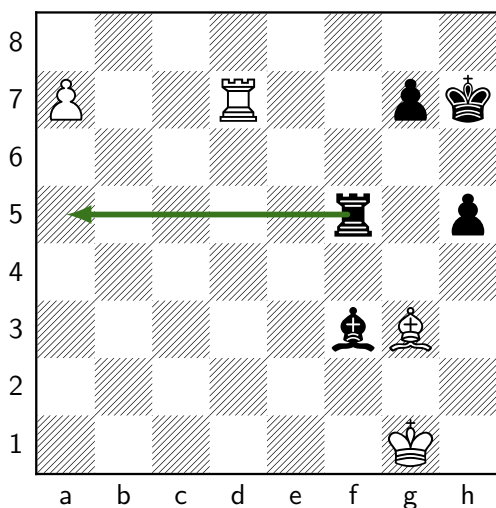
Figure 43: **Move 49, White: Qe5+.** (a) Our position; Melanie plays Qe5+ (g5e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3925 (ply 89) [structural distance $d=32$, QAP obj=23]: the master played Rc5; under μ our Qg5 plays the role of that game's Rc7 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa7, our Bc4 as their Ph7; transferred move g5e5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2485 (ply 85) [structural distance $d=31$, QAP obj=18]: the master played Rd4; under μ our Qg5 plays the role of that game's Rd2 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Ph6; transferred move g5e5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3575 (ply 101) [structural distance $d=32$, QAP obj=13]: the master played Nc5; under μ our Qg5 plays the role of that game's Ne3 (a **cross-type** correspondence); transferred move g5e5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qe5+ (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



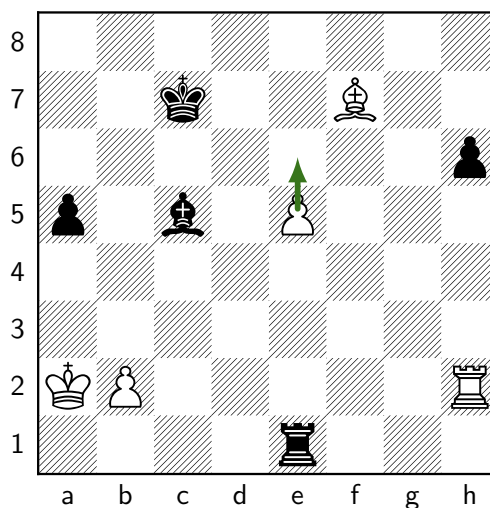
(a)



(b)

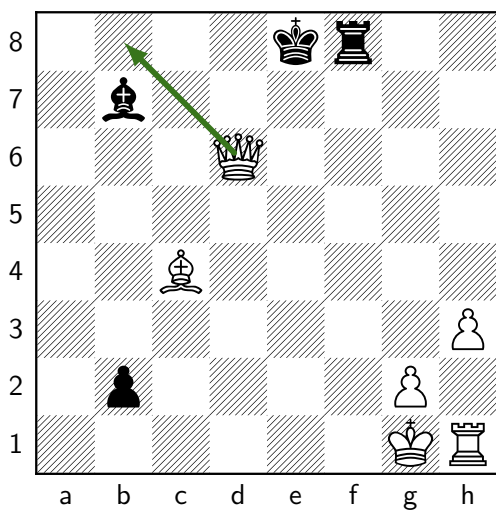


(c)

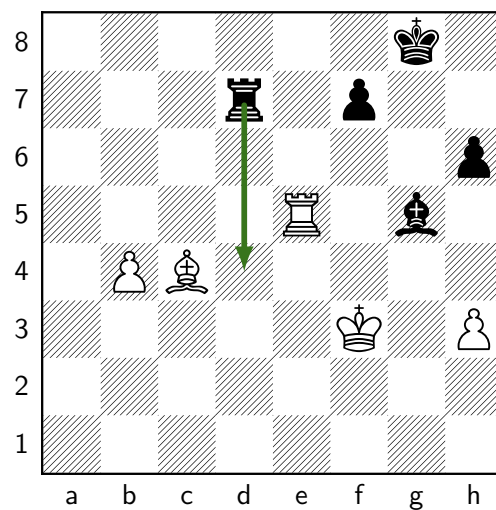


(d)

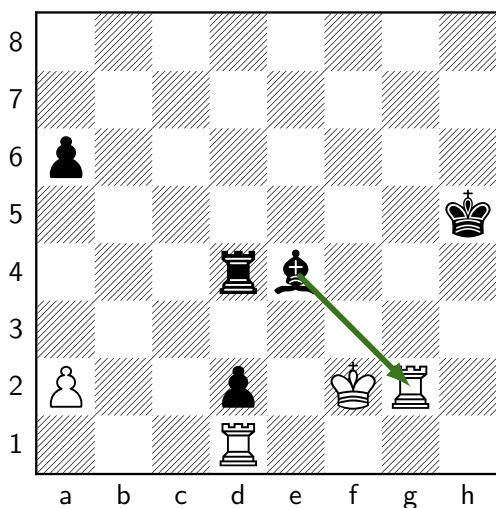
Figure 44: **Move 50, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (e5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#477` (ply 101) [structural distance $d=29$, QAP obj=13]: the master played Rd3+; under μ our Qe5 plays the role of that game's Rd5 (a **cross-type** correspondence); cross-type roles: our Bc4 as their Ne8, our bb7 as their rb7; transferred move e5d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#991` (ply 115) [structural distance $d=30$, QAP obj=27]: the master played Ra5; under μ our Bc4 plays the role of that game's Rf4 (a **cross-type** correspondence); cross-type roles: our bb7 as their rd7, our Rh1 as their Ph5; transferred move c4b3. (d) Retrieved from `ccr1_top_p0002#1987` (ply 92) [structural distance $d=27$, QAP obj=25]: the master played e6; under μ our Pg2 plays the role of that game's Pe5; cross-type roles: our bb7 as their re1, our Rh1 as their Pb2; transferred move g2g4. Synthesis: 1 retrieved analog transferred to Qd6+ (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



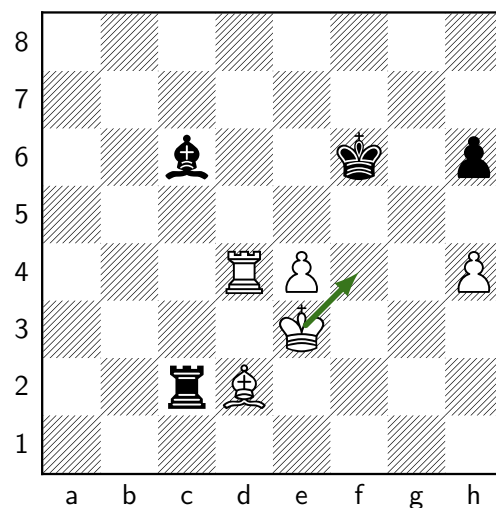
(a)



(b)

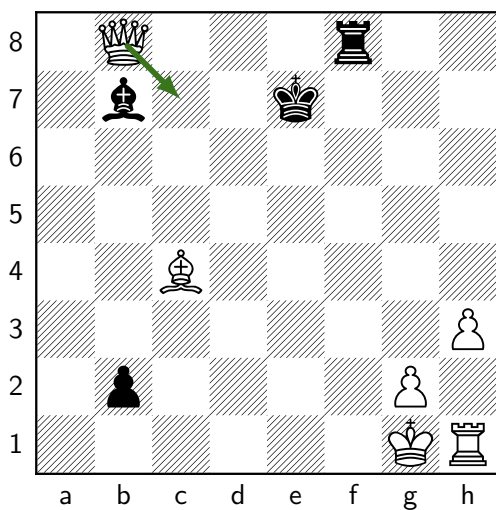


(c)

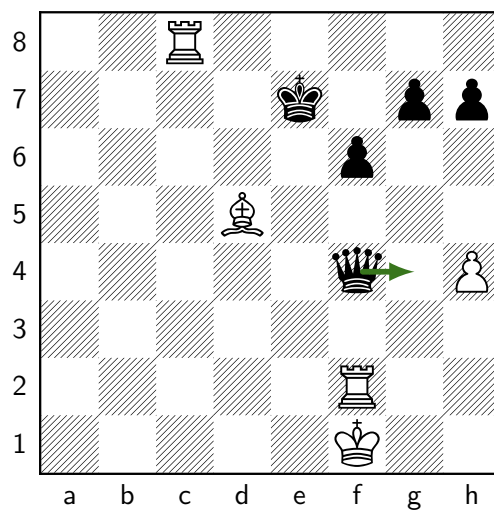


(d)

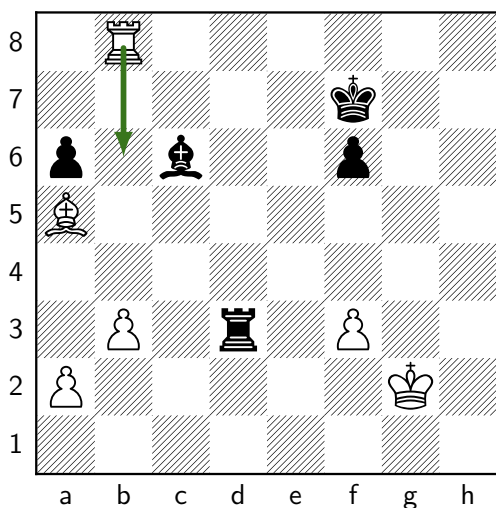
Figure 45: **Move 51, White: Qb8+.** (a) Our position; Melanie plays Qb8+ (d6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2485` (ply 85) [structural distance $d=29$, QAP obj=26]: the master played Rd4; under μ our Rh1 plays the role of that game’s Rd2; cross-type roles: our Qd6 as their re5, our rf8 as their Bg5; transferred move h1h2. (c) Retrieved from `ccr1_top_p0003#2780` (ply 91) [structural distance $d=32$, QAP obj=24]: the master played Bxg2; under μ our Qd6 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Bc4 as their Rd4, our Rh1 as their pa2; transferred move d6f8. (d) Retrieved from `ccr1_top_p0003#2217` (ply 98) [structural distance $d=31$, QAP obj=16]: the master played Kf4; under μ our Kg1 plays the role of that game’s Ke3; cross-type roles: our Qd6 as their rc2, our rf8 as their Bd2; transferred move g1h2. Synthesis: no retrieved analog transferred to Qb8+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



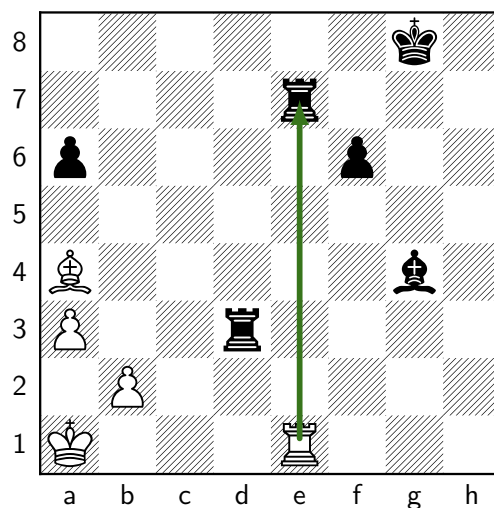
(a)



(b)

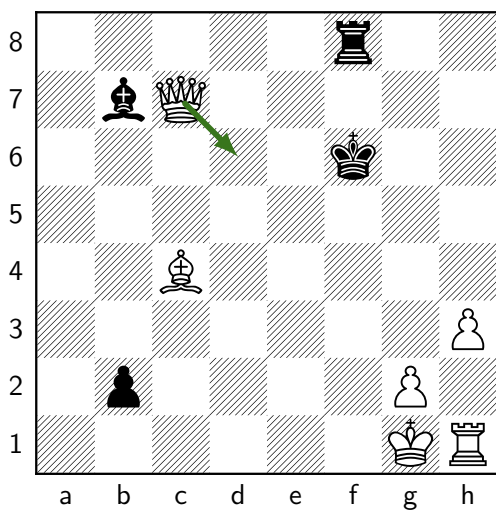


(c)

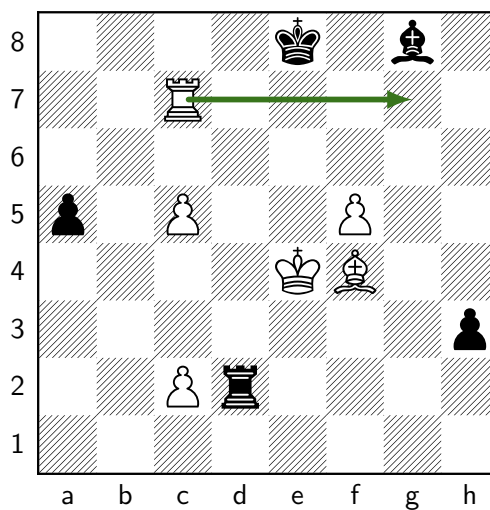


(d)

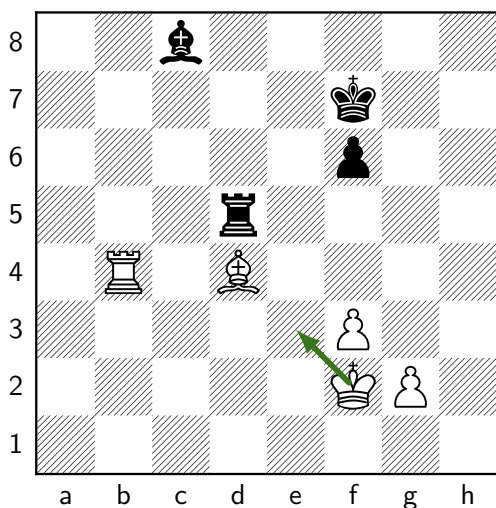
Figure 46: **Move 52, White: Qc7+.** (a) Our position; Melanie plays Qc7+ (b8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2527` (ply 91) [structural distance $d=37$, QAP obj=19]: the master played Qg4; under μ our Qb8 plays the role of that game's Qf5; cross-type roles: our Bc4 as their Pg7; transferred move b8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1179` (ply 110) [structural distance $d=34$, QAP obj=16]: the master played Rb6; under μ our Qb8 plays the role of that game's Rb8 (a **cross-type** correspondence); cross-type roles: our Bc4 as their Pb3, our Rh1 as their Ba5; transferred move b8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1625` (ply 92) [structural distance $d=35$, QAP obj=11]: the master played Rxe7; under μ our Qb8 plays the role of that game's Re1 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pb2, our Ph3 as their rd3; transferred move b8c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qc7+ (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



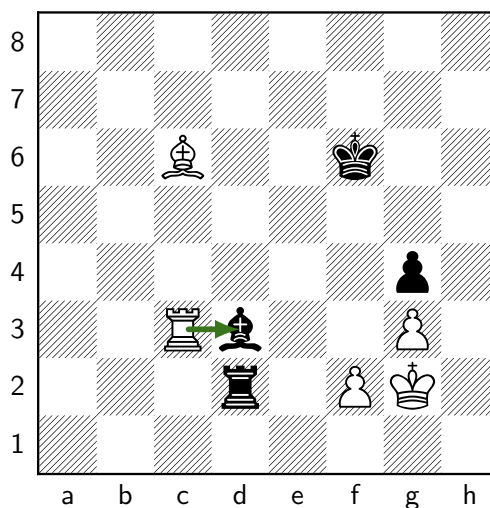
(a)



(b)

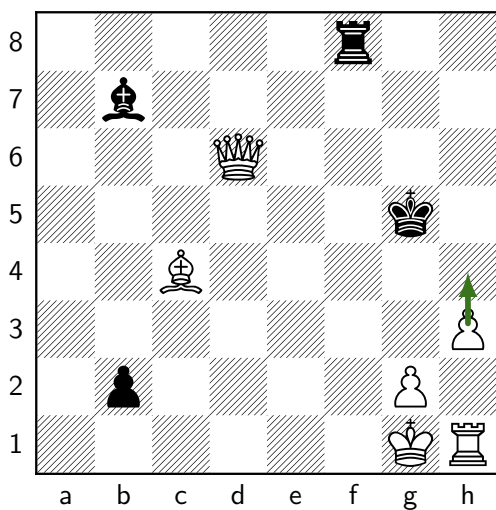


(c)

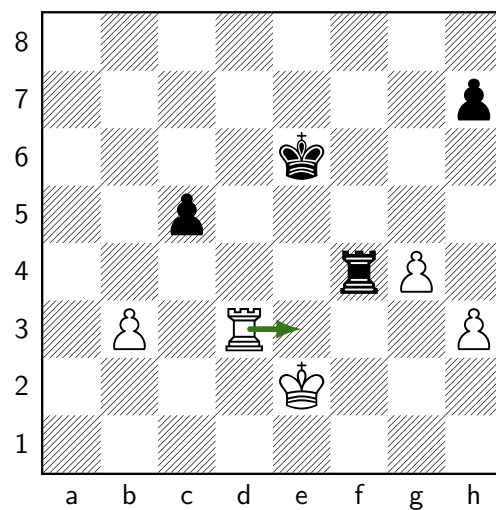


(d)

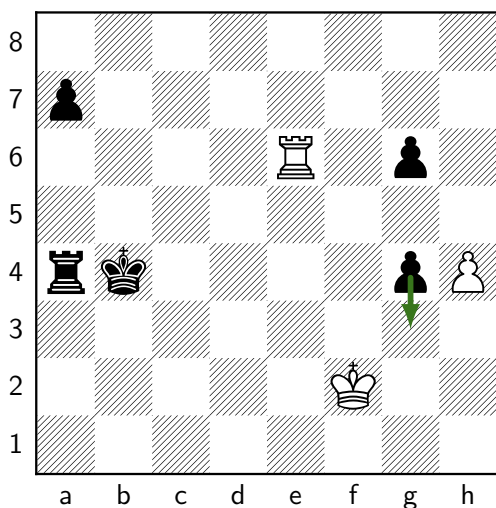
Figure 47: **Move 53, White: Qd6+.** (a) Our position; Melanie plays Qd6+ (c7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#977` (ply 106) [structural distance $d=32$, QAP obj=29]: the master played Rg7; under μ our Bc4 plays the role of that game's Rc7 (a **cross-type** correspondence); cross-type roles: our bb7 as their rd2, our rf8 as their ph3; transferred move c4b5. (c) Retrieved from `ccr1_top_p0000#1362` (ply 118) [structural distance $d=31$, QAP obj=26]: the master played Ke3; under μ our Kg1 plays the role of that game's Kf2; cross-type roles: our Rh1 as their Pg2, our Qc7 as their rd5; transferred move g1f1. (d) Retrieved from `ccr1_top_p0003#2960` (ply 96) [structural distance $d=29$, QAP obj=26]: the master played Rxd3; under μ our Qc7 plays the role of that game's Rc3 (a **cross-type** correspondence); cross-type roles: our bb7 as their rd2, our rf8 as their pg4; transferred move c7f4. Synthesis: no retrieved analog transferred to Qd6+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



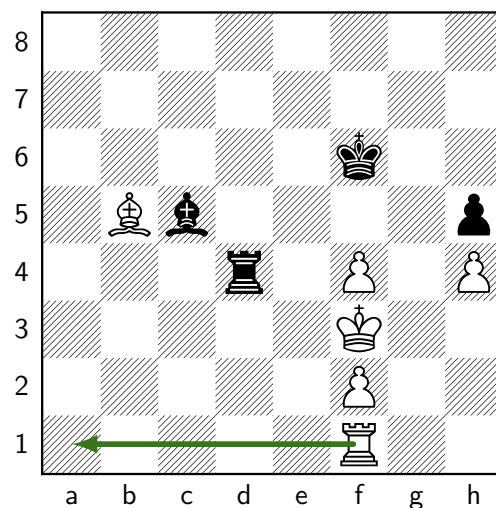
(a)



(b)

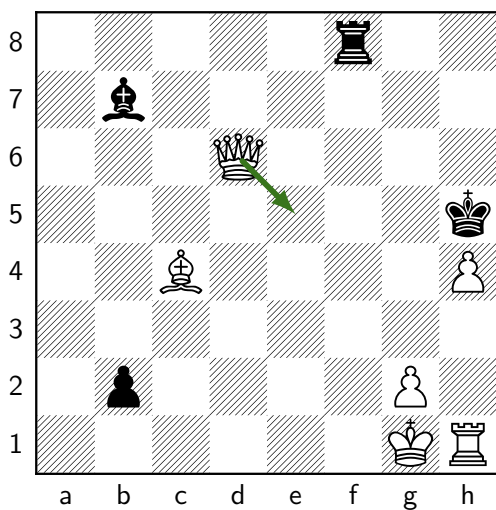


(c)

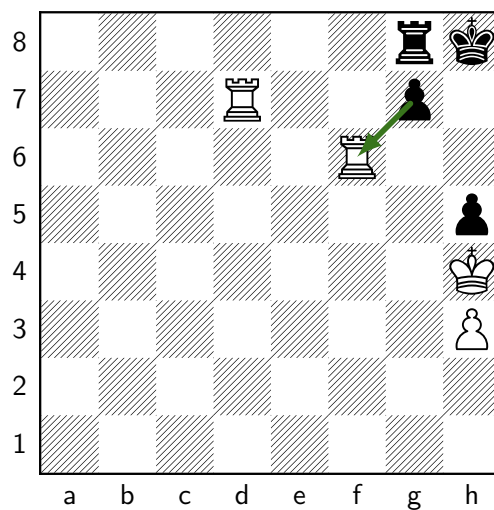


(d)

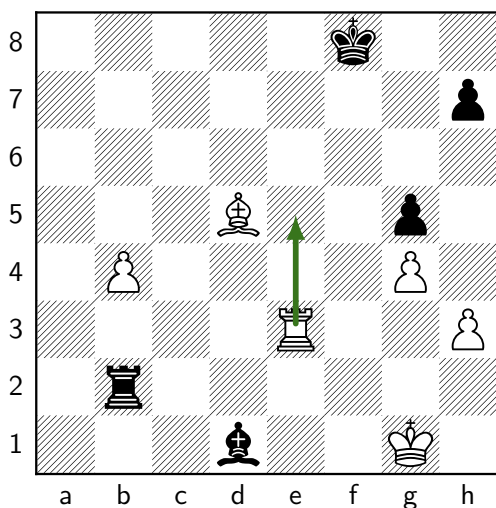
Figure 48: **Move 54, White: h4+ (h3h4)**. (a) Our position; Melanie plays h4+ (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#854` (ply 122) [structural distance $d=25$, QAP obj=8]: the master played **Re3+**; under μ our *Ph3* plays the role of that game’s *Rd3* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *Pb3*, our *Bc4* as their *Ph3*; transferred move **h3h4 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#3172` (ply 97) [structural distance $d=27$, QAP obj=3]: the master played **g3+**; under μ our *Ph3* plays the role of that game’s *Pg5*; cross-type roles: our *rf8* as their *Pa7*, our *Qd6* as their *re6*; transferred move **h3h4 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#94` (ply 98) [structural distance $d=26$, QAP obj=2]: the master played **Ra1**; under μ our *Ph3* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Rh1* as their *Pf2*, our *bb7* as their *rd4*; transferred move **h3h4 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to h4+ (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



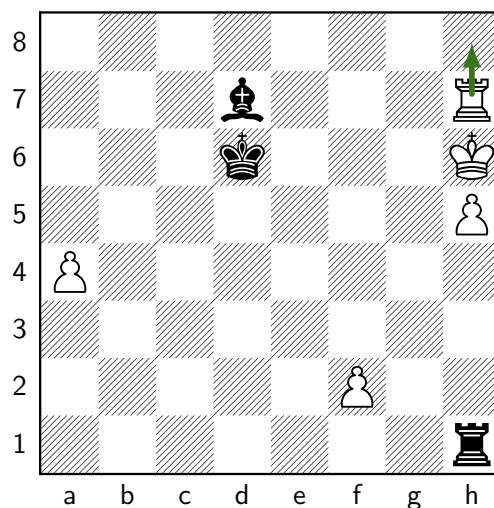
(a)



(b)

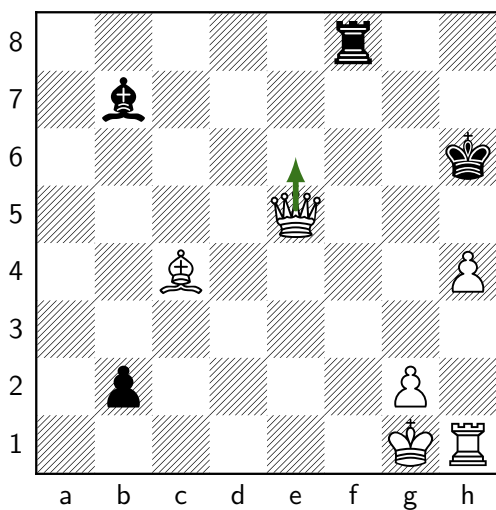


(c)

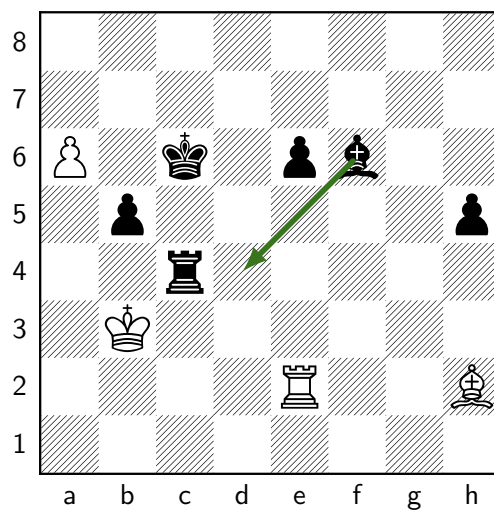


(d)

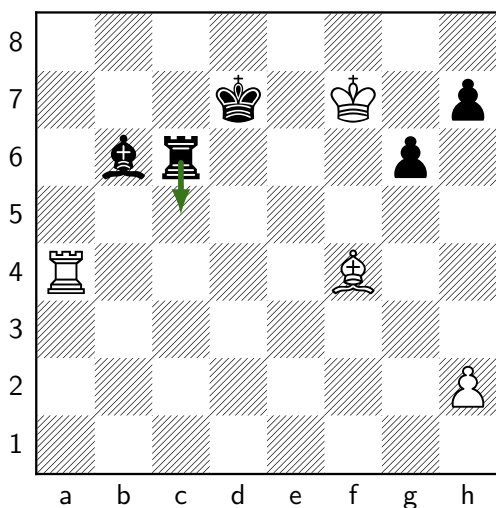
Figure 49: **Move 55, White: Qe5+.** (a) Our position; Melanie plays **Qe5+** (**d6e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1847` (ply 97) [structural distance $d=26$, QAP obj=12]: the master played **gxf6**; under μ our *Qd6* plays the role of that game's *Pg2* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *rd7*; transferred move **d6e5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1576` (ply 118) [structural distance $d=28$, QAP obj=1]: the master played **Re5**; under μ our *Qd6* plays the role of that game's *Re3* (a **cross-type** correspondence); cross-type roles: our *rf8* as their *bd1*, our *Bc4* as their *Pg4*; transferred move **d6e5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3899` (ply 102) [structural distance $d=30$, QAP obj=24]: the master played **Rh8**; under μ our *Qd6* plays the role of that game's *Rh7* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *rh1*, our *Rh1* as their *Pf2*; transferred move **d6f8**. Synthesis: 2 retrieved analogs transferred to **Qe5+** (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



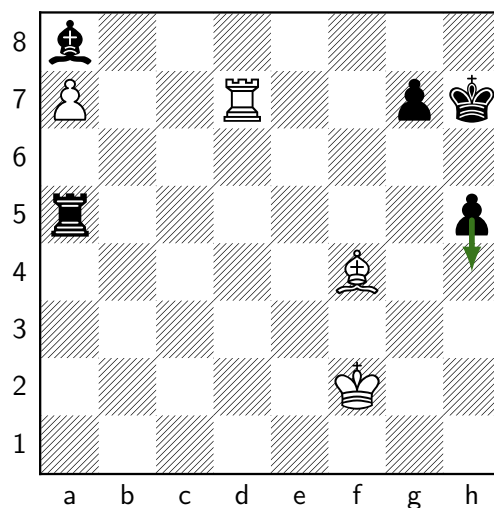
(a)



(b)

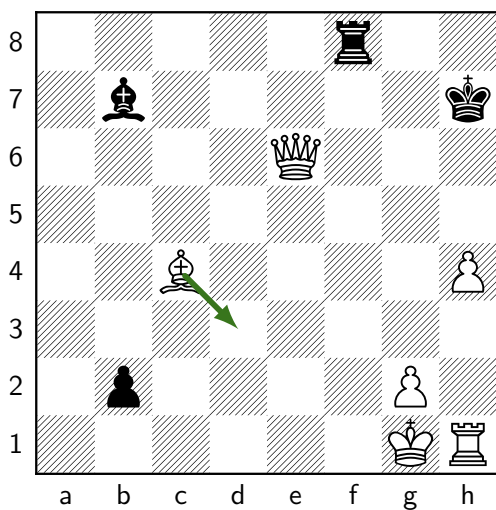


(c)

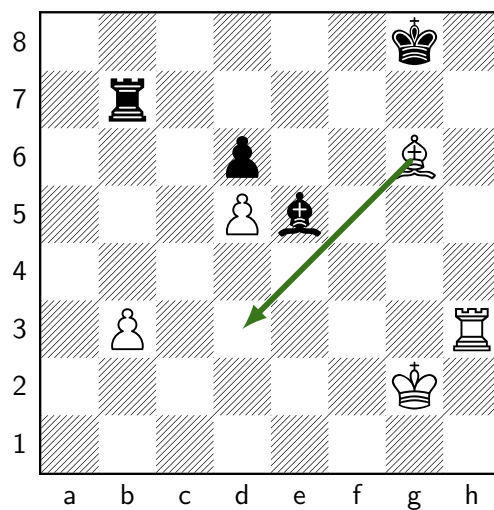


(d)

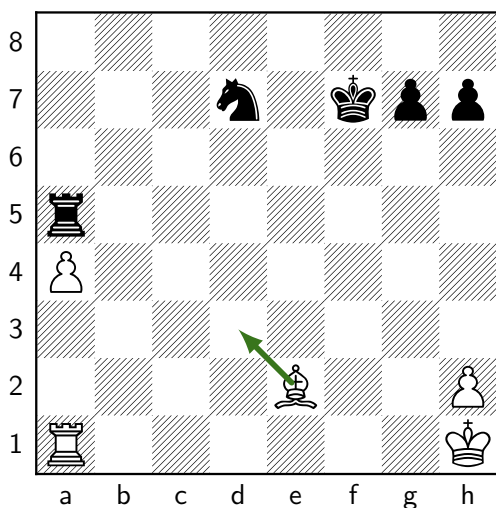
Figure 50: **Move 56, White: Qe6+.** (a) Our position; Melanie plays **Qe6+** (**e5e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2423` (ply 107) [structural distance $d=25$, QAP obj=2]: the master played **Bd4**; under μ our **Qe5** plays the role of that game's **Bf3** (a **cross-type** correspondence); cross-type roles: our **Bc4** as their **Ph5**; transferred move **e5e6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1805` (ply 111) [structural distance $d=27$, QAP obj=2]: the master played **Rc5**; under μ our **Qe5** plays the role of that game's **Rc3** (a **cross-type** correspondence); cross-type roles: our **bb7** as their **ra4**, our **rf8** as their **bf4**; transferred move **e5e6** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#991` (ply 119) [structural distance $d=27$, QAP obj=34]: the master played **h4**; under μ our **Rh1** plays the role of that game's **Ph4** (a **cross-type** correspondence); cross-type roles: our **Ph4** as their **Ba8**, our **bb7** as their **rd7**; transferred move **h1h3**. Synthesis: 2 retrieved analogs transferred to **Qe6+** (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



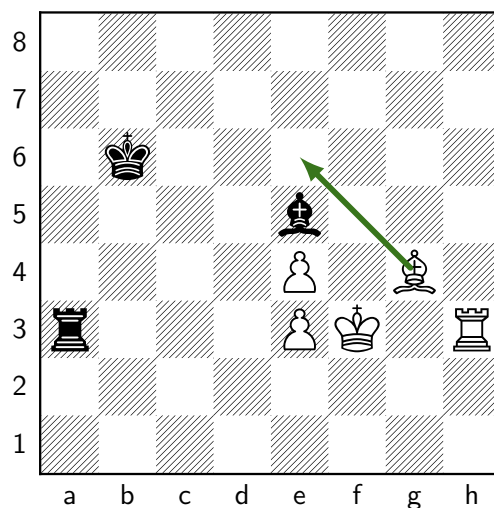
(a)



(b)

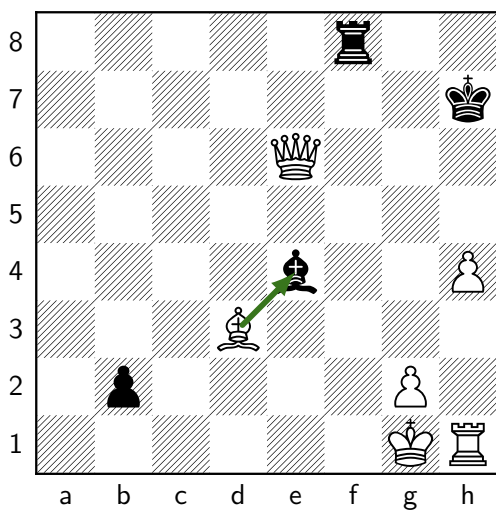


(c)

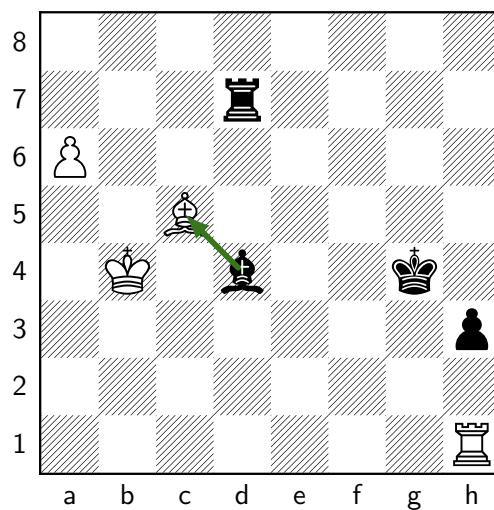


(d)

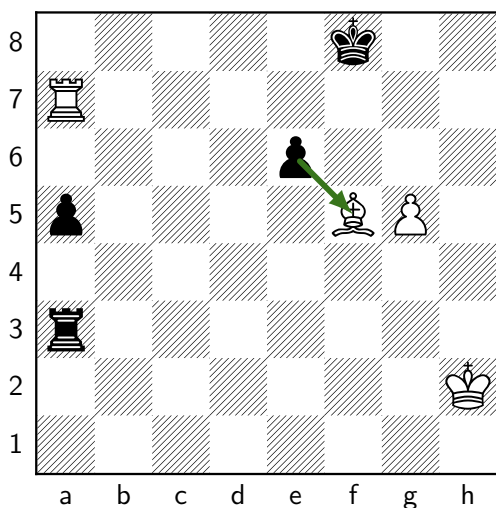
Figure 51: **Move 57, White: Bd3+.** (a) Our position; Melanie plays Bd3+ (c4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2126` (ply 96) [structural distance $d=32$, QAP obj=16]: the master played Bd3; under μ our *Bc4* plays the role of that game’s *Bg6*; cross-type roles: our *Qe6* as their *Rh3*, our *Rh1* as their *Pd5*; transferred move c4d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2238` (ply 98) [structural distance $d=31$, QAP obj=1]: the master played Bd3; under μ our *Bc4* plays the role of that game’s *Be2*; cross-type roles: our *Qe6* as their *Ra1*, our *rf8* as their *Ph2*; transferred move c4d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2676` (ply 124) [structural distance $d=30$, QAP obj=37]: the master played Be6; under μ our *Rh1* plays the role of that game’s *Bg4* (a **cross-type** correspondence); cross-type roles: our *bb7* as their *ra3*, our *Qe6* as their *Rh3*; transferred move h1h3. Synthesis: 2 retrieved analogs transferred to Bd3+ (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



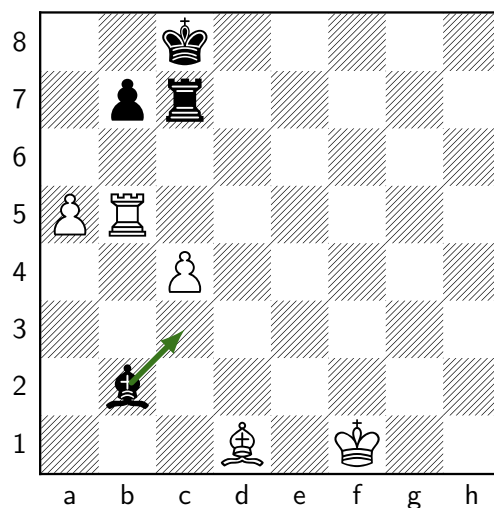
(a)



(b)

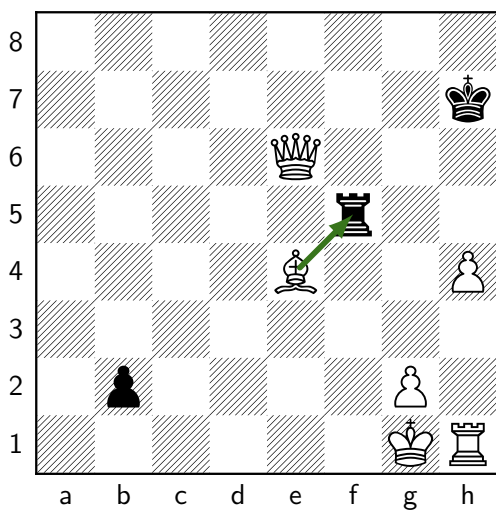


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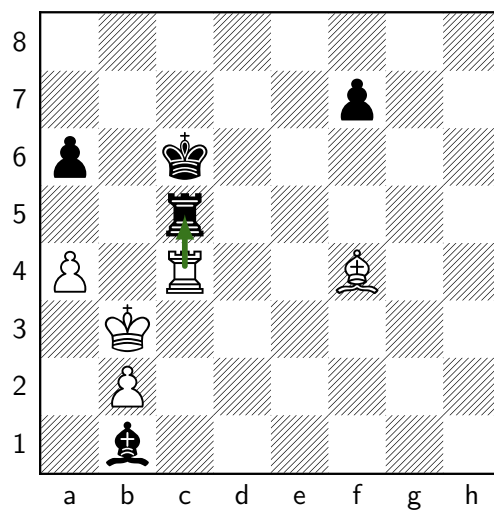


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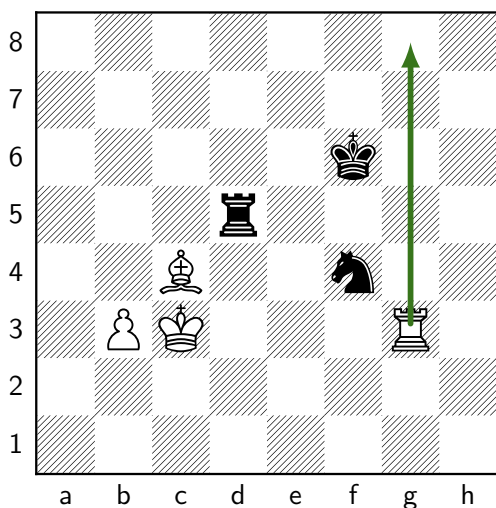
Figure 52: **Move 58, White: Bxe4+.** (a) Our position; Melanie plays Bxe4+ (d3e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2188` (ply 131) [structural distance $d=34$, QAP obj=40]: the master played Bxc5+; under μ our Bd3 plays the role of that game’s Bd5; cross-type roles: our Qe6 as their Rd7, our rf8 as their pa6; transferred move d3e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#355` (ply 107) [structural distance $d=40$, QAP obj=26]: the master played exf5; under μ our Bd3 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our Rh1 as their pg5, our Qe6 as their Ra3; transferred move d3e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#276` (ply 105) [structural distance $d=42$, QAP obj=10]: the master played Bc3; under μ our Bd3 plays the role of that game’s Bb7; cross-type roles: our Qe6 as their Pb7, our rf8 as their pa5; transferred move d3e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxe4+ (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



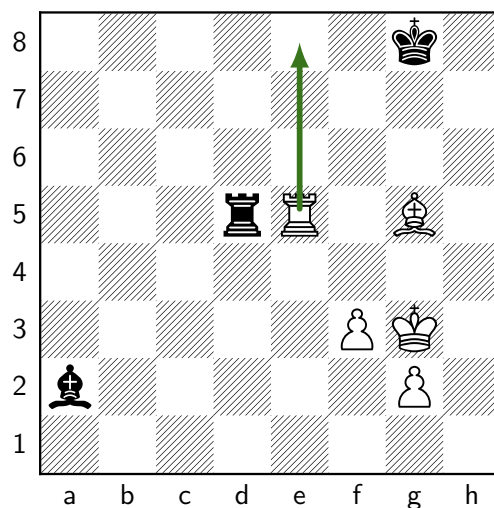
(a)



(b)



(c)



(d)

Figure 53: **Move 59, White: Bxf5+.** (a) Our position; Melanie plays Bxf5+ (e4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2730` (ply 100) [structural distance $d=30$, QAP obj=41]: the master played Rxc5+; under μ our *Be4* plays the role of that game's *Rc4* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *bb1*, our *Rh1* as their *Pb2*; transferred move e4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1617` (ply 124) [structural distance $d=34$, QAP obj=20]: the master played Rg8; under μ our *Rh1* plays the role of that game's *Rg3*; cross-type roles: our *Qe6* as their *Pb3*, our *pb2* as their *nf4*; transferred move h1h3. (d) Retrieved from `ccr1_top_p0003#1197` (ply 128) [structural distance $d=34$, QAP obj=20]: the master played Re8+; under μ our *Qe6* plays the role of that game's *Re5* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *ba2*, our *Rh1* as their *Pg2*; transferred move e6c8. Synthesis: 1 retrieved analog transferred to Bxf5+ (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

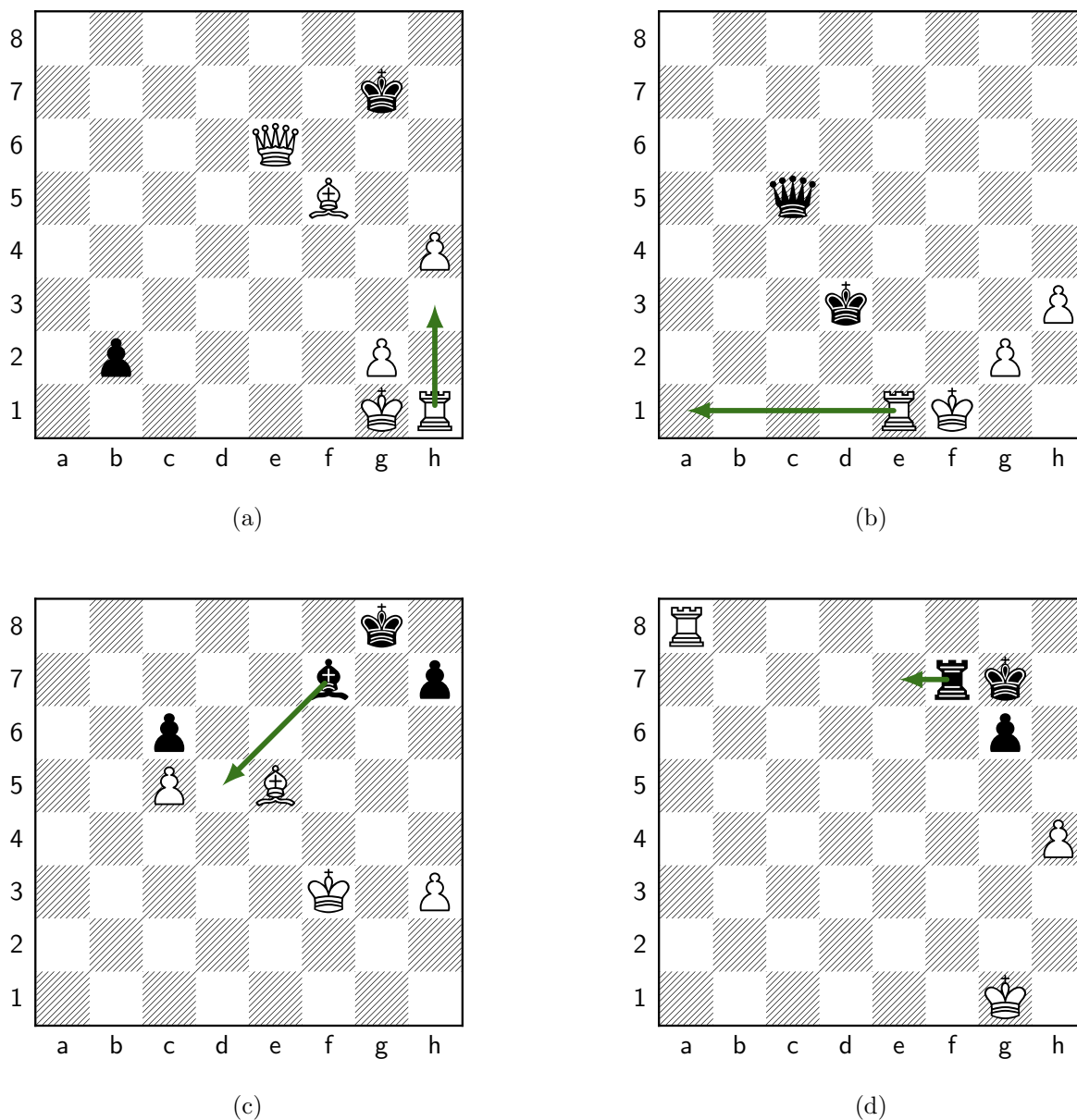
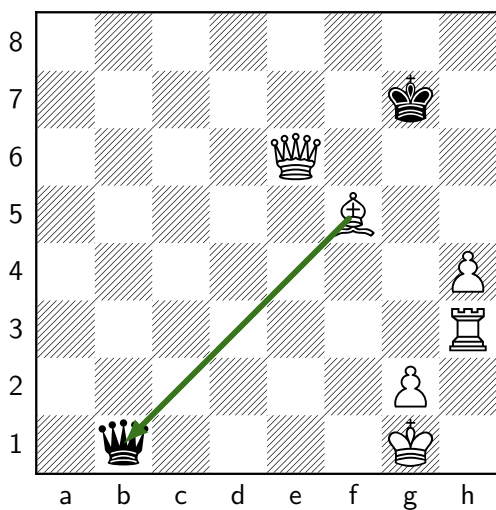
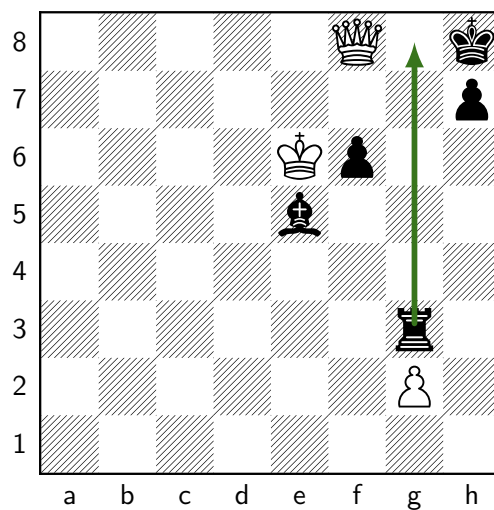


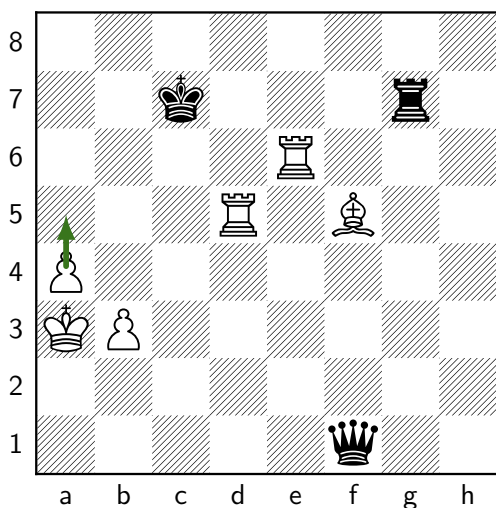
Figure 54: **Move 60, White: Rh3.** (a) Our position; Melanie plays Rh3 (**h1h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1557` (ply 132) [structural distance $d=23$, QAP obj=3]: the master played **Ra1**; under μ our *Rh1* plays the role of that game’s *Re1*; cross-type roles: our *Bf5* as their *Ph3*, our *Pg2* as their *qc5*; transferred move **h1h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3022` (ply 119) [structural distance $d=23$, QAP obj=2]: the master played **Bd5+**; under μ our *Rh1* plays the role of that game’s *Bf2* (a **cross-type** correspondence); cross-type roles: our *Bf5* as their *Pc6*, our *Ph4* as their *be5*; transferred move **h1h3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2597` (ply 117) [structural distance $d=22$, QAP obj=2]: the master played **Re7**; under μ our *Rh1* plays the role of that game’s *Rf2*; cross-type roles: our *Pg2* as their *ra8*, our *Bf5* as their *ph4*; transferred move **h1h3** (**endorses this move**). Synthesis: 14 retrieved analogs transferred to **Rh3** (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



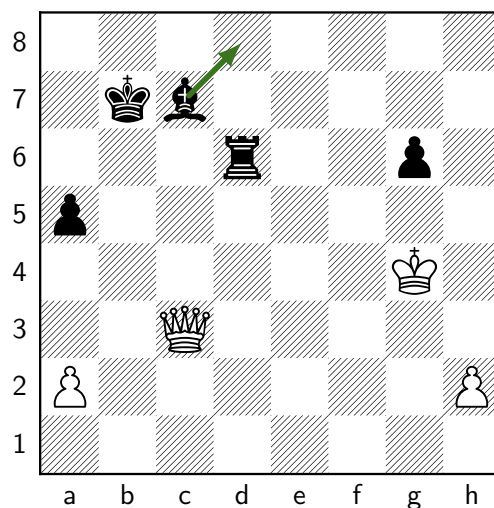
(a)



(b)

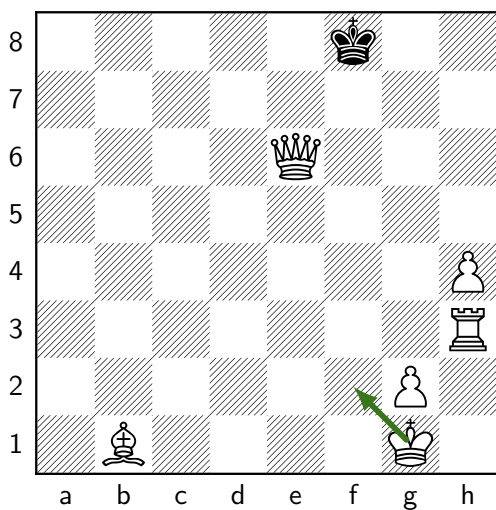


(c)

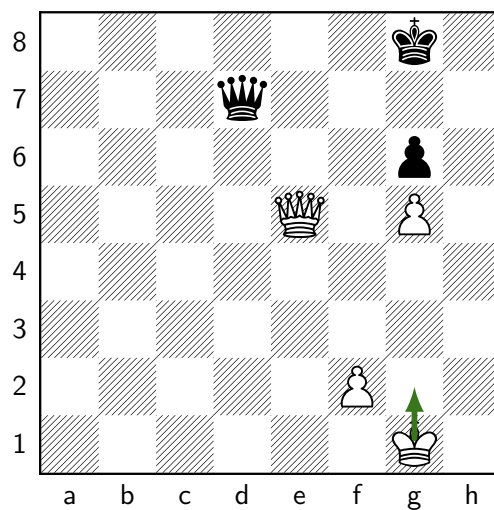


(d)

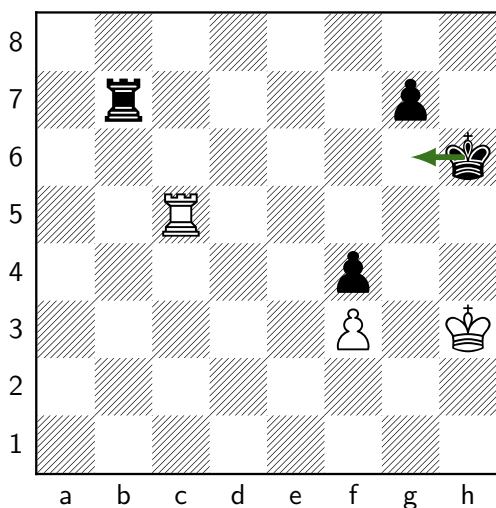
Figure 55: **Move 61, White: Bxb1.** (a) Our position; Melanie plays Bxb1 (f5b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#529` (ply 107) [structural distance $d=33$, QAP obj=29]: the master played Rg8; under μ our Bf5 plays the role of that game's Rg6 (a **cross-type** correspondence); cross-type roles: our Rh3 as their Pf6, our Qe6 as their Be5; transferred move f5b1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#41` (ply 120) [structural distance $d=31$, QAP obj=4]: the master played a5; under μ our Bf5 plays the role of that game's Pa4 (a **cross-type** correspondence); cross-type roles: our Qe6 as their Rd5, our Ph4 as their Bf5; transferred move f5b1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3825` (ply 105) [structural distance $d=23$, QAP obj=3]: the master played Bd8; under μ our Bf5 plays the role of that game's Bc2; cross-type roles: our Qe6 as their Rd6, our Rh3 as their Pg6; transferred move f5b1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bxb1 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



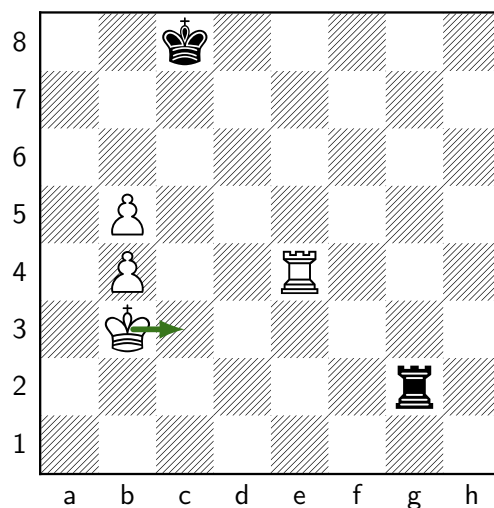
(a)



(b)

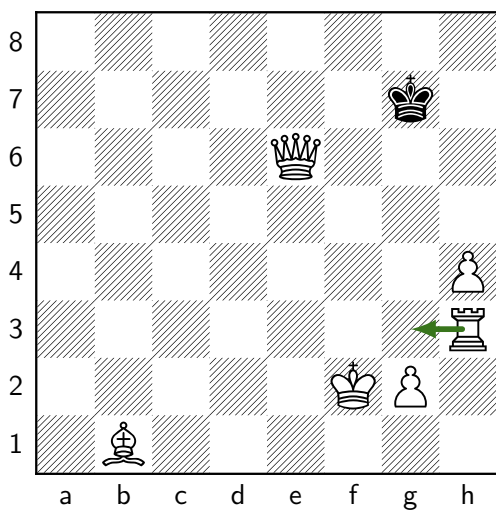


(c)

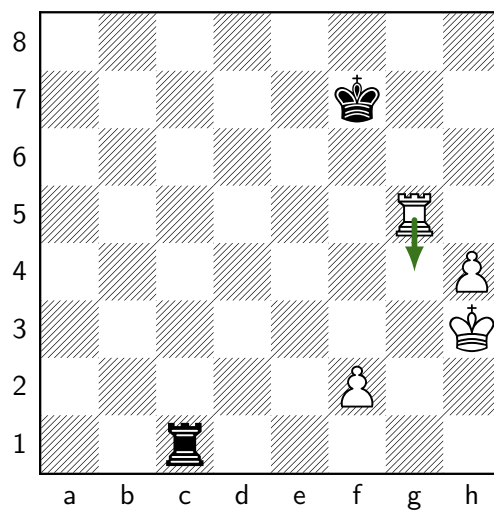


(d)

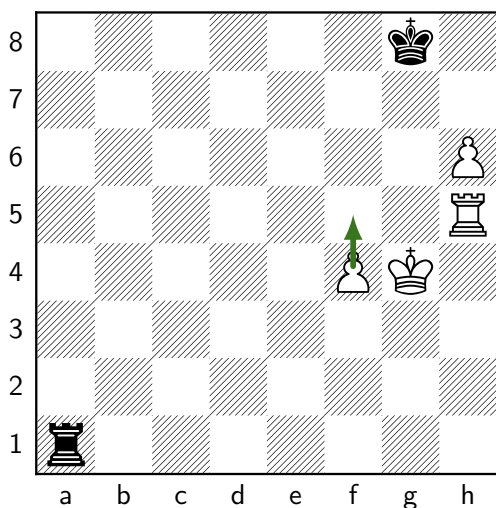
Figure 56: **Move 62, White: Kf2**. (a) Our position; Melanie plays Kf2 (g1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3283 (ply 112) [structural distance $d=15$, QAP obj=1]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Rh3 as their Pg5, our Bb1 as their pg6; transferred move g1f2 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2018 (ply 121) [structural distance $d=15$, QAP obj=1]: the master played Kg6; under μ our Kg1 plays the role of that game’s Kh3; cross-type roles: our Ph4 as their rc5, our Qe6 as their Pf4; transferred move g1f2 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3614 (ply 120) [structural distance $d=16$, QAP obj=1]: the master played Kc3; under μ our Kg1 plays the role of that game’s Kb3; cross-type roles: our Bb1 as their rg2, our Ph4 as their Re4; transferred move g1f2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kf2 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



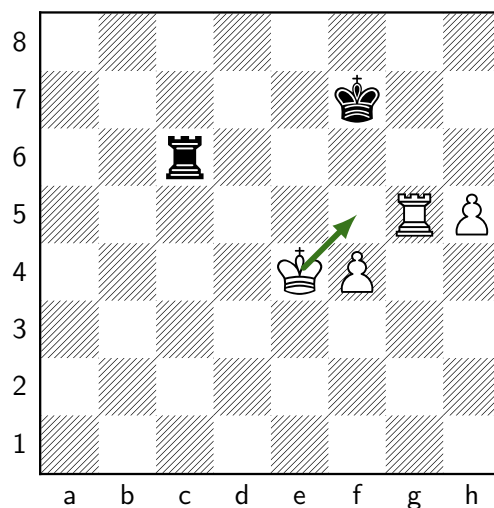
(a)



(b)



(c)



(d)

Figure 57: **Move 63, White: Rg3+.** (a) Our position; Melanie plays Rg3+ (h3g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3862` (ply 114) [structural distance $d=14$, QAP obj=2]: the master played Rg4; under μ our Rh3 plays the role of that game's Rg5; cross-type roles: our Ph4 as their rc1, our Bb1 as their Pf2; transferred move h3g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1797` (ply 114) [structural distance $d=12$, QAP obj=0]: the master played f5; under μ our Rh3 plays the role of that game's Pf4 (a **cross-type** correspondence); cross-type roles: our Pg2 as their ra1, our Qe6 as their Rh5; transferred move h3g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1542` (ply 134) [structural distance $d=13$, QAP obj=2]: the master played Kf5; under μ our Kf2 plays the role of that game's Ke4; cross-type roles: our Bb1 as their rc6; transferred move f2f3. Synthesis: 2 retrieved analogs transferred to Rg3+ (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

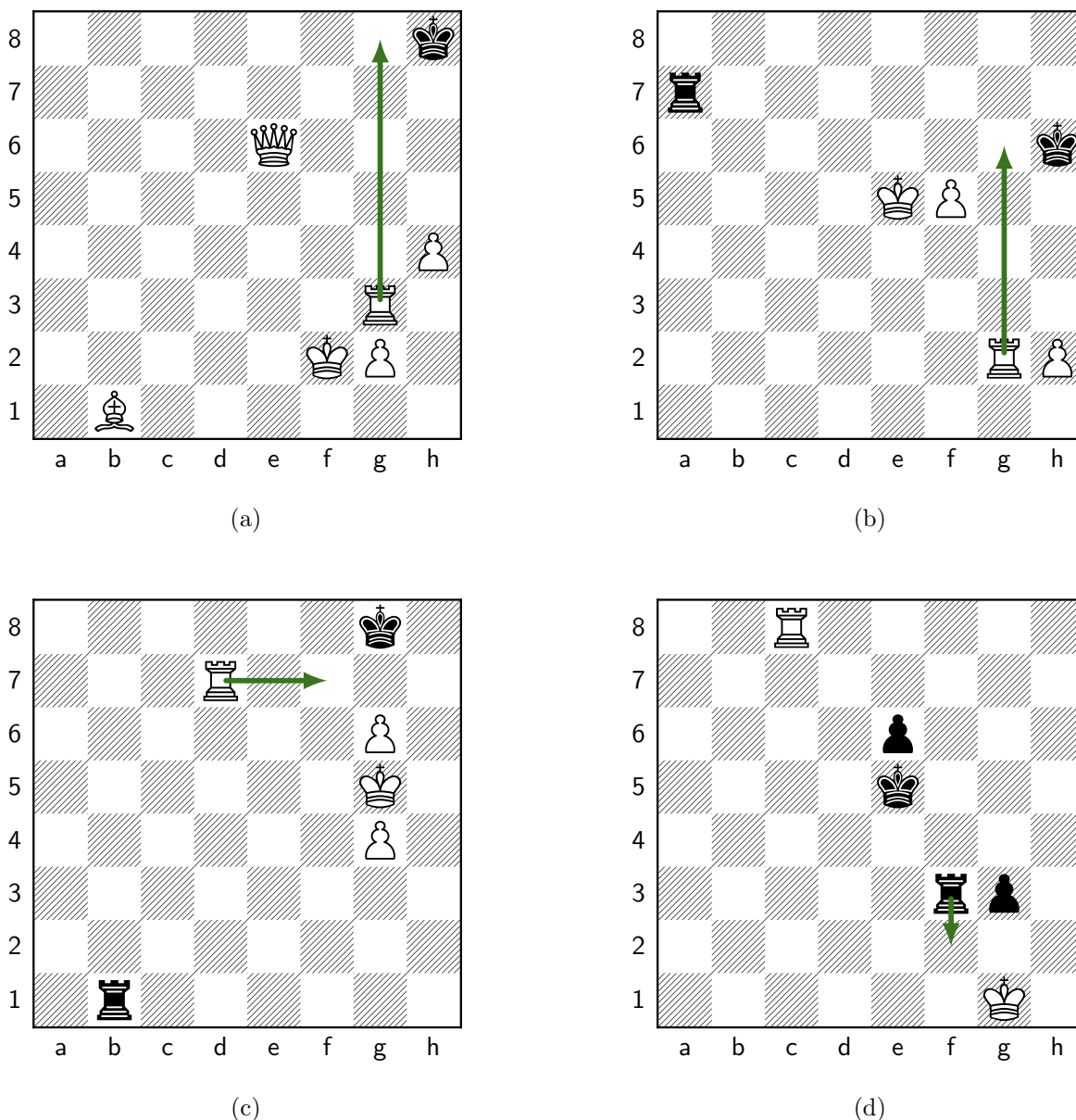


Figure 58: **Move 64, White: Rg8#.** (a) Our position; Melanie plays Rg8# (g3g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1562` (ply 114) [structural distance $d=13$, QAP obj=1]: the master played Rg6+; under μ our Rg3 plays the role of that game’s Rg2; cross-type roles: our Bb1 as their Pf5, our Pg2 as their ra7; transferred move g3g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2860` (ply 130) [structural distance $d=15$, QAP obj=0]: the master played Rf7; under μ our Rg3 plays the role of that game’s Rd7; cross-type roles: our Bb1 as their rb1, our Qe6 as their Pg6; transferred move g3g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#587` (ply 119) [structural distance $d=15$, QAP obj=0]: the master played Rf2; under μ our Rg3 plays the role of that game’s Rf6; cross-type roles: our Pg2 as their rc8, our Bb1 as their Pg3; transferred move g3g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rg8# (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.